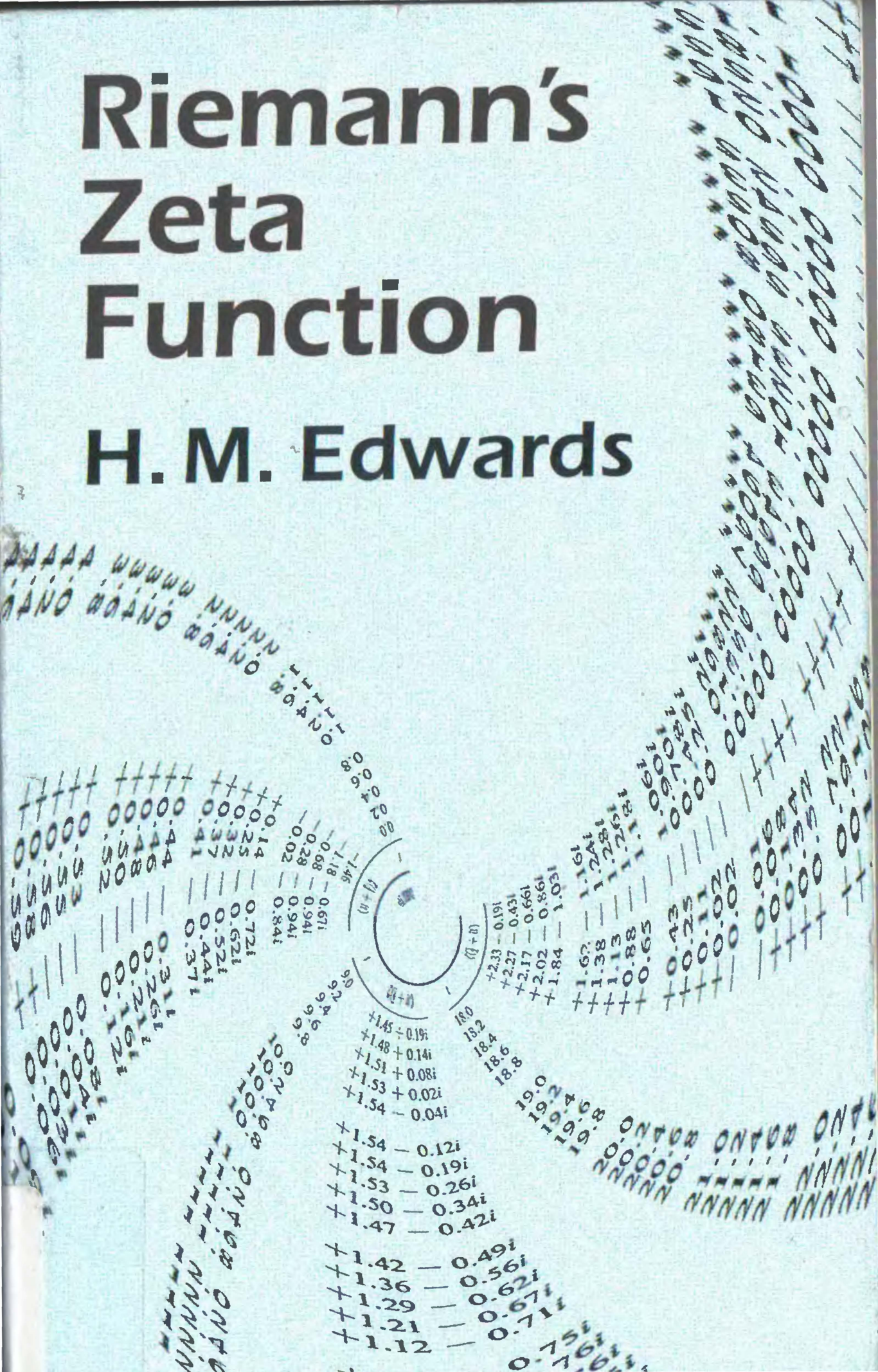
Riemann's Zeta Function

H. M. Edwards



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Preface

My primary objective in this book is to make a point, not about analytic number theory, but about the way in which mathematics is and ought to be studied. Briefly put, I have tried to say to students of mathematics that they should *read the classics* and beware of secondary sources.

This is a point which Eric Temple Bell makes repeatedly in his biographies of great mathematicians in *Men of Mathematics*. In case after case, Bell points out that the men of whom he writes learned their mathematics not by studying in school or by reading textbooks, but by going straight to the sources and reading the best works of the masters who preceded them. It is a point which in most fields of scholarship at most times in history would have gone without saying.

No mathematical work is more clearly a classic than Riemann's memoir *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse,* published in 1859. Much of the work of many of the great mathematicians since Riemann—men like Hadamard, von Mangoldt, de la Vallée Poussin, Landau, Hardy, Littlewood, Siegel, Polya, Jensen, Lindelöf, Bohr, Selberg, Artin, Hecke, to name just a few of the most important—has stemmed directly from the ideas contained in this eight-page paper. According to legend, the person who acquired the copy of Riemann's collected works from the library of Adolph Hurwitz after Hurwitz's death found that the book would automatically fall open to the page on which the Riemann hypothesis was stated.

Yet it is safe to say both that the dictum "read the classics" is not much heard among contemporary mathematicians and that few students read *Ueber die Anzahl* . . . today. On the contrary, the mathematics of previous generations is generally considered to be unrigorous and naïve, stated in obscure terms which can be vastly simplified by modern terminology, and full of false starts and misstatements which a student would be best

advised to avoid. Riemann in particular is avoided because of his reputation for lack of rigor (his "Dirichlet principle" is remembered more for the fact that Weierstrass pointed out that its proof was inadequate than it is for the fact that it was after all correct and that with it Riemann revolutionized the study of Abelian functions), because of his difficult style, and because of a general impression that the valuable parts of his work have all been gleaned and incorporated into subsequent more rigorous and more readable works.

These objections are all valid. When Riemann makes an assertion, it may be something which the reader can verify himself, it may be something which Riemann has proved or intends to prove, it may be something which was not proved rigorously until years later, it may be something which is still unproved, and, alas, it may be something which is not true unless the hypotheses are strengthened. This is especially distressing for a modern reader who is trained to digest each statement before going on to the next. Moreover, Riemann's style is extremely difficult. His tragically brief life was too occupied with mathematical creativity for him to devote himself to elegant exposition or to the polished presentation of all of his results. His writing is extremely condensed and *Ueber die anzahl* . . . in particular is simply a resumé of very extensive researches which he never found the time to expound upon at greater length; it is the only work he ever published on number theory, although Siegel found much valuable new material on number theory in Riemann's private papers. Finally, it is certainly true that most of Riemann's best ideas have been incorporated in later, more readable works.

Nonetheless, it is just as true that one should read the classics in this case as in any other. No secondary source can duplicate Riemann's insight. Riemann was so far ahead of his time that it was 30 years before anyone else began really to grasp his ideas—much less to have their own ideas of comparable value. In fact, Riemann was so far ahead of his time that the results which Siegel found in the private papers were a major contribution to the field when they were published in 1932, seventy years after Riemann discovered them. Any simplification, paraphrasing, or reworking of Riemann's ideas runs a grave risk of missing an important idea, of obscuring a point of view which was a source of Riemann's insight, or of introducing new technicalities or side issues which are not of real concern. There is no mathematician since Riemann whom I would trust to revise his work.

The perceptive reader will of course have noted the paradox here of a secondary source denouncing secondary sources. I might seem to be saying, "Do not read this book." But he will also have seen the answer to the

paradox. What I am saying is: Read the classics, not just Riemann, but all the major contributions to analytic number theory that I discuss in this book. The purpose of a secondary source is to make the primary sources accessible to you. If you can read and understand the primary sources without reading this book, more power to you. If you read this book without reading the primary sources you are like a man who carries a sack lunch to a banquet.

Acknowledgments

It is a pleasure to acknowledge my indebtedness to many individuals and institutions that have aided me in the writing of this book. The two institutions which have supported me during this time are the Washington Square College of New York University and the School of General Studies of the Australian National University; I have been proud to be a part of these particular schools because of their affirmation of the importance of undergraduate teaching in an age when the pressures toward narrow overspecialization are so great.

I am grateful to many libraries for the richness of the resources which they make available to us all. Of the many I have used during the course of the preparation of this book, I think particularly of the following institutions which provided access to relatively rare documents: The New York Public Library, the Courant Institute of Mathematical Sciences, the University of Illinois, Sydney University, the Royal Society of Adelaide, the Linnean Society of New South Wales, the Public Library of New South Wales, and the Australian National University. I am especially grateful to the University Library in Göttingen for giving me access to Riemann's *Nachlass* and for permitting me to photocopy the portions of it relevant to this book.

Among the individuals I would like to thank for their comments on the manuscript are Gabriel Stolzenberg of Northeastern University, David Lubell of Adelphi, Bruce Chandler of New York, Ian Richards of Minnesota, Robert Spira of Michigan State, and Andrew Coppel of the Australian National University. J. Barkley Rosser and D. H. Lehmer were very helpful in providing information on their researches. Carl Ludwig Siegel was very hospitable and generous with his time during my brief visit to Göttingen. And, finally, I am deeply grateful to Wilhelm Magnus for his understanding of my objectives and for his encouragement, which sustained me through many long days when it seemed that the work would never be done.

Chapter 1

Riemann's Paper

1.1 THE HISTORICAL CONTEXT OF THE PAPER

This book is a study of Bernhard Riemann's epoch-making 8-page paper "On the Number of Primes Less Than a Given Magnitude,"† and of the subsequent developments in the theory which this paper inaugurated. This first chapter is an examination and an amplification of the paper itself, and the remaining 11 chapters are devoted to some of the work which has been done since 1859 on the questions which Riemann left unanswered.

The theory of which Riemann's paper is a part had its beginnings in Euler's theorem, proved in 1737, that the sum of the reciprocals of the prime numbers

$$\frac{1}{2}+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+\frac{1}{11}+\frac{1}{13}+\frac{1}{17}+\cdots \tag{1}$$

is a divergent series. This theorem goes beyond Euclid's ancient theorem that there are infinitely many primes [E2] and shows that the primes are rather *dense* in the set of all integers—denser than the squares, for example, in that the sum of the reciprocals of the square numbers converges.

Euler in fact goes beyond the mere statement of the divergence of (1) to say that since $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots$ diverges like the logarithm and since the series (1) diverges like‡ the logarithm of $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\cdots$, the series (1)

†The German title is *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse.* An English translation of the paper is given in the Appendix.

‡This is true by dint of the Euler product formula which gives $\sum(1/n)=\prod(1-p^{-1})^{-1}$ (see Section 1.2); hence $\log\sum(1/n)=-\sum\log(1-p^{-1})=\sum(p^{-1}+\frac{1}{2}p^{-2}+\frac{1}{3}p^{-3}+\cdots)=\sum(1/p)+$ convergent.

must diverge like the log of the log, which Euler writes [E4] as

$$\frac{1}{2}+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+\frac{1}{11}+\cdots=\log(\log\infty). \tag{2}$$

It is not clear exactly what Euler understood this equation to mean—if indeed he understood it as anything other than a mnemonic—but an obvious interpretation of it would be

$$\sum_{p<x} 1/p \sim \log(\log x) \qquad (x\longrightarrow\infty), \tag{2'}$$

where the left side denotes the sum of $1/p$ over all primes p less than x and where the sign $\sim$ means that the relative error is arbitrarily small for x sufficiently large or, what is the same, that the ratio of the two sides approaches one as $x\longrightarrow\infty$. Now

$$\log(\log x)=\int_1^{\log x}\frac{du}{u}=\int_e^x\frac{dv}{v\log v}$$

so (2') says that the integral of $1/v$ relative to the measure $dv/\log v$ diverges in the same way as the integral of $1/v$ relative to the point measure which assigns weight 1 to primes and weight 0 to all other points. In this sense (2') can be regarded as saying that the density of primes is roughly $1/\log v$. However, there is no evidence that Euler thought about the density of primes, and his methods were not adequate to prove the formulation (2') of his statement (2).

Gauss states† in a letter [G2] written in 1849 that he had observed as early as 1792 or 1793 that the density of prime numbers appears on the average to be $1/\log x$ and he says that each new tabulation of primes which was published in the ensuing years had tended to confirm his belief in the accuracy of this approximation. However, he does not mention Euler's formula (2) and he gives no analytical basis for the approximation, which he presents solely as an empirical observation. He gives, in particular, Table I.

TABLE I[a]

x	Count of primes $<x$	$\int\frac{dn}{\log n}$	Difference
500,000	41,556	41,606.4	50.4
1,000,000	78,501	78,627.5	126.5
1,500,000	114,112	114,263.1	151.1
2,000,000	148,883	149,054.8	171.8
2,500,000	183,016	183,245.0	229.0
3,000,000	216,745	216,970.6	225.6

[a]From Gauss [G2].

†For some corroboration of Gauss's claim see his collected works [G3].

Gauss does not say exactly what he means by the symbol $\int(dn/\log n)$, but the data given in Table II, taken from D.N. Lehmer [L9], would indicate that he means n to be a continuous variable integrated from 2 to x, that is, $\int_2^x (dt/\log t)$. Note that Lehmer's count† of primes, which can safely be assumed to be accurate, differs from Gauss's information and that the difference is in *favor* of Gauss's estimate for the larger values of x.

TABLE II[a]

x	Count of primes $< x$	$\int_2^x \frac{dt}{\log t}$	Difference
500,000	41,538	41,606	68
1,000,000	78,498	78,628	130
1,500,000	114,155	114,263	108
2,000,000	148,933	149,055	122
2,500,000	183,072	183,245	173
3,000,000	216,816	216,971	155

[a]Data from Lehmer [L9].

Around 1800 Legendre published in his *Theorie des Nombres* [L11] an empirical formula for the number of primes less than a given value which amounted more or less to the same statement, namely, that the density of primes is $1/\log x$. Although Legendre made some slight attempt to prove his formula, his argument amounts to nothing more than the statement that if the density of primes is assumed to have the form

$$1/(A_1x^{m_1} + A_2x^{m_2} + \cdots)$$

where $m_1 > m_2 > \ldots$, then m_1 cannot be positive [because then the sum (1) would converge]; hence m_1 must be "infinitely small" and the density must be of the form

$$1/(A \log x + B).$$

He then determines A and B empirically. Legendre's formula was well known in the mathematical world and was mentioned prominently by Abel [A2], Dirichlet [D3], and Chebyshev [C2] during the period 1800–1850.

The first significant results beyond Euler's were obtained by Chebyshev around 1850. Chebyshev proved that the relative error in the approximation

$$\pi(x) \sim \int_2^x \frac{dt}{\log t}, \tag{3}$$

†Lehmer insists on counting 1 as a prime. To conform to common usage his counts have therefore been reduced by one in Table II.

where $\pi(x)$ denotes the number of primes less than x, is less than 11% for all sufficiently large x; that is, he proved† that

$$(0.89)\int_2^x \frac{dt}{\log t} < \pi(x) < (1.11)\int_2^x \frac{dt}{\log t}$$

for all sufficiently large x. He proved, moreover, that no approximation of Legendre's form

$$\pi(x) \sim x/(A \log x + B)$$

can be better than the approximation (3) and that if the ratio of $\pi(x)$ to $\int_2^x (dt/\log t)$ approaches a limit as $x \to \infty$, then this limit must be 1. It is clear that Chebyshev was attempting to prove that the relative error in the approximation (3) approaches zero as $x \to \infty$, but it was not until almost 50 years later that this theorem, which is known as the "prime number theorem," was proved. Although Chebyshev's work was published in France well before Riemann's paper, Riemann does not refer to Chebyshev in his paper. He does refer to Dirichlet, however, and Dirichlet, who was acquainted with Chebyshev (see Chebyshev's report on his trip to Western Europe [C5, Vol. 5, p. 245 and pp. 254–255]) would probably have made Riemann aware of Chebyshev's work. Riemann's unpublished papers do contain several of Chebyshev's formulas, indicating that he had studied Chebyshev's work, and contain at least one direct reference to Chebyshev (see Fig. 1).

The real contribution of Riemann's 1859 paper lay not in its results but in its methods. The principal result was a formula‡ for $\pi(x)$ as the sum of an infinite series in which $\int_2^x (dt/\log t)$ is by far the largest term. However, Riemann's proof of this formula was inadequate; in particular, it is by no means clear from Riemann's arguments that the infinite series for $\pi(x)$ even *converges*, much less that its largest term $\int_2^x (dt/\log t)$ dominates it for large x. On the other hand, Riemann's methods, which included the study of the function $\zeta(s)$ as a function of a complex variable, the study of the complex zeros of $\zeta(s)$, Fourier inversion, Möbius inversion, and the representation of functions such as $\pi(x)$ by "explicit formulas" such as his infinite series, all have had important parts in the subsequent development of the theory.

For the first 30 years after Riemann's paper was published, there was

†Chebyshev did not state his result in this form. This form can be obtained from his estimate of the number of primes between l and L (see Chebyshev [C3, Section 6]) by fixing l, letting $L \to \infty$, and using $\int_2^L (\log t)^{-1}\, dt \sim L/\log L$.

‡See Section 1.17. Note that $Li(x) = \int_2^x (dt/\log t) + \text{const.}$

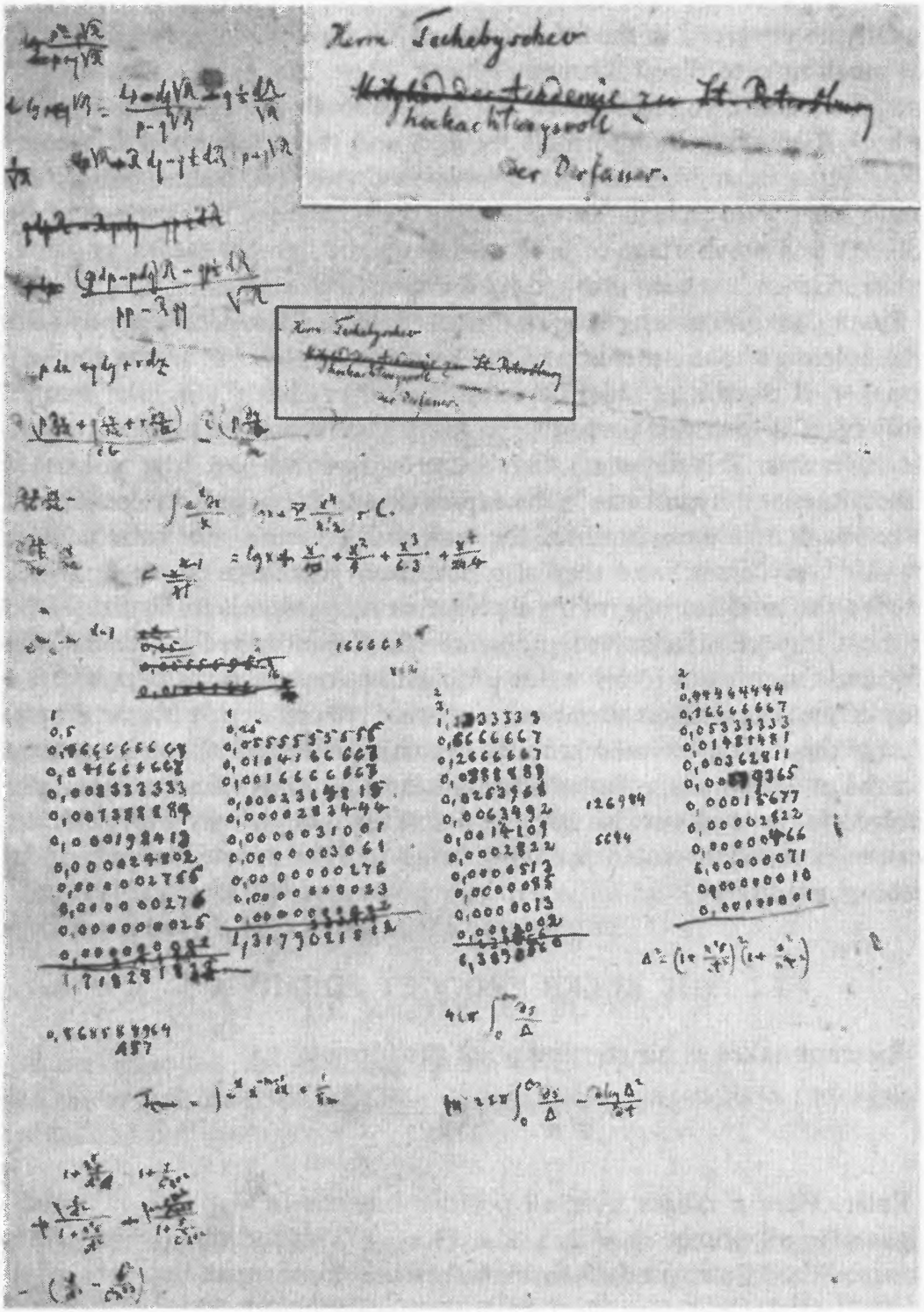


Fig. 1. A scrap sheet used to hold some other loose sheets in Riemann's papers. The note seems to prove that Riemann was aware of Chebyshev's work and intended to send him an offprint of his own paper. In all likelihood Riemann was practicing his penmanship in forming Roman, rather than German, letters to write a dedication to Chebyshev. (Reproduced with the permission of the Niedersächsische Staats- und Universitätsbibliothek, Handschriftenabteilung, Göttingen.)

virtually no progress† in the field. It was as if it took the mathematical world that much time to digest Riemann's ideas. Then, in a space of less than 10 years, Hadamard, von Mangoldt, and de la Vallée Poussin succeeded in proving both Riemann's main formula for $\pi(x)$ and the prime number theorem (3), as well as a number of other related theorems. In all these proofs Riemann's ideas were crucial. Since that time there has been no shortage of new problems and no shortage of progress in analytic number theory, and much of this progress has been inspired by Riemann's ideas.

Finally, no discussion of the historical context of Riemann's paper would be complete without a mention of the Riemann hypothesis. In the course of the paper, Riemann says that he considers it "very likely" that the complex zeros of $\zeta(s)$ all have real part equal to $\frac{1}{2}$, but that he has been unable to prove that this is true. This statement, that the zeros have real part $\frac{1}{2}$, is now known as the "Riemann hypothesis." The experience of Riemann's successors with the Riemann hypothesis has been the same as Riemann's—they also consider its truth "very likely" and they also have been unable to prove it. Hilbert included the problem of proving the Riemann hypothesis in his list [H9] of the most important unsolved problems which confronted mathematics in 1900, and the attempt to solve this problem has occupied the best efforts of many of the best mathematicians of the twentieth century. It is now unquestionably the most celebrated problem in mathematics and it continues to attract the attention of the best mathematicians, not only because it has gone unsolved for so long but also because it appears tantalizingly vulnerable and because its solution would probably bring to light new techniques of far-reaching importance.

1.2 THE EULER PRODUCT FORMULA

Riemann takes as his starting point the formula

$$\sum_n \frac{1}{n^s} = \prod_p \frac{1}{\left(1 - \frac{1}{p^s}\right)} \tag{1}$$

of Euler. Here n ranges over all positive integers ($n = 1, 2, 3, \ldots$) and p ranges over all primes ($p = 2, 3, 5, 7, 11, \ldots$). This formula, which is now known as the "Euler product formula," results from expanding each of the

†A major exception to this statement was Mertens's theorem [M5] of 1874 stating that (2′) is true in the strong sense that the difference of the two sides approaches a limit as $x \longrightarrow \infty$, namely, Euler's constant plus $\sum_p [\log(1 - p^{-1}) + p^{-1}]$. Another perhaps more natural statement of Mertens's theorem is

$$\lim_{x \to \infty} \log x \prod_{p<x} (1 - p^{-1}) = e^{-\gamma},$$

where γ is Euler's constant. See, for example, Hardy and Wright [H7].

factors on the right

$$\frac{1}{\left(1-\frac{1}{p^s}\right)} = 1 + \frac{1}{p^s} + \frac{1}{(p^2)^s} + \frac{1}{(p^3)^s} + \cdots$$

and observing that their product is therefore a sum of terms of the form

$$\frac{1}{(p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r})^s},$$

where $p_1, \ldots, p_r$ are distinct primes and $n_1, n_2, \ldots, n_r$ are natural numbers, and then using the fundamental theorem of arithmetic (every integer can be written in essentially only one way as a product of primes) to conclude that this sum is simply $\sum(1/n)^s$. Euler used this formula principally as a formal identity and principally for integer values of s (see, for example, Euler [E5]).

Dirichlet also based his work† in this field on the Euler product formula. Since Dirichlet was one of Riemann's teachers and since Riemann refers to Dirichlet's work in the first paragraph of his paper, it seems certain that Riemann's use of the Euler product formula was influenced by Dirichlet. Dirichlet, unlike Euler, used the formula (1) with s as a real variable and, also unlike Euler, he proved‡ rigorously that (1) is true for all real $s > 1$.

Riemann, as one of the founders of the theory of functions of a complex variable, would naturally be expected to consider s as a *complex* variable. It is easy to show that both sides of the Euler product formula converge for complex s in the halfplane Re $s > 1$, but Riemann goes much further and shows that even though both sides of (1) diverge for other values of s, the function they define is meaningful for *all* values of s except for a pole at $s = 1$. This extension of the range of s requires a few facts about the factorial function which will be covered in the next section.

1.3 THE FACTORIAL FUNCTION

Euler extended the factorial function $n! = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1$ from the natural numbers n to all real numbers greater than -1 by observing that¶

$$n! = \int_0^\infty e^{-x} x^n \, dx \qquad (n = 1, 2, 3, \ldots) \tag{1}$$

†Dirichlet's major contribution to the theory was his proof that if m is relatively prime to n, then the congruence $p \equiv m \pmod{n}$ has infinitely many prime solutions p. He was also interested in questions concerning the density of the distribution of primes, but he did not have significant success with these questions.

‡Dirichlet [D3]. Since the terms p^{-s} are all positive, there is nothing subtle or difficult about this proof—it is essentially a reordering of absolutely convergent series—but it has the important effect of transforming (1) from a formal identity true for various values of s to an analytical formula true for all real $s > 1$.

¶However Euler wrote the integral in terms of $y = e^{-x}$ as $n! = \int_0^1 (\log 1/y)^n \, dy$ (see Euler [E3]).

(integration by parts) and by observing that the integral on the right converges for noninteger values of n, provided only that $n > -1$. Gauss [G1] introduced the notation†

$$\Pi(s) = \int_0^\infty e^{-x} x^s \, dx \qquad (s > -1) \tag{2}$$

for Euler's integral on the right side of (1). Thus $\Pi(s)$ is defined for all real numbers s greater than -1, in fact for all complex numbers s in the halfplane $\operatorname{Re} s > -1$, and $\Pi(s) = s!$ whenever s is a natural number. There is another representation of $\Pi(s)$ which was also known‡ to Euler, namely,

$$\Pi(s) = \lim_{N\to\infty} \frac{1 \cdot 2 \cdots N}{(s+1)(s+2)\cdots(s+N)} (N+1)^s. \tag{3}$$

This formula is valid for all s for which (2) defines $\Pi(s)$, that is, for all s in the halfplane $\operatorname{Re} s > -1$. On the other hand, it is not difficult to show [use formula (4) below] that the limit (3) exists for *all* values of s, real or complex, provided only that the denominator is not zero, that is, provided only that s is not a negative integer. In short, formula (3) extends the definition of $\Pi(s)$ to all values of s other than $s = -1, -2, -3, \ldots$.

In addition to the fact that the two definitions (2) and (3) of $\Pi(s)$ coincide for real $s > -1$, the following facts will be used without proof:

$$\Pi(s) = \prod_{n=1}^{\infty} \frac{n^{1-s}(n+1)^s}{s+n} = \prod_{n=1}^{\infty} \left(1 + \frac{s}{n}\right)^{-1} \left(1 + \frac{1}{n}\right)^s, \tag{4}$$

$$\Pi(s) = s\Pi(s-1), \tag{5}$$

$$\frac{\pi s}{\Pi(s)\Pi(-s)} = \sin \pi s, \tag{6}$$

$$\Pi(s) = 2^s \Pi\left(\frac{s}{2}\right) \Pi\left(\frac{s-1}{2}\right) \pi^{-1/2}. \tag{7}$$

For the proofs of these facts the reader is referred to any book which deals with factorial function or the "Γ-function," for example, Edwards [E1, pp. 421–425]. Identity (4) is a simple reformulation of formula (3). Using it one can prove that $\Pi(s)$ *is an analytic function of the complex variable* s *which has simple poles at* $s = -1, -2, -3, \ldots$. *It has no zeros.* Identity (5) is

†Unfortunately, Legendre subsequently introduced the notation $\Gamma(s)$ for $\Pi(s-1)$. Legendre's reasons for considering $(n-1)!$ instead of $n!$ are obscure (perhaps he felt it was more natural to have the first pole occur at $s = 0$ rather than at $s = -1$) but, whatever the reason, this notation prevailed in France and, by the end of the nineteenth century, in the rest of the world as well. Gauss's original notation appears to me to be much more natural and Riemann's use of it gives me a welcome opportunity to reintroduce it.

‡See Euler [E3, E8].

called the "functional equation of the factorial function"; together with $\Pi(0) = 1$ [from (4)] it gives $\Pi(n) = n!$ immediately. Identity (6) is essentially the product formula for the sine; when $s = \frac{1}{2}$ it combines with (5) to give the important value $\Pi(-\frac{1}{2}) = \pi^{1/2}$. Identity (7) is known as the *Legendre relation.* It is the case $n = 2$ of a more general identity

$$\frac{\Pi(s)}{n^s \Pi\left(\frac{s}{n}\right)\Pi\left(\frac{s-1}{n}\right)\cdots\Pi\left(\frac{s-n+1}{n}\right)} = \left[\frac{2\pi n}{(2\pi)^n}\right]^{1/2}$$

which will not be needed.

1.4 THE FUNCTION $\zeta(s)$

It is interesting to note that Riemann does not speak of the "analytic continuation" of the function $\sum n^{-s}$ beyond the halfplane Re $s > 1$, but speaks rather of finding a formula for it which "remains valid for all s." This indicates that he viewed the problem in terms more analogous to the extension of the factorial function by formula (3) of the preceding section than to a piece-by-piece extension of the function in the manner that analytic continuation is customarily taught today. The view of analytic continuation in terms of chains of disks and power series convergent in each disk descends from Weierstrass and is quite antithetical to Riemann's basic philosophy that analytic functions should be dealt with *globally*, not locally in terms of power series.

Riemann derives his formula for $\sum n^{-s}$ which "remains valid for all s" as follows. Substitution of nx for x in Euler's integral for $\Pi(s-1)$ gives

$$\int_0^\infty e^{-nx} x^{s-1}\, dx = \frac{\Pi(s-1)}{n^s}$$

($s > 0$, $n = 1, 2, 3, \ldots$). Riemann sums this over n and uses $\sum_{n=1}^\infty r^{-n} = (r-1)^{-1}$ to obtain†

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx = \Pi(s-1) \sum_{n=1}^\infty \frac{1}{n^s} \tag{1}$$

($s > 1$). Convergence of the improper integral on the left and the validity of

†This formula, with $s = 2n$, occurs in a paper [A1] of Abel which was included in the 1839 edition of Abel's collected works. It seems very likely that Riemann would have been aware of this. A very similar formula

$$\int_0^\infty (e^x - 1)^{-1} e^{-x} x^\rho\, dx = \Pi(\rho) \sum_{n=2}^\infty n^{-1-\rho}$$

is the point of departure of Chebyshev's 1848 paper [C2].

the interchange of summation and integration are not difficult to establish.

Next he considers the contour integral

$$\int_{+\infty}^{+\infty} \frac{(-x)^s}{e^x - 1} \frac{dx}{x}.$$

The limits of integration are intended to indicate a path of integration which begins at $+\infty$, moves to the left down the positive real axis, circles the origin once in the positive (counterclockwise) direction, and returns up the positive real axis to $+\infty$. The definition of $(-x)^s$ is $(-x)^s = \exp[s \log(-x)]$, where the definition of $\log(-x)$ conforms to the usual definition of $\log z$ for z not on the negative real axis as the branch which is real for positive real z; thus $(-x)^s$ is not defined on the positive real axis and, strictly speaking, the path of integration must be taken to be slightly above the real axis as it descends from $+\infty$ to 0 and slightly below the real axis as it goes from 0 back to $+\infty$. When this integral is written in the form

$$\int_{+\infty}^{\delta} \frac{(-x)^s\,dx}{(e^x - 1)x} + \int_{|x|=\delta} \frac{(-x)^s\,dx}{(e^x - 1)x} + \int_{\delta}^{+\infty} \frac{(-x)^s\,dx}{(e^x - 1)x},$$

the middle term is $2\pi i$ times the average value of $(-x)^s(e^x - 1)^{-1}$ on the circle $|x| = \delta$ [because on this circle $i\,d\theta = (dx/x)$]. Thus the middle term approaches zero as $\delta \longrightarrow 0$ provided $s > 1$ [because $x(e^x - 1)^{-1}$ is nonsingular near $x = 0$]. The other two terms can then be combined to give

$$\begin{aligned}\int_{+\infty}^{+\infty} \frac{(-x)^s}{e^x - 1} \cdot \frac{dx}{x} &= \lim_{\delta \to 0} \left\{ \int_{+\infty}^{\delta} \frac{\exp[s(\log x - i\pi)]\,dx}{(e^x - 1)x} \right. \\ &\qquad \left. + \int_{\delta}^{+\infty} \frac{\exp[s(\log x + i\pi)]\,dx}{(e^x - 1)x} \right\} \\ &= (e^{i\pi s} - e^{-i\pi s}) \int_0^{\infty} \frac{x^{s-1}\,dx}{e^x - 1}\end{aligned}$$

which combines with the previous formula (1) to give

$$\int_{+\infty}^{+\infty} \frac{(-x)^s}{e^x - 1} \cdot \frac{dx}{x} = 2i \sin(\pi s)\Pi(s - 1) \sum_{n=1}^{\infty} \frac{1}{n^s}$$

or, finally, when both sides are multiplied by $\Pi(-s)s/2\pi is$ and identity (6) of the preceding section is used,

$$\frac{\Pi(-s)}{2\pi i} \int_{+\infty}^{+\infty} \frac{(-x)^s}{e^x - 1} \cdot \frac{dx}{x} = \sum_{n=1}^{\infty} \frac{1}{n^s}. \tag{2}$$

In other words, if $\zeta(s)$ is defined by the formula†

$$\zeta(s) = \frac{\Pi(-s)}{2\pi i} \int_{+\infty}^{+\infty} \frac{(-x)^s}{e^x - 1} \cdot \frac{dx}{x}, \tag{3}$$

†This formula is misstated by the editors of Riemann's works in the notes; they put the factor π on the wrong side of their equation.

then, for real values of s greater than one, $\zeta(s)$ is equal to Dirichlet's function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}. \tag{4}$$

However, formula (3) for $\zeta(s)$ "remains valid for all s." In fact, since the integral in (3) clearly converges for all values of s, real or complex (because e^x grows much faster than x^s as $x \to \infty$), and since the function it defines is complex analytic (because convergence is uniform on compact domains), the function $\zeta(s)$ of (3) is defined and analytic at all points with the possible exception of the points $s = 1, 2, 3, \ldots$, where $\Pi(-s)$ has poles. Now at $s = 2, 3, 4, \ldots$, formula (4) shows that $\zeta(s)$ has no pole [hence the integral in (3) must have a zero which cancels the pole of $\Pi(-s)$ at these points, a fact which also follows immediately from Cauchy's theorem], and at $s = 1$ formula (4) shows that $\lim \zeta(s) = \infty$ as $s \downarrow 1$, hence that $\zeta(s)$ has a simple [because the pole of $\Pi(-s)$ is simple] pole at $s = 1$. Thus *formula (3) defines a function $\zeta(s)$ which is analytic at all points of the complex s-plane except for a simple pole at $s = 1$.* This function coincides with $\sum n^{-s}$ for real values of $s > 1$ and in fact, by analytic continuation, throughout the halfplane $\operatorname{Re} s > 1$.

The function $\zeta(s)$ is known as the Riemann zeta function.

1.5 VALUES OF $\zeta(s)$

The function $x(e^x - 1)^{-1}$ is analytic near $x = 0$; therefore it can be expanded as a power series

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n x^n}{n!} \tag{1}$$

valid near zero [in fact valid in the disk $|x| < 2\pi$ which extends to the nearest singularities $x = \pm 2\pi i$ of $x(e^x - 1)^{-1}$]. The coefficients B_n of this expansion are by definition the *Bernoulli numbers;* the first few are easily determined to be

$$\begin{aligned} B_0 &= 1, & B_1 &= -\tfrac{1}{2}, \\ B_2 &= \tfrac{1}{6}, & B_3 &= 0, \\ B_4 &= -\tfrac{1}{30}, & B_5 &= 0, \\ B_6 &= \tfrac{1}{42}, & B_7 &= 0, \\ B_8 &= -\tfrac{1}{30}, & B_9 &= 0. \end{aligned}$$

The odd Bernoulli numbers B_{2n+1} are all zero† after the first, and the even Bernoulli numbers B_{2n} can be determined successively, but there is no simple

†This can be proved directly by noting that $(-t)(e^{-t} - 1)^{-1} + (-t/2) = (-te^t + t - t)(1 - e^t)^{-1} - (t/2) = t(e^t - 1)^{-1} + (t/2)$, that is, $t(e^t - 1)^{-1} + (t/2)$ is an even function. For alternative proofs see the note of Section 1.6 and formula (10) of Section 6.2.

computational formula for them. (See Euler [E6] for a list of the values of $(-1)^{n-1} B_{2n}$ up to B_{30}.)

When $s = -n$ $(n = 0, 1, 2, \ldots)$, this expansion (1) can be used in the defining equation of $\zeta(s)$ to obtain

$$\begin{aligned}\zeta(-n) &= \frac{\Pi(n)}{2\pi i}\int_{+\infty}\frac{(-x)^{-n}}{e^x-1}\cdot\frac{dx}{x}\\ &= \frac{\Pi(n)}{2\pi i}\int_{|x|=\delta}\left(\sum_m \frac{B_m x^m}{m!}\right)\frac{(-x)^{-n}}{x}\cdot\frac{dx}{x}\\ &= \sum_m \Pi(n)\frac{B_m}{m!}(-1)^n\cdot\frac{1}{2\pi}\int_0^{2\pi} x^{m-n-1}\,d\theta\\ &= n!\frac{B_{n+1}}{(n+1)!}(-1)^n = (-1)^n\frac{B_{n+1}}{n+1}.\end{aligned}$$

Riemann does not give this formula for $\zeta(-n)$, but he does state the particular consequence $\zeta(-2) = \zeta(-4) = \zeta(-6) = \cdots = 0$. He was surely aware, however, not only of the values†

$$\zeta(0) = -1/2, \qquad \zeta(-1) = -1/12, \qquad \zeta(-3) = 1/120,$$

etc., which it implies, but also of the values

$$\zeta(2) = \pi^2/6,\ \zeta(4) = \pi^4/90, \ldots,$$

and, in general,

$$\zeta(2n) = \frac{(2\pi)^{2n}(-1)^{n+1}B_{2n}}{2\cdot(2n)!} \tag{2}$$

which had been found by Euler [E6]. There is no easy way to deduce this famous formula of Euler's from Riemann's integral formula for $\zeta(s)$ [(3) of Section 1.4] and it may well have been this problem of deriving (2) anew which led Riemann to the discovery‡ of the functional equation of the zeta function which is the subject of the next section.

1.6 FIRST PROOF OF THE FUNCTIONAL EQUATION

For negative real values of s, Riemann evaluates the integral

$$\zeta(s) = \frac{\Pi(-s)}{2\pi i}\int_{+\infty}^{+\infty}\frac{(-x)^s}{e^x-1}\cdot\frac{dx}{x} \tag{1}$$

†The editors of Riemann's collected works give the erroneous value $\zeta(0) = \frac{1}{2}$.

‡Actually the functional equation occurs in Euler's works [E7] in a slightly different form, and it is entirely possible that Riemann found it there. (See also Hardy [H5, pp. 23–26].) In any case, Euler had nothing but an empirical (!) proof of the functional equation and Riemann, in a reversal of his usual role, gave the first rigorous proof of a statement which had been made, but not adequately proved, by someone else.

as follows. Let D denote the domain in the s-plane which consists of all points other than those which lie within ϵ of the positive real axis or within ϵ of one of the singularities $x = \pm 2\pi in$ of the integrand of (1). Let ∂D be the boundary of D oriented in the usual way. Then, ignoring for the moment the fact that D is not compact, Cauchy's theorem gives

$$\frac{\Pi(-s)}{2\pi i}\int_{\partial D}\frac{(-x)^s}{e^x-1}\cdot\frac{dx}{x}=0. \tag{2}$$

Now one component of this integral is the integral (1) with the orientation reversed, whereas the others are integrals over the circles $|x \pm 2\pi in| = \epsilon$ oriented clockwise. Thus when the circles are oriented in the usual counterclockwise sense, (2) becomes

$$-\zeta(s)-\sum\frac{\Pi(-s)}{2\pi i}\int_{|x\pm 2\pi in|=\epsilon}\frac{(-x)^s}{e^x-1}\cdot\frac{dx}{x}=0. \tag{3}$$

The integrals over the circles can be evaluated by setting $x = 2\pi in + y$ for $|y| = \epsilon$ to find

$$\begin{aligned}\frac{\Pi(-s)}{2\pi i}&\int_{|y|=\epsilon}\frac{(-2\pi in-y)^s}{e^{2\pi in+y}-1}\frac{dy}{2\pi in+y}\\ &=-\frac{\Pi(-s)}{2\pi i}\int_{|y|=\epsilon}(-2\pi in-y)^{s-1}\cdot\frac{y}{e^y-1}\cdot\frac{dy}{y}\\ &=-\Pi(-s)(-2\pi in)^{s-1}\end{aligned}$$

by the Cauchy integral formula. Summing over all integers n other than $n = 0$ and using (3) then gives

$$\begin{aligned}\zeta(s)&=\sum_{n=1}^{\infty}\Pi(-s)[(-2\pi in)^{s-1}+(2\pi in)^{s-1}]\\ &=\Pi(-s)(2\pi)^{s-1}[i^{s-1}+(-i)^{s-1}]\sum_{n=1}^{\infty}n^{s-1}.\end{aligned}$$

Finally, using the simplification

$$\begin{aligned}i^{s-1}+(-i)^{s-1}&=\frac{1}{i}[e^{s\log i}-e^{s\log(-i)}]\\ &=\frac{1}{i}[e^{s\pi i/2}-e^{-s\pi i/2}]=2\sin\frac{s\pi}{2},\end{aligned}$$

one obtains the desired formula

$$\zeta(s)=\Pi(-s)(2\pi)^{s-1}\,2\sin(s\pi/2)\zeta(1-s). \tag{4}$$

This relationship between $\zeta(s)$ and $\zeta(1-s)$ is known as the *functional equation of the zeta function.*

In order to prove rigorously that (4) holds for $s < 0$, it suffices to modify the above argument by letting D_n be the intersection of D with the disk $|s| \leq (2n+1)\pi$ and letting $n \to \infty$; then the integral (2) splits into two parts, one

being an integral over the circle $|s| = (2n + 1)\pi$ with the points within ϵ of the positive real axis deleted, and the other being an integral whose limit as $n \to \infty$ is the left side of (3). The first of these two parts approaches zero because the length of the path of integration is less than $2\pi(2n + 1)\pi$, because the factor $(e^x - 1)^{-1}$ is bounded on the circle $|s| = (2n + 1)\pi$, and because the modulus of $(-x)^s/x$ on this circle is $|x|^{s-1} \leq [(2n + 1)\pi]^{-\delta-1}$ for $s \leq -\delta < 0$. Thus the second part, which by Cauchy's theorem is the negative of the first part, also approaches zero, which implies (3) and hence (4).

This completes the proof of the functional equation (4) in the case $s < 0$. However, both sides of (4) are analytic functions of s, so this suffices to prove (4) for all values of s [except for $s = 0, 1, 2, \ldots$, where† one or more of the terms of (4) have poles].

For $s = 1 - 2n$ the functional equation plus the identity

$$\zeta(-(2n - 1)) = (-1)^{2n-1}\frac{B_{2n}}{2n}$$

of the previous section gives

$$(-1)^{2n-1}\frac{B_{2n}}{2n} = \Pi(2n - 1)(2\pi)^{-2n}2(-1)^n\zeta(2n)$$

and hence Euler's famous formula for $\zeta(2n)$ [(2) of Section 1.5].

Riemann uses two of the basic identities of the factorial function [(6) and (7) of Section 1.3] to rewrite the functional equation (4) in the form

$$\zeta(s) = \pi^{-1/2}2^{-s}\Pi\left(-\frac{s}{2}\right)\Pi\left(-\frac{s+1}{2}\right)2^s\pi^{s-1}\frac{\pi s/2}{\Pi\left(\frac{s}{2}\right)\Pi\left(-\frac{s}{2}\right)}\zeta(1 - s)$$

and hence in the form

$$\Pi\left(\frac{s}{2} - 1\right)\pi^{-s/2}\zeta(s) = \Pi\left(\frac{1-s}{2} - 1\right)\pi^{-(1-s)/2}\zeta(1 - s). \tag{5}$$

In words, then, *the function on the left side of* (5) *is unchanged by the substitution* $s = 1 - s$.

Riemann appears to consider this symmetrical statement (5) as the natural statement of the functional equation, because he gives‡ an alternative proof

†When $s = 2n + 1$, the fact that $\zeta(s)$ has no pole at $2n + 1$ implies, since Π has a pole at $-2n - 1$ and $\sin(s\pi/2)$ has no zero at $2n + 1$, that $\zeta(-2n) = 0$ and hence, by the formula for $\zeta(-2n)$ of the preceding section, that the odd Bernoulli numbers $B_3, B_5, B_7, \ldots$ are all zero.

‡Since the second proof renders the first proof wholly unnecessary, one may ask why Riemann included the first proof at all. Perhaps the first proof shows the argument by which he originally discovered the functional equation or perhaps it exhibits some properties which were important in his understanding of ζ.

which exhibits this symmetry in a more satisfactory way. This second proof is given in the next section.

1.7 SECOND PROOF OF THE FUNCTIONAL EQUATION

Riemann first observes that the change of variable $x = n^2\pi x$ in Euler's integral for $\Pi(s/2 - 1)$ gives

$$\frac{1}{n^s}\pi^{-s/2}\Pi\left(\frac{s}{2} - 1\right) = \int_0^\infty e^{-n^2\pi x}x^{s/2}\frac{dx}{x} \qquad (\mathrm{Re}\, s > 1).$$

Thus summation over n gives

$$(1) \qquad \Pi\left(\frac{s}{2} - 1\right)\pi^{-s/2}\zeta(s) = \int_0^\infty \psi(x)x^{s/2}\cdot\frac{dx}{x} \qquad (\mathrm{Re}\, s > 1),$$

where† $\psi(x) = \sum_{n=1}^\infty \exp(-n^2\pi x)$. The symmetrical form of the functional equation is the statement that the function (1) is unchanged by the substitution $s = 1 - s$. To prove directly that the integral on the right side of (1) is unchanged by this substitution Riemann uses the *functional equation of the theta function* in a form taken from Jacobi,‡ namely, in the form

$$(2) \qquad \frac{1 + 2\psi(x)}{1 + 2\psi(1/x)} = \frac{1}{\sqrt{x}}.$$

[Since $\psi(x)$ approaches zero very rapidly as $x \to \infty$, this shows in particular that $\psi(x)$ is like $\frac{1}{2}(x^{-1/2} - 1)$ for x near zero and hence that the integral on the right side of (1) is convergent for $s > 1$. Once this has been established, the validity of (1) for $s > 1$ can be proved by an elementary argument using absolute convergence to justify the interchange of summation and integration.] Using (2), Riemann reformulates the integral on the right side of (1) as

$$\begin{aligned}\int_0^\infty \psi(x)x^{s/2}\frac{dx}{x} &= \int_1^\infty \psi(x)x^{s/2}\cdot\frac{dx}{x} - \int_\infty^1 \psi\left(\frac{1}{x}\right)x^{-s/2}\cdot\frac{dx}{x} \\ &= \int_1^\infty \psi(x)x^{s/2}\cdot\frac{dx}{x} + \int_1^\infty \left[x^{1/2}\psi(x) + \frac{x^{1/2}}{2} - \frac{1}{2}\right]x^{-s/2}\frac{dx}{x} \\ &= \int_1^\infty \psi(x)[x^{s/2} + x^{(1-s)/2}]\frac{dx}{x} \\ &\quad + \frac{1}{2}\int_1^\infty [x^{-(s-1)/2} - x^{-s/2}]\frac{dx}{x}.\end{aligned}$$

†This function $\psi(x)$ has nothing whatsoever to do with the function $\psi(x)$ which appears in Chapter 3.

‡Riemann refers to Section 65 of Jacobi's treatise "Fundmenta Nova Theoriae Functionum Ellipticarum." Although the needed formula is not given explicitly there, Jacobi in another place [J1] shows how the needed formula follows from formula (6) of Section 65. Jacobi attributes the formula to Poisson. For a proof of the formula see Section 10.4.

Now $\int_1^\infty x^{-a}\,(dx/x) = 1/a$ for $a > 0$ so the second integral is

$$\frac{1}{2}\left[\frac{1}{(s-1)/2} - \frac{1}{s/2}\right] = \frac{1}{s(s-1)}$$

for $s > 1$. Thus for $s > 1$ the formula

$$\text{(3)} \qquad \Pi\left(\frac{s}{2} - 1\right)\pi^{-s/2}\zeta(s) = \int_1^\infty \psi(x)[x^{s/2} + x^{(1-s)/2}]\,\frac{dx}{x} - \frac{1}{s(1-s)}$$

holds. But, because $\psi(x)$ decreases more rapidly than any power of x as $x \to \infty$, the integral in this formula converges for all† s. Since both sides are analytic, the same equation holds for all s. Because the right side is obviously unchanged by the substitution $s = 1 - s$, this proves the functional equation of the zeta function.

1.8 THE FUNCTION $\xi(s)$

The function $\Pi((s/2) - 1)\pi^{-s/2}\,\zeta(s)$, which occurs in the symmetrical form of the functional equation, has poles at $s = 0$ and $s = 1$. [This follows immediately from (3) of the preceding section.] Riemann multiplies it by $s(s-1)/2$ and defines‡

$$\text{(1)} \qquad \xi(s) = \Pi(s/2)(s-1)\pi^{-s/2}\,\zeta(s).$$

Then $\xi(s)$ is an entire function—that is, an analytic function of s which is defined for all values of s—and the functional equation of the zeta function is equivalent to $\xi(s) = \xi(1-s)$.

Riemann next derives the following representation of $\xi(s)$. Equation (3) of the preceding section gives

$$\begin{aligned}\xi(s) &= \frac{1}{2} - \frac{s(1-s)}{2}\int_1^\infty \psi(s)(x^{s/2} + x^{(1-s)/2})\,\frac{dx}{x} \\ &= \frac{1}{2} - \frac{s(1-s)}{2}\int_1^\infty \frac{d}{dx}\left\{\psi(x)\left[\frac{x^{s/2}}{s/2} + \frac{x^{(1-s)/2}}{(1-s)/2}\right]\right\}dx \\ &\quad + \frac{s(1-s)}{2}\int_1^\infty \psi'(x)\left[\frac{x^{s/2}}{s/2} + \frac{x^{(1-s)/2}}{(1-s)/2}\right]dx\end{aligned}$$

†Note that this gives, therefore, another formula for $\zeta(s)$ which is "valid for all s" other than $s = 0, 1$; that is, it gives an alternative proof of the fact that $\zeta(s)$ can be analytically continued.

‡Actually Riemann uses the letter ξ to denote the function which it is now customary to denote by Ξ, namely, the function $\Xi(t) = \xi(\frac{1}{2} + it)$, where ξ is defined as above. I follow Landau, and almost all subsequent writers, in rejecting Riemann's change of variable $s = \frac{1}{2} + it$ in formula (1) as being confusing. In fact, there is reason to believe that Riemann himself was confused by it [see remarks concerning $\xi(0)$ in Section 1.16].

$$= \frac{1}{2} + \frac{s(1-s)}{2}\psi(1)\left[\frac{2}{s} + \frac{2}{1-s}\right]$$

$$+ \int_1^\infty \psi'(x)[(1-s)x^{s/2} + sx^{(1-s)/2}]\,dx$$

$$= \frac{1}{2} + \psi(1) + \int_1^\infty x^{3/2}\psi'(x)[(1-s)x^{[(s-1)/2]-1} + sx^{-(s/2)-1}]\,dx$$

$$= \frac{1}{2} + \psi(1) + \int_1^\infty \frac{d}{dx}[x^{3/2}\psi'(x)(-2x^{(s-1)/2} - 2x^{-s/2})]\,dx$$

$$-\int_1^\infty \frac{d}{dx}[x^{3/2}\psi'(x)]\,[-2x^{(s-1)/2} - 2x^{-s/2}]\,dx$$

$$= \frac{1}{2} + \psi(1) - \psi'(1)[-2-2]$$

$$+\int_1^\infty \frac{d}{dx}[x^{3/2}\psi'(x)]\,(2x^{(s-1)/2} + 2x^{-s/2})\,dx.$$

Now differentiation of

$$2\psi(x) + 1 = x^{-1/2}[2\psi(1/x) + 1]$$

easily gives

$$\tfrac{1}{2} + \psi(1) + 4\psi'(1) = 0,$$

and using this puts the formula in the final form

$$\xi(s) = 4\int_1^\infty \frac{d[x^{3/2}\psi'(x)]}{dx}\,x^{-1/4}\cosh\left[\frac{1}{2}\left(s - \frac{1}{2}\right)\log x\right]dx \tag{2}$$

or, as Riemann writes it,

$$\xi\left(\frac{1}{2} + it\right) = 4\int_1^\infty \frac{d[x^{3/2}\psi'(x)]}{dx}\,x^{-1/4}\cos\left(\frac{t}{2}\log x\right)dx.$$

If $\cosh[\frac{1}{2}(s-\frac{1}{2})\log x]$ is expanded in the usual power series $\cosh y = \frac{1}{2}(e^y + e^{-y}) = \sum y^{2n}/(2n)!$, formula (2) shows that

$$\xi(s) = \sum_{n=0}^{\infty} a_{2n}(s - \tfrac{1}{2})^{2n} \tag{3}$$

where

$$a_{2n} = 4\int_1^\infty \frac{d[x^{3/2}\psi'(x)]}{dx}\,x^{-1/4}\frac{(\frac{1}{2}\log x)^{2n}}{(2n)!}\,dx.$$

Riemann states that this series representation of $\xi(s)$ as an even function of $s - \frac{1}{2}$ "converges very rapidly," but he gives no explicit estimates and he does not say what role this series plays in the assertions which he makes next.

The two paragraphs which follow the formula (2) for $\xi(s)$ are the most difficult portion of Riemann's paper. Their goal is essentially to prove that

$\xi(s)$ can be expanded as an infinite product

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right), \tag{4}$$

where ρ ranges† over the roots of the equation $\xi(\rho) = 0$. Now any *polynomial* $p(s)$ can be expanded as a finite product $p(s) = p(0) \prod_{\rho} [1 - (s/\rho)]$, where ρ ranges over the roots of the equation $p(\rho) = 0$ [except that the product formula for $p(s)$ is slightly different if $p(0) = 0$]; hence the product formula (4) states that $\xi(s)$ is *like a polynomial of infinite degree.* (Similarly, Euler thought of $\sin x$ as a "polynomial of infinite degree" when he conjectured, and finally proved, the formula $\sin \pi x = \pi x \prod_{n=1}^{\infty} [1 - (x/n)^2]$.) On the other hand, the statement that the series (3) converges "very rapidly" is also a statement that $\xi(s)$ is like a polynomial of infinite degree—a finite number of terms gives a very good approximation in any finite part of the plane. Thus there is some relationship between the series (3) and the product formula (4)—in fact it is *precisely* the rapid decrease of the coefficients a_n which Hadamard (in 1893) proved was necessary and sufficient for the validity of the product formula—but the steps of the argument by which Riemann went from the one to the other are obscure, to say the very least.

The next section contains a discussion of the distribution of the roots ρ of $\xi(\rho) = 0$, and the following section returns to the discussion of the product formula for $\xi(s)$.

1.9 THE ROOTS ρ OF ξ

In order to prove the convergence of the product $\xi(s) = \xi(0) \prod_{\rho} [1 - (s/\rho)]$, Riemann needed, of course, to investigate the distribution of the roots ρ of $\xi(\rho) = 0$. He begins by observing that the Euler product formula

$$\zeta(s) = \prod_{p} (1 - p^{-s})^{-1} \qquad (\operatorname{Re} s > 1)$$

shows immediately that $\zeta(s)$ has no zeros in the halfplane $\operatorname{Re} s > 1$ (because a convergent infinite product can be zero only if one of its factors is zero). Since $\xi(s) = \Pi(s/2)(s - 1)\pi^{-s/2}\zeta(s)$ and since the factors other than $\zeta(s)$ have only the simple zero at $s = 1$, it follows that none of the roots ρ of $\xi(\rho) = 0$ lie in the halfplane $\operatorname{Re} s > 1$. Since $1 - \rho$ is a root if and only if ρ is, this implies that none of the roots lie in the halfplane $\operatorname{Re} s < 0$ either, and hence that *all the roots* ρ *of* $\xi(\rho) = 0$ *lie in the strip* $0 \leq \operatorname{Re} \rho \leq 1$.

He then goes on to say that the number of roots ρ whose imaginary parts

†Here, and in the many formulas in the remainder of the book which involve sums or products over the roots ρ, it is understood that multiple roots—if there are any—are to be counted with multiplicities.

lie between 0 and T is approximately

$$\frac{T}{2\pi}\log\frac{T}{2\pi}-\frac{T}{2\pi} \tag{1}$$

and that the relative† error in this approximation is of the order of magnitude $1/T$. His "proof" of this is simply to say that the number of roots in this region is equal to the integral of $\xi'(s)\,ds/2\pi i\xi(s)$ around the boundary of the rectangle $\{0 \le \operatorname{Re} s \le 1,\ 0 \le \operatorname{Im} s \le T\}$ and that this integral is equal to (1) with a relative error T^{-1}. Unfortunately he gives no hint whatsoever of the method he used to estimate the integral. He himself was a master at evaluating and estimating definite integrals (see, for example, Section 1.14 or 7.4) and it is quite possible that he assumed that his readers would be able to carry out their own estimation of this integral, but if so he was wrong; it was not until 1905 that von Mangoldt succeeded in proving that Riemann's estimate was correct (see Section 6.7).

Riemann's next statement is even more baffling. He states that the number of roots *on the line* $\operatorname{Re} s = \frac{1}{2}$ is also "about" (1). He does not make precise the sense in which this approximation is true, but it is generally assumed that he meant that the relative error in the approximation of the number of zeros of $\xi(\frac{1}{2}+it)$ for $0 \le t \le T$ by (1) approaches zero as $T \to \infty$. He gives no indication of a proof at all, and no one since Riemann has been able to prove (or disprove) this statement. It was proved in 1914 that $\xi(\frac{1}{2}+it)$ has infinitely many real roots (Hardy [H3]), in 1921 that the number of real roots between 0 and T is at least KT for some positive constant K and all sufficiently large T (Hardy and Littlewood [H6]), in 1942 that this number is in fact at least $KT\log T$ for some positive K and all large T (Selberg, [S1]), and in 1914 that the number of complex roots t of $\xi(\frac{1}{2}+it)=0$ in the range $\{0 \le \operatorname{Re} t \le T,\ -\epsilon \le \operatorname{Im} t \le \epsilon\}$ is equal, for any $\epsilon > 0$, to (1) with a relative error which approaches zero as $T \to \infty$ (Bohr and Landau, [B8]). However, these partial results are still far from Riemann's statement. We can only guess what lay behind this statement (see Siegel [S4 p. 67], Titchmarsh [T8, pp. 213–214], or Section 7.8 of this book), but we do know that it led Riemann to conjecture an even stronger statement, namely, that *all* the roots lie on $\operatorname{Re} s = \frac{1}{2}$.

This is of course the famous "Riemann hypothesis." He says he considers it "very likely" that the roots all do lie on $\operatorname{Re} s = \frac{1}{2}$, but says that he was not able to prove it (which would seem to imply, incidentally, that he did feel he had rigorous proofs of the preceding two statements). Since it is not necessary for his main goal, which is the proof of his formula for the number of primes less than a given magnitude, he simply leaves the matter there—where it has remained ever since—and goes on to the product formula for $\xi(s)$.

†Titchmarsh, in an unfortunate lapse which he did not catch in the 21 years between the publication of his two books on the zeta function, failed to realize that Riemann meant the *relative* error and believed that Riemann had made a mistake at this point. See Titchmarsh [T8, p. 213].

1.10 THE PRODUCT REPRESENTATION OF $\xi(s)$

A recurrent theme in Riemann's work is the *global characterization of analytic functions by their singularities.*† Since the function $\log \xi(s)$ has logarithmic singularities at the roots ρ of $\xi(s)$ and no other singularities, it has the same singularities as the formal sum

$$\sum_{\rho} \log\left(1 - \frac{s}{\rho}\right). \tag{1}$$

Thus if this sum converges and if the function it defines is in some sense as well behaved near ∞ as $\log \xi(s)$ is, then it should follow that the sum (1) differs from $\log \xi(s)$ by at most an additive constant; setting $s = 0$ gives the value $\log \xi(0)$ for this constant, and hence exponentiation gives

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right) \tag{2}$$

as desired. This is essentially the proof of the product formula (2) which Riemann sketches.

There are two problems associated with the sum (1). The first is the determination of the imaginary parts of the logarithms it contains. Riemann passes over this point without comment and, indeed, it is not a very serious problem. For any fixed s the ambiguity in the imaginary part of $\log[1 - (s/\rho)]$ disappears for large ρ; hence the sum (1) is defined except for a (finite) multiple of $2\pi i$ which drops out when one exponentiates (2). Furthermore, one can ignore the imaginary parts altogether; the real parts of the terms of (1) are unambiguously defined and their sum is a harmonic function which differs from $\operatorname{Re} \log \xi(s)$ by a harmonic function without singularities, and if this difference function can be shown to be constant, it will follow that its harmonic conjugate is constant also.

The second problem associated with the sum (1) is its convergence. It is in fact a conditionally convergent sum, and the *order* of the series must be specified in order for the sum to be well determined. Roughly speaking the natural order for the terms would be the order of increasing $|\rho|$, or perhaps of increasing $|\rho - \frac{1}{2}|$, but specifically it suffices merely to stipulate that each

†See, for example, the Inauguraldissertation, especially article 20 (*Werke*, pp. 37–39) or part 3 of the introduction to the article "Theorie der Abel'schen Functionen," which is entitled "Determination of a function of a complex variable by boundary values and singularities [R1]." See also Riemann's introduction to Paper XI of the collected works, where he writes "... our method, which is based on the determination of functions by means of their singularities (*Unstetigkeiten und Unendlichwerden*) ... [R1]." Finally, see Ahlfors [A3], the section at the end entitled "Riemann's point of view."

term be paired with its "twin" $\rho \leftrightarrow 1 - \rho$,

$$\sum_{\mathrm{Im}\,\rho>0} \left[\log\left(1 - \frac{s}{\rho}\right) + \log\left(1 - \frac{s}{1-\rho}\right)\right], \tag{3}$$

because this sum converges absolutely. The proof of the absolute convergence of (3) is roughly as follows.

To prove the absolute convergence of

$$\sum_{\mathrm{Im}\,\rho>0} \log\left[\left(1 - \frac{s}{\rho}\right)\left(1 - \frac{s}{1-\rho}\right)\right] = \sum_{\mathrm{Im}\,\rho>0} \log\left[1 - \frac{s(1-s)}{\rho(1-\rho)}\right],$$

it suffices to prove the absolute convergence of

$$\sum_{\mathrm{Im}\,\rho>0} \frac{1}{\rho(1-\rho)}. \tag{4}$$

(In other words, to prove the absolute convergence of a product $\prod(1 + a_i)$, it suffices to prove the absolute convergence of the sum $\sum a_i$.) But the estimate of the distribution of the roots ρ given in the preceding section indicates that their density is roughly

$$d\left(\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi}\right) = \frac{1}{2\pi}\log\frac{T}{2\pi}\,dT.$$

Hence

$$\sum_{\mathrm{Im}\,\rho>0} \frac{1}{\rho(1-\rho)} \sim \int^{\infty} \frac{1}{T^2}\frac{1}{2\pi}\log\frac{T}{2\pi}\,dT < \infty,$$

or, in short, the terms are like T^{-2} and their density is like $\log T$ so their sum converges. As will be seen in Chapter 2, the only serious difficulty in making this into a rigorous proof of the absolute convergence of (3) is the proof that the vertical density of the roots ρ is in some sense a constant times $\log(T/2\pi)$. Riemann merely states this fact without proof.

Riemann then goes on to say that the function defined by (3) grows only as fast as $s \log s$ for large s; hence, because it differs from $\log \xi(s)$ by an even function of $s - \frac{1}{2}$ [and because $\log \xi(s)$ also grows like $s \log s$ for large s], this difference must be constant because it can contain no terms in $(s - \frac{1}{2})^2$, $(s - \frac{1}{2})^4, \ldots$. It will be shown in Chapter 2 that the steps in this argument can all be filled in more or less as Riemann indicates, but it must be admitted that Riemann's sketch is so abbreviated as to make it virtually useless in constructing a proof of (2).

The first proof of the product representation (2) of $\xi(s)$ was published by Hadamard [H1] in 1893.

1.11 THE CONNECTION BETWEEN $\zeta(s)$ AND PRIMES

The essence of the relationship between $\zeta(s)$ and prime numbers is the Euler product formula

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}} \qquad (\operatorname{Re} s > 1) \tag{1}$$

in which the product on the right is over all prime numbers p. Taking the log of both sides and using the series $\log(1 - x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \cdots$ puts this in the form

$$\log \zeta(s) = \sum_p \Big[\sum_n (1/n) p^{-ns} \Big] \qquad (\operatorname{Re} s > 1).$$

Since the double series on the right is absolutely convergent for $\operatorname{Re} s > 1$, the order of summation is unimportant and the sum can be written simply

$$\log \zeta(s) = \sum_p \sum_n (1/n) p^{-ns} \qquad (\operatorname{Re} s > 1). \tag{2}$$

It will be convenient in what follows to write this sum as a Stieltjes integral

$$\log \zeta(s) = \int_0^\infty x^{-s}\, dJ(x) \qquad (\operatorname{Re} s > 1) \tag{3}$$

where $J(x)$ is† the function which begins at 0 for $x = 0$ and increases by a jump of 1 at primes p, by a jump of $\frac{1}{2}$ at prime squares p^2, by a jump of $\frac{1}{3}$ at prime cubes, etc. As is usual in the theory of Stieltjes integrals, the value of $J(x)$ at each jump is defined to be halfway between its new value and its old value. Thus $J(x)$ is zero for $0 \le x < 2$, is $\frac{1}{2}$ for $x = 2$, is 1 for $2 < x < 3$, is $1\frac{1}{2}$ for $x = 3$, is 2 for $3 < x < 4$, is $2\frac{1}{4}$ for $x = 4$, is $2\frac{1}{2}$ for $4 < x < 5$, is 3 for $x = 5$, is $3\frac{1}{2}$ for $5 < x < 7$, etc. A formula for $J(x)$ is

$$J(x) = \frac{1}{2}\Big[\sum_{p^n < x} \frac{1}{n} + \sum_{p^n \le x} \frac{1}{n} \Big].$$

Riemann did not, of course, have the vocabulary of Stieltjes integration available to him, and he stated (3) in the slightly different form

$$\log \zeta(s) = s \int_0^\infty J(x) x^{-s-1}\, dx \qquad (\operatorname{Re} s > 1) \tag{4}$$

which can be obtained from (3) by integration by parts. [As $x \downarrow 0$, clearly $x^{-s}J(x) = 0$ because $J(x) \equiv 0$ for $x < 2$. On the other hand, $J(x) < x$ for all x, so $x^{-s}J(x) \longrightarrow 0$ as $x \longrightarrow \infty$ for $\operatorname{Re} s > 1$.] The integral in (4) can be con-

†Riemann denotes this function $f(x)$, and most other writers denote it $\Pi(x)$. Since $f(x)$ now is commonly used to denote a generic function and since $\Pi(x)$ in this book denotes the factorial function, I have taken the liberty of introducing a new notation $J(x)$ for this function.

sidered to be an ordinary Riemann integral and the formula itself can be derived without using Stieltjes integration by setting

$$p^{-ns} = s\int_{p^n}^{\infty} x^{-s-1}\,dx \qquad (\operatorname{Re} s > 1)$$

in (2), which is Riemann's derivation of (4).

Formulas (2)–(4) should all be thought of as minor variations of the Euler product formula (1) which is the basic idea connecting $\zeta(s)$ and primes.

1.12 FOURIER INVERSION

Riemann was a master of Fourier analysis and his work in developing this theory must certainly be counted among his greatest contributions to mathematics. It is not surprising, therefore, that he immediately applies Fourier inversion to the formula

$$\frac{\log \zeta(s)}{s} = \int_0^{\infty} J(x)x^{-s-1}\,dx \qquad (\operatorname{Re} s > 1) \tag{1}$$

to conclude

$$J(x) = \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} \log \zeta(s)x^s\,\frac{ds}{s} \qquad (a > 1). \tag{2}$$

Then using an alternative formula for $\log \zeta(s)$, he obtains an alternative formula for $J(x)$ which is the main result of the paper.

[The improper integral in (2) is only conditionally convergent and an "order of summation" must be specified. Here it is understood that the integral in (2) means the limit as $T \to \infty$ of the integral over the vertical line segment from $a - iT$ to $a + iT$. More generally, conditionally convergent integrals and series are very common in Fourier analysis, and it is always understood that such integrals and series are summed in their "natural order"; for example,

$$\sum_{n=-\infty}^{\infty} c_n e^{inx} \qquad \text{means} \qquad \lim_{N\to\infty} \sum_{n=-N}^{N} c_n e^{inx},$$

$$\int_{-\infty}^{\infty} f(y)e^{iyx}\,dy \qquad \text{means} \qquad \lim_{T\to\infty} \int_{-T}^{T} f(y)e^{iyx}\,dy,$$

etc. This is analogous to the convention that discontinuous functions such as $J(x)$ assume the middle value $J(x) = \frac{1}{2}[J(x-\epsilon) + J(x+\epsilon)]$ at any jump x, that divergent integrals such as $\operatorname{Li}(x)$ (see Section 1.14 below) are taken to mean the Cauchy principal value, and that the product $\prod [1 - (s/\rho)]$ is ordered in such a way as to pair ρ with $1 - \rho$, or, later on, ordered by $|\operatorname{Im} \rho|$.]

In deriving (2) from (1) Riemann makes use of "Fourier's theorem," by which he means† the Fourier inversion formula

$$\phi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty}\phi(\lambda)e^{i(x-\lambda)\mu}\,d\lambda\right]d\mu. \tag{3}$$

Otherwise stated, "Fourier's theorem" is the statement that in order to write a given function $\phi(x)$ as a superposition of exponentials

$$\phi(x) = \int_{-\infty}^{\infty}\Phi(\mu)e^{i\mu x}\,d\mu,$$

it is necessary and sufficient (under suitable conditions) that the "coefficients" $\Phi(\mu)$ of the expansion be defined by

$$\Phi(\mu) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\phi(\lambda)e^{-i\lambda\mu}\,d\lambda.$$

This statement of Fourier's theorem brings out the analogy with Fourier series

$$f(x) = \sum_{-\infty}^{\infty} a_n e^{inx} \Longleftrightarrow a_n = \frac{1}{2\pi}\int_0^{2\pi} f(\lambda)e^{-in\lambda}\,d\lambda,$$

and in fact theorem (3) for Fourier integrals follows formally from a passage to the limit in the theorem for Fourier series.

To derive (2) from (1), let $s = a + i\mu$, where a is a constant $a > 1$ and μ is a real variable, let $\lambda = \log x$, and let $\phi(x) = 2\pi J(e^x)e^{-ax}$. Then (1) becomes

$$\begin{aligned}\frac{\log\zeta(a+i\mu)}{a+i\mu} &= \int_{-\infty}^{\infty} J(e^\lambda)e^{-(a+i\mu)\lambda}\,d\lambda \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\phi(\lambda)e^{-i\mu\lambda}\,d\lambda \qquad (a>1),\end{aligned}$$

and when this function is taken to be $\Phi(\mu)$, Fourier's theorem gives

$$\begin{aligned}2\pi J(e^x)e^{-ax} &= \int_{-\infty}^{\infty}\frac{\log\zeta(a+i\mu)}{a+i\mu}e^{i\mu x}\,d\mu, \\ J(y) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\log\zeta(a+i\mu)}{a+i\mu}y^{a+i\mu}\,d\mu,\end{aligned}$$

from which (2) follows immediately.

Riemann completely ignores the question of the applicability of Fourier's theorem to the function $J(e^x)e^{-ax}$ and states simply that (2) holds in "complete generality." However, $J(e^x)e^{-ax}$ is a very well-behaved function—it has simple well-behaved jumps, it is identically zero for $x < 0$, and it goes to zero

†See Riemann [R2, p. 86].

faster than $e^{-(a-1)x}$ as $x \to \infty$—and the very simplest theorems† on Fourier integrals suffice to prove rigorously Riemann's statement that (2) holds in complete generality.

1.13 METHOD FOR DERIVING THE FORMULA FOR $J(x)$

The two formulas for $\xi(s)$, namely,

$$\xi(s) = \Pi\left(\frac{s}{2}\right)\pi^{-s/2}(s-1)\zeta(s) \qquad \text{and} \qquad \xi(s) = \xi(0)\prod_{\rho}\left(1 - \frac{s}{\rho}\right),$$

combine to give

$$\begin{aligned}\log \zeta(s) &= \log \xi(s) - \log \Pi\left(\frac{s}{2}\right) + \frac{s}{2}\log \pi - \log(s-1) \\ &= \log \xi(0) + \sum_{\rho}\log\left(1 - \frac{s}{\rho}\right) - \log \Pi\left(\frac{s}{2}\right) \\ &\quad + \frac{s}{2}\log \pi - \log(s-1).\end{aligned}$$

Riemann's formula for $J(x)$, which is the main result of his paper, is obtained essentially by substituting this formula for $\log \zeta(s)$ in the formula

$$J(x) = \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} \log \zeta(s) x^s \frac{ds}{s} \qquad (a > 1)$$

of the preceding section and integrating termwise. However, because a direct substitution leads to divergent integrals [the term $(s/2)\log \pi$, for example, leads to an integral which is a constant times $(i)^{-1}\int x^s\, ds = e^a \int e^{iu \log x}\, du$ which oscillates and does not converge even conditionally], Riemann first integrates by parts to obtain

$$(1) \qquad J(x) = -\frac{1}{2\pi i}\cdot\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left[\frac{\log \zeta(s)}{s}\right]x^s\, ds \qquad (a > 1)$$

before substituting the above expression for $\log \zeta(s)$. The validity of the integration by parts by which (1) is obtained depends merely on showing that

$$(2) \qquad \lim_{T\to\infty}\frac{\log \zeta(a \pm iT)}{a \pm iT}x^{a\pm iT} = 0,$$

†See, for example, Taylor [T2]. The particular form of Fourier inversion that Riemann uses here—which is essentially Fourier analysis on the multiplicative group of positive reals rather than on the additive group of all real numbers—is often called Mellin inversion. Riemann's work precedes that of Mellin by 40 years.

which follows easily from the inequality

$$|\log \zeta(a \pm iT)| = \left|\sum_n \sum_p (1/n)p^{-n(a\pm iT)}\right|$$

$$\leq \sum_n \sum_p (1/n)p^{-na} = \log \zeta(a) = \text{const} \tag{3}$$

because this shows that the numerator in (2) is bounded while the denominator goes to infinity.

The substitution of

$$\log \zeta(s) = \log \xi(0) + \sum_\rho \log\left(1 - \frac{s}{\rho}\right) - \log \Pi\left(\frac{s}{2}\right)$$

$$+ \frac{s}{2} \log \pi - \log(s - 1)$$

into (1) expresses $J(x)$ as a sum of five terms (the integral of a finite sum is always the sum of the integrals provided the latter converge) and the derivation of Riemann's formula for $J(x)$ depends now on the evaluation of these five definite integrals.

It should be noted that for any fixed s there is some ambiguity in the definition of $\log[1 - (s/\rho)]$ for those roots ρ which are not large relative to s. In order to remove this ambiguity in $\operatorname{Re} s > 1$ let $\log[1 - (s/\rho)]$ be defined to be $\log(s - \rho) - \log(-\rho)$; this is meaningful because none† of the ρ's are real and greater than or equal to 0. In this way $\log[1 - (s/\rho)]$ is unambiguously defined throughout $\operatorname{Re} s > 1$ and, in particular, on the path of integration $\operatorname{Re} s = a > 1$.

1.14 THE PRINCIPAL TERM OF $J(x)$

It will be seen below that the principal term in the formula for $J(x)$ is the term corresponding to the term $-\log(s - 1)$ of the expansion of $\log \zeta(s)$. This term is

$$\frac{1}{2\pi i} \frac{1}{\log x} \int_{a-i\infty}^{a+i\infty} \frac{d}{ds}\left[\frac{\log(s-1)}{s}\right] x^s \, ds \qquad (a > 1).$$

Riemann shows that for $x > 1$ the value of this definite integral is the logarithmic integral

$$\operatorname{Li}(x) = \lim_{\epsilon \downarrow 0}\left[\int_0^{1-\epsilon} \frac{dt}{\log t} + \int_{1+\epsilon}^{x} \frac{dt}{\log t}\right];$$

†See Section 2.3, or observe that the series $1 - 2^{-s} + 3^{-s} - 4^{-s} + 5^{-s} - \cdots$ converges to a positive number for $s > 0$ and that this number is

$$\zeta(s) - 2 \cdot 2^{-s}(1 + 2^{-s} + 3^{-s} + \cdots) = (1 - 2^{1-s})\zeta(s).$$

that is, it is the Cauchy principal value of the divergent integral $\int_0^x (dt/\log t)$. His argument is as follows:

Fix $x > 1$ and consider the function of β defined by

$$F(\beta) = \frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left\{\frac{\log[(s/\beta)-1]}{s}\right\}x^s\,ds$$

so that the desired number is $F(1)$. The definition of $F(\beta)$ can be extended to all real or complex numbers β other than real numbers $\beta \leq 0$ by taking $a > \operatorname{Re}\beta$ and defining $\log[(s/\beta)-1]$ to be $\log(s-\beta) - \log\beta$, where, as usual, $\log z$ is defined for all z other than real $z \leq 0$ by the condition that it be real for real $z > 0$. The integral $F(\beta)$ converges absolutely because

$$\left|\frac{d}{ds}\left\{\frac{\log[(s/\beta)-1]}{s}\right\}\right| \leq \frac{|\log[(s/\beta)-1]|}{|s|^2} + \frac{1}{|s(s-\beta)|}$$

is integrable while x^s oscillates on the line of integration. Because

$$\frac{d}{d\beta}\left\{\frac{\log[(s/\beta)-1]}{s}\right\} = \frac{1}{(\beta-s)\beta},$$

differentiation under the integral sign and integration by parts give

$$\begin{aligned} F'(\beta) &= \frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left[\frac{1}{(\beta-s)\beta}\right]x^s\,ds \\ &= -\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\frac{x^s}{(\beta-s)\beta}\,ds. \end{aligned}$$

This last integral can be evaluated by applying Fourier inversion to the formula

$$\frac{1}{s-\beta} = \int_1^\infty x^{-s}x^{\beta-1}\,dx \qquad [\operatorname{Re}(s-\beta) > 0],$$

$$\frac{1}{a+i\mu-\beta} = \int_0^\infty e^{-i\lambda\mu}e^{\lambda(\beta-a)}\,d\lambda \qquad [a > \operatorname{Re}\beta],$$

to obtain

$$\text{(1)} \qquad \int_{-\infty}^{\infty}\frac{1}{a+i\mu-\beta}e^{i\mu x}\,d\mu = \begin{cases} 2\pi e^{x(\beta-a)} & \text{if } x > 0, \\ 0 & \text{if } x < 0. \end{cases}$$

from which it follows that

$$\text{(2)} \qquad \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\frac{1}{s-\beta}y^s\,ds = \begin{cases} y^\beta & \text{if } y > 1, \\ 0 & \text{if } y < 1, \end{cases}$$

provided $a > \operatorname{Re}\beta$. Since $x > 1$ by assumption, this gives $F'(\beta) = x^\beta/\beta$.

Now let C^+ denote the contour in the complex t-plane which consists of the line segment from 0 to $1-\epsilon$ (where ϵ is a small positive number), followed by the semicircle in the upper halfplane $\operatorname{Im} t \geq 0$ from $1-\epsilon$ to $1+\epsilon$, fol-

lowed by the line segment from $1 + \epsilon$ to x, and let

$$G(\beta) = \int_{C+} \frac{t^{\beta-1}}{\log t}\, dt.$$

Then

$$G'(\beta) = \int_{C+} t^{\beta-1}\, dt = \frac{t^{\beta}}{\beta}\Big|_0^x = F'(\beta).$$

Now $G(\beta)$ is defined and analytic for $\operatorname{Re}\beta > 0$ (if $\operatorname{Re}\beta < 0$, then the integral which defines G diverges at $t = 0$) as is $F(\beta)$; hence they differ by a constant (which might depend on x) throughout $\operatorname{Re}\beta > 0$. Riemann states that this constant can be evaluated by holding $\operatorname{Re}\beta$ fixed and letting $\operatorname{Im}\beta \to +\infty$ in both $F(\beta)$ and $G(\beta)$, but he does not carry out this evaluation.

To evaluate the limit of $G(\beta)$, set $\beta = \sigma + i\tau$, where σ is fixed and $\tau \to \infty$. The change of variable $t = e^u$, $u = \log t$ puts $G(\beta)$ in the form

$$\int_{i\delta-\infty}^{i\delta+\log x} \frac{e^{\beta u}}{u}\, du + \int_{i\delta+\log x}^{\log x} \frac{e^{\beta u}}{u}\, du,$$

where the path of integration has been altered slightly using Cauchy's theorem. The changes of variable $u = i\delta + v$ in the first integral and $u = \log x + iw$ in the second put this in the form

$$G(\beta) = e^{i\delta\sigma}e^{-\delta\tau}\int_{-\infty}^{\log x} \frac{e^{\sigma v}}{i\delta + v}e^{i\tau v}\, dv - ix^{\beta}\int_0^{\delta} \frac{e^{-\tau w}e^{\sigma i w}}{\log x + iw}\, dw.$$

Both integrals in this expression approach zero as $\tau \to \infty$, the first because $e^{-\delta\tau} \to 0$ and the second because $e^{-\tau w} \to 0$ except at $w = 0$. Thus *the limit of $G(\beta)$ as $\tau \to \infty$ is zero.* (Note, however, that this argument would not be valid if C^+ were changed to follow the lower semicircle because then $e^{-\delta\tau}$ would be replaced by $e^{\delta\tau}$ and $e^{-\tau w}$ would be replaced by $e^{\tau w}$.)

To evaluate the limit of $F(\beta)$ let

$$H(\beta) = \frac{1}{2\pi i}\frac{1}{\log x}\int_{a-t\infty}^{a+t\infty} \frac{d}{ds}\left\{\frac{\log[1 - (s/\beta)]}{s}\right\}x^s\, ds$$

where $a > \operatorname{Re}\beta$ and where $\log[1 - (s/\beta)]$ is defined for all complex numbers β other than real numbers $\beta \geq 0$ to be $\log(s - \beta) - \log(-\beta)$. The difference $H(\beta) - F(\beta)$ is defined for all complex numbers β other than the real axis, and in the upper halfplane $\operatorname{Im}\beta > 0$ it is equal to

$$\begin{aligned} H(\beta) - F(\beta) &= \frac{1}{2\pi i}\frac{1}{\log x}\int_{a-t\infty}^{a+t\infty} \frac{d}{ds}\left[\frac{\log\beta - \log(-\beta)}{s}\right]x^s\, ds, \\ &= \frac{1}{2\pi i}\frac{1}{\log x}\int_{a-t\infty}^{a+t\infty} \frac{d}{ds}\left[\frac{i\pi}{s}\right]x^s\, ds \\ &= -\frac{1}{2\pi i}\int_{a-t\infty}^{a+t\infty} \frac{i\pi}{s}x^s\, ds = -i\pi \end{aligned}$$

by the case $\beta = 0$ of (2). Thus $F(\beta) = H(\beta) + i\pi$ throughout the upper halfplane, and it will suffice to evaluate the limit of $H(\beta)$ as $\tau \to \infty$ ($\beta = \sigma + i\tau$). Now $1 - (s/\beta) \to 1$; hence its log goes to zero and it appears plausible therefore that $H(\beta)$ also goes to zero. This can be proved by carrying out the differentiation

$$\frac{d}{ds}\left\{\frac{\log[1-(s/\beta)]}{s}\right\} = -\frac{\log[1-(s/\beta)]}{s^2} + \frac{1}{s(s-\beta)}$$
$$= -\frac{\log[1-(s/\beta)]}{s^2} + \frac{1}{\beta(s-\beta)} - \frac{1}{\beta s},$$

multiplying by $x^s\, ds/2\pi i$, and integrating from $a - i\infty$ to $a + i\infty$ (in the usual sense, namely, the limit as $T \to \infty$ of the integral from $a - iT$ to $a + iT$). Because of the s^2 in the denominator of the first integral, it is not difficult to show, using the Lebesgue bounded convergence theorem (see Edwards [E1]), that the limit of this integral as $\tau \to \infty$ is the integral of the limit, namely, zero. The remaining two integrals can be evaluated using (2) to find they are $x^\beta/\beta - x^0/\beta = (x^\beta - 1)/\beta$. Since the numerator is bounded and $|\beta| \to \infty$, this approaches zero; hence $H(\beta)$ approaches zero and $F(\beta)$ therefore approaches $i\pi$. Hence $F(\beta) = G(\beta) + i\pi$ in the halfplane $\operatorname{Re}\beta > 0$. Thus the desired number $F(1)$ is

$$F(1) = \int_0^{1-\epsilon} \frac{dt}{\log t} + \int_{1-\epsilon}^{1+\epsilon} \frac{(t-1)}{\log t} \cdot \frac{dt}{t-1} + \int_{1+\epsilon}^{x} \frac{dt}{\log t} + i\pi,$$

where the second integral is over the semicircle in the upper halfplane; as $\epsilon \downarrow 0$, the quotient $(t-1)/\log t$ approaches 1 along this semicircle, and hence the integral approaches $\int_{1-\epsilon}^{1+\epsilon} dt/(t-1) = -i\pi$. Thus the limit as $\epsilon \downarrow 0$ of the above formula is

$$F(1) = \operatorname{Li}(x)$$

as was to be shown.

1.15 THE TERM INVOLVING THE ROOTS ρ

Consider next the term in the formula for $J(x)$ arising from the term $\sum \log[1 - (s/\rho)]$ in the formula for $\log \zeta(s)$, namely,

$$(1) \qquad -\frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty} \frac{d}{ds}\left\{\frac{\sum \log[1-(s/\rho)]}{s}\right\} x^s\, ds.$$

If the operation of summation over ρ can be interchanged with the differentiation and the integration, *then* this is equal to $-\Sigma H(\rho)$, where $H(\rho)$ is defined as in the preceding section. Now it was shown that $H(\rho) \equiv G(\rho)$ for ρ in the first quadrant ($\operatorname{Re}\rho > 0$, $\operatorname{Im}\rho > 0$) and in exactly the same way it can

be shown that for ρ in the fourth quadrant (Re $\rho > 0$, Im $\rho < 0$) the value of $H(\rho)$ is equal to the integral $G(\rho)$ except that the integral must be over the contour C^- which goes over the lower semicircle from $1 - \epsilon$ to $1 + \epsilon$ rather than over the upper semicircle as C^+ did. Thus, pairing terms of the sum over ρ in the usual way, the integral (1) would be

$$-\sum_{\mathrm{Im}\rho>0}\left(\int_{C^+}\frac{t^{\rho-1}}{\log t}\,dt+\int_{C^-}\frac{t^{-\rho}}{\log t}\,dt\right) \tag{2}$$

if it could be evaluated termwise. Now if β is real and positive, then the change of variable $u = t^\beta$, which implies $\log t = \log u/\beta$, $dt/t = du/u\beta$, gives

$$\int_{C^+}\frac{t^{\beta-1}}{\log t}\,dt=\int_0^{x^\beta}\frac{du}{\log u}=\mathrm{Li}(x^\beta)-i\pi,$$

where the second integral is over a path which passes above the singularity at $u = 1$. Since the integral on the left converges throughout the halfplane Re $\beta > 0$, this formula gives the analytic continuation of $\mathrm{Li}(x^\beta)$ to this halfplane (when x is, as always, a fixed number $x > 1$). In the same way

$$\int_{C^-}\frac{t^{\beta-1}}{\log t}\,dt=\mathrm{Li}(x^\beta)+i\pi,$$

and (2) becomes

$$-\sum_{\mathrm{Im}\,\rho>0}[\mathrm{Li}(x^\rho)+\mathrm{Li}(x^{1-\rho})]. \tag{3}$$

Thus, if termwise evaluation is valid, the desired integral (1) is equal to (3).

Riemann states that termwise evaluation is valid and that (3) is indeed the desired value (1) but that the series (3) is only conditionally convergent—even though the terms ρ, $1 - \rho$ are paired—and that it must be summed in the order of increasing† Im ρ. He concedes that the validity of this termwise evaluation of (1) requires "a more exact discussion of the function ξ," but says that this is "easy" and passes on to the next point.

One other small remark about the sum (3) is necessary. The computations above assume Re $\rho > 0$, but it has not been shown that this is true for all roots ρ. Although Hadamard later proved that there are no roots ρ on the line Re $\rho = 0$ (see Section 4.2), Riemann has not excluded this possibility and he is therefore not justified in ignoring the point as he does.

†It is interesting to note that Riemann writes $\rho = \frac{1}{2} + i\alpha$ and says first that the sum (3) is over all *positive* values of α in order of size before then adding parenthetically that it is over all α's with Re$(\alpha) > 0$ in order of size. Thus he admits, albeit parenthetically, the possibility that the Riemann hypothesis is false.

1.16 THE REMAINING TERMS

One of the three remaining terms in the formula for $J(x)$, namely, the term arising from $(s/2)\log\pi$, drops out when it is divided by s and differentiated with respect to s. The term arising from the constant $\log\xi(0)$ is

$$-\frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left(\frac{\log\xi(0)}{s}\right)x^s\,ds$$
$$=\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\frac{\log\xi(0)}{s}x^s\,ds=\log\xi(0)$$

using (2) of Section 1.14 in the case $\beta=0$. Now $\xi(0)=\Pi(0)\pi^{-0}(0-1)\zeta(0)=-\zeta(0)=\frac{1}{2}$ so $\log\xi(0)=-\log 2$ is the numerical value of this term.

Riemann writes $\log\xi(0)$ instead of $-\log 2$, but since he uses ξ to denote a different function—namely, the function $\xi(\frac{1}{2}+it)$ of t—his $\xi(0)$ denotes $\xi(\frac{1}{2})\neq\frac{1}{2}$ and thus his formula is in error. It is hard to guess what the source of this trivial error might be, other than to say that it arises from some confusion between the product formula

$$\xi(s)=\xi(0)\prod_{\rho}[1-(s/\rho)]$$

in the form it is given above and the product formula

$$\xi\left(\frac{1}{2}+it\right)=\xi(0)\prod\left(1-\frac{\frac{1}{2}+it}{\frac{1}{2}+i\alpha}\right)$$
$$=\xi(0)\prod\left(\frac{i\alpha-it}{\frac{1}{2}+i\alpha}\right)$$
$$=\xi(0)\prod\left(\frac{i\alpha}{\frac{1}{2}+i\alpha}\right)\prod\left(1-\frac{it}{i\alpha}\right)$$
$$=\xi(0)\prod\left(1-\frac{\frac{1}{2}}{\frac{1}{2}+i\alpha}\right)\prod\left(1-\frac{t}{\alpha}\right)$$
$$=\xi\left(\frac{1}{2}\right)\prod_{\mathrm{Re}\,\alpha>0}\left(1-\frac{t^2}{\alpha^2}\right)$$

in the form given by Riemann, and a concomitant confusion of the integral

$$\frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left\{\frac{\log[1-(s/\rho)]}{s}\right\}x^s\,ds$$

which he evaluates, with the integral

$$\frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left\{\frac{\log[1-(s-\frac{1}{2})/i\alpha]}{s}\right\}x^s\,ds$$

which differs from it by a constant. Whatever the source of the error, Riemann makes the same error in the letter quoted by the editors in the notes which follow the paper in the collected works, and his unpublished papers [R1a] include a computation of $\log\xi(\frac{1}{2})$ to several decimal places, so it was definitely not a typographical error as the editors of the collected works suppose. The error was noticed by Genocchi [G4] during Riemann's lifetime.

This leaves only one term

$$(1) \qquad \frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left[\frac{\log \Pi(s/2)}{s}\right]x^s\,ds$$

to be evaluated. Now by formula (4) of Section 1.3

$$\log \Pi\left(\frac{s}{2}\right) = \sum_{n=1}^{\infty}\left[-\log\left(1+\frac{s}{2n}\right)+\frac{s}{2}\log\left(1+\frac{1}{n}\right)\right].$$

Using this formula in (1) and assuming that termwise integration is valid puts (1) in the form

$$-\sum_{n=1}^{\infty}\frac{1}{2\pi i}\frac{1}{\log x}\int_{a-i\infty}^{a+i\infty}\frac{d}{ds}\left\{\frac{\log[1+(s/2n)]}{s}\right\}x^s\,ds = -\sum_{n=1}^{\infty}H(-2n),$$

where H is as in Section 1.14. The previous formulas for $H(\beta)$ apply only in the halfplane $\operatorname{Re}\beta > 0$. To obtain a formula for H in $\operatorname{Re}\beta < 0$ set

$$E(\beta) = -\int_x^{\infty}\frac{t^{\beta-1}}{\log t}\,dt.$$

Then $E(\beta)$ converges for $\operatorname{Re}\beta < 0$ and satisfies

$$E'(\beta) = -\int_x^{\infty} t^{\beta-1}\,dt = \frac{x^{\beta}}{\beta} = F'(\beta) = H'(\beta)$$

so $E(\beta)$ differs from $H(\beta)$ by a constant throughout $\operatorname{Re}\beta < 0$. Since both E and H approach zero as $\beta \to -\infty$, the constant is zero and $E \equiv H$. Thus (1) becomes

$$\sum_{n=1}^{\infty}\int_x^{\infty}\frac{t^{-2n-1}}{\log t}\,dt = \int_x^{\infty}\frac{1}{t\log t}(\Sigma t^{-2n})\,dt = \int_x^{\infty}\frac{dt}{t(t^2-1)\log t}$$

provided termwise integration is valid. The proof that termwise integration is valid, which Riemann (tacitly) leaves to the reader, can be given as follows.

Note first that the series

$$\frac{d}{ds}\left[\frac{\log \Pi(s/2)}{s}\right] = -\sum_{n=1}^{\infty}\frac{d}{ds}\left\{\frac{\log[1+(s/2n)]}{s}\right\}$$

converges uniformly in any disk $|s| \le K$. [For large n the series expansion $\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots$ can be used, and the summand on the right contains only terms in $n^{-2}, n^{-3}, \ldots$.] This justifies the termwise differentiation and also justifies termwise integration over any finite interval

$$(2) \qquad \frac{1}{2\pi i}\frac{1}{\log x}\int_{a-iT}^{a+iT}\frac{d}{ds}\left[\frac{\log \Pi(s/2)}{s}\right]x^s\,ds$$
$$= -\sum_{n=1}^{\infty}\frac{1}{2\pi i}\frac{1}{\log x}\int_{a-iT}^{a+iT}\frac{d}{ds}\left\{\frac{\log[1+(s/2n)]}{s}\right\}x^s\,ds.$$

To estimate the nth term of the sum on the right set $v = (s-a)/2n$, $b = a/2n$,

$c = T/2n$, so $s = 2n(v + b)$ and the nth term is minus

$$\frac{1}{2\pi i}\frac{1}{\log x}\int_{-ic}^{ic}\frac{d}{2n\,dv}\left[\frac{\log(1+v+b)}{2n(v+b)}\right]x^{2nv+a}2n\,dv$$
$$= \frac{1}{2\pi i}\frac{x^a}{2n\log x}\int_{-ic}^{ic}\frac{d}{dv}\left[\frac{\log(1+v+b)}{v+b}\right]x^{2nv}\,dv.$$

Integration by parts puts this in the form

$$= \frac{1}{2\pi i}\frac{x^a}{2n\log x}\cdot\frac{1}{2n\log x}\left(\left\{\frac{d}{dv}\left[\frac{\log(1+v+b)}{v+b}\right]x^{2nv}\right\}_{v=-ic}^{v=ic}\right.$$
$$\left.-\int_{-ic}^{ic}\frac{d^2}{dv^2}\left[\frac{\log(1+v+b)}{v+b}\right]x^{2nv}\,dv\right).$$

Now b is a real number $0 \le b \le a$, the function

$$\frac{d}{dv}\left[\frac{\log(1+v+b)}{v+b}\right] = \frac{1}{(v+b)(v+b+1)} - \frac{\log(1+v+b)}{(v+b)^2}$$

is bounded on the imaginary axis, and its derivative is absolutely integrable over $(-i\infty, i\infty)$, from which it follows that the modulus of the nth term of the series on the right side of (2) is at most a constant times n^{-2} *for all* T. Thus the series converges uniformly in T and one can pass to the limit $T \longrightarrow \infty$ termwise, as was to be shown.

This completes the evaluation of the terms in the formula for $J(x)$. Combining them gives the final result

$$(3)\qquad J(x) = \mathrm{Li}(x) - \sum_{\mathrm{Im}\rho>0}[\mathrm{Li}(x^\rho) + \mathrm{Li}(x^{1-\rho})] + \int_x^\infty \frac{dt}{t(t^2-1)\log t} + \log\xi(0) \qquad (x > 1)$$

which is Riemann's formula [except that, as noted above, $\log\xi(0)$ equals $\log(\frac{1}{2})$ and not $\log\xi(\frac{1}{2})$ as in Riemann's notation it should]. This analytic formula for $J(x)$ is the principal result of the paper.

1.17 THE FORMULA FOR $\pi(x)$

Of course Riemann's goal was to obtain a formula not for $J(x)$ but for the function $\pi(x)$, that is, for the number of primes less than any given magnitude x. Since the number of prime squares less than x is obviously equal to the number of primes less than $x^{1/2}$, that is, equal to $\pi(x^{1/2})$, and since in the same way the number of prime nth powers p^n less than x is $\pi(x^{1/n})$, it follows that J and π are related by the formula

$$(1)\qquad J(x) = \pi(x) + \frac{1}{2}\pi(x^{1/2}) + \frac{1}{3}\pi(x^{1/3}) + \cdots + \frac{1}{n}\pi(x^{1/n}) + \cdots.$$

The series in this formula is finite for any given x because $x^{1/n} < 2$ for n sufficiently large, which implies $\pi(x^{1/n}) = 0$. Riemann inverts this relationship by means of the Möbius inversion formula† (see Section 10.9) to obtain

$$(2) \qquad \pi(x) = J(x) - \frac{1}{2}J(x^{1/2}) - \frac{1}{3}J(x^{1/3}) - \frac{1}{5}J(x^{1/5}) + \frac{1}{6}J(x^{1/6}) + \cdots + \frac{\mu(n)}{n}J(x^{1/n}) + \cdots,$$

where $\mu(n)$ is 0 if n is divisible by a prime square, 1 if n is a product of an even number of distinct primes, and -1 if n is a product of an odd number of distinct primes. The series (2) is a finite series for any fixed x and when combined with the analytical formula for $J(x)$

$$(3) \qquad J(x) = \mathrm{Li}(x) - \sum_{\rho} \mathrm{Li}(x^{\rho}) - \log 2 + \int_x^{\infty} \frac{dt}{t(t^2-1)\log t} \qquad (x > 1),$$

it gives an analytical formula for $\pi(x)$ as desired.

The formula for $\pi(x)$ which results from substituting (3) in the (finite) series (2) consists of three kinds of terms, namely, those which do not grow as x grows [arising from the last two terms of (3)], those which grow as x grows but which oscillate in sign [the terms arising from $\mathrm{Li}(x^{\rho})$ which Riemann calls "periodic"], and those which grow steadily as x grows [the terms arising from $\mathrm{Li}(x)$]. If all but the last type are ignored, the terms in the formula for $\pi(x)$ are just

$$\mathrm{Li}(x) - \tfrac{1}{2}\mathrm{Li}(x^{1/2}) - \tfrac{1}{3}\mathrm{Li}(x^{1/3}) - \tfrac{1}{5}\mathrm{Li}(x^{1/5}) + \tfrac{1}{6}\mathrm{Li}(x^{1/6}) - \tfrac{1}{7}\mathrm{Li}(x^{1/7}) + \cdots.$$

Now *empirically* this is found to be a good approximation to $\pi(x)$. In fact, the first term alone is essentially Gauss's approximation

$$\pi(x) \sim \int_2^x \frac{dt}{\log t} = \mathrm{Li}(x) - \mathrm{Li}(2)$$

[$\mathrm{Li}(2) = 1.04\ldots$] and the first two terms indicate that

$$\pi(x) \sim \mathrm{Li}(x) - \tfrac{1}{2}\mathrm{Li}(x^{1/2})$$

†Very simply this inversion is effected by performing successively for each prime $p = 2, 3, 5, 7, 11, \ldots$ the operation of replacing the functions $f(x)$ on each side of the equation with the functions $f(x) - (1/p)f(x^{1/p})$. This gives successively

$$J(x) - \tfrac{1}{2}J(x^{1/2}) = \pi(x) + \tfrac{1}{3}\pi(x^{1/3}) + \tfrac{1}{5}\pi(x^{1/5}) + \cdots,$$

$$J(x) - \tfrac{1}{2}J(x^{1/2}) - \tfrac{1}{3}J(x^{1/3}) + \tfrac{1}{6}J(x^{1/6}) = \pi(x) + \tfrac{1}{5}\pi(x^{1/5}) + \tfrac{1}{7}\pi(x^{1/7}) + \cdots,$$

etc., where at each step the sum on the left consists of those terms of the right side of (2) for which the factors of n contain *only* the primes already covered and the sum on the right consists of those terms of the right side of (1) for which the factors of n contain *none* of the primes already covered. Once p is sufficiently large, the latter are all zero except for $\pi(x)$.

which gives, for example,

$$\pi(10^6) \sim 78{,}628 - \tfrac{1}{2}\cdot 178 = 78{,}539$$

which is better than Gauss's approximation and which becomes still better if the third term is used. The extent to which Riemann's suggested approximation

$$\pi(x) \sim \mathrm{Li}(x) + \sum_{n=2}^{\infty} \frac{\mu(n)}{n} \mathrm{Li}(x^{1/n}) \tag{4}$$

is better than $\pi(x) \sim \mathrm{Li}(x)$ is stunningly illustrated by one of Lehmer's tables [L9], an extract of which is given in Table III.

TABLE III[a]

x	Riemann's error	Gauss's error
1,000,000	30	130
2,000,000	−9	122
3,000,000	0	155
4,000,000	33	206
5,000,000	−64	125
6,000,000	24	228
7,000,000	−38	179
8,000,000	−6	223
9,000,000	−53	187
10,000,000	88	339

[a]From Lehmer [L9].

Of course Riemann did not have such extensive empirical data at his disposal, but he seems well aware of the fact that (4) is a better approximation, as well as a more natural approximation, to $\pi(x)$.

Riemann was also well aware, however, of the defects of the approximation (4) and of his analysis of it. Although he has succeeded in giving an exact analytical formula for the error

$$\pi(x) - \sum_{n=1}^{N} \frac{\mu(n)}{n} \mathrm{Li}(x^{1/n}) = \sum_{n=1}^{N} \sum_{\rho} \mathrm{Li}(x^{\rho/n}) + \text{lesser terms}$$

(where N is large enough that $x^{1/(N+1)} < 2$) he has no estimate at all of the size of these "periodic" terms $\sum \sum \mathrm{Li}(x^{\rho/n})$. Actually, the empirical fact that they are as small as Lehmer found them to be is somewhat surprising in view of the fact that the series $\sum [\mathrm{Li}(x^{\rho}) + \mathrm{Li}(x^{1-\rho})]$ is only conditionally convergent—hence the smallness of its sum for any x depends on wholesale cancellation of signs among the terms—and in view of the fact that the in-

dividual terms $\mathrm{Li}(x^\rho)$ grow in magnitude like $|x^\rho/\log x^\rho| = x^{\mathrm{Re}\rho}/|\rho|\log x$ (see Section 5.5) so that many of them grow at least as fast as $x^{1/2}/\log x \sim 2\,\mathrm{Li}(x^{1/2}) > \mathrm{Li}(x^{1/3})$ and would therefore be expected to be as significant for large x as the term $-\frac{1}{2}\,\mathrm{Li}(x^{1/2})$ and more significant than any of the following terms of (4). On these subjects Riemann restricts himself to the statement that it would be interesting in later counts of primes to study the effect of the particular "periodic" terms on their distribution.

In short, although formulas (2) and (3) combine to give an analytical formula for $\pi(x)$, the validity of the new approximation (4) to $\pi(x)$ to which it leads is based, like that of the old approximation $\pi(x) \sim \mathrm{Li}(x)$, solely on empirical evidence.

1.18 THE DENSITY *dJ*

A simple formulation of the main result

$$(1) \qquad J(x) = \mathrm{Li}(x) - \sum_\rho \mathrm{Li}(x^\rho) - \log 2 + \int_x^\infty \frac{dt}{t(t^2-1)\log t}$$

can be obtained by differentiating to find

(2)

$$dJ = \left[\frac{1}{\log x} - \sum_{\mathrm{Re}\alpha>0} \frac{2\cos(\alpha \log x)}{x^{1/2}\log x} - \frac{1}{x(x^2-1)\log x}\right]dx \qquad (x>1),$$

where α ranges over all values such that $\rho = \frac{1}{2} + i\alpha$—in other words $\alpha = -i(\rho - \frac{1}{2})$, where ρ ranges over the roots—so that

$$x^{\rho-1} + x^{-\rho} = x^{-1/2}[x^{i\alpha} + x^{-i\alpha}] = 2x^{-1/2}\cos(\alpha \log x).$$

[The Riemann hypothesis is that the α's are all real. In writing formula (2) in this form Riemann is clearly thinking of the α's as being real since otherwise the natural form would be $x^{\rho-1} + x^{\beta-1} = 2x^{\beta-1}\cos(\gamma \log x)$, where $\rho = \beta + i\gamma$.]

By the definition of J, the measure dJ is dx times the density of primes plus $\frac{1}{2}$ the density of prime squares, plus $\frac{1}{3}$ the density of prime cubes plus, etc. Thus $1/\log x$ should be considered to be an approximation not to the density of primes as Gauss suggested but rather to dJ, that is, to the density of primes *plus* $\frac{1}{2}$ the density of prime squares, plus, etc.

Given two large numbers $a < b$ the approximation obtained by taking a finite number of the α's

$$(3) \qquad J(b) - J(a) \sim \int_a^b \frac{dt}{\log t} - 2\sum \int_a^b \frac{\cos(\alpha \log t)\,dt}{t^{1/2}\log t}$$

should be a fairly good approximation because the omitted term $\int dx/x(x^2-1)\log x$

is entirely negligible and because the integrals involving the large α's oscillate very rapidly for large x and therefore should make very small contributions. In fact, the basic formula (1) implies immediately that the error in (3) approaches the negligible omitted term as more and more of the α's are included in the sum.

It is in the sense of investigating the number of α's which are significant in (3) that Riemann meant to investigate empirically the influence of the "periodic" terms on the distribution of primes. So far as I know, no such investigation has ever been carried out.

1.19 QUESTIONS UNRESOLVED BY RIEMANN

Riemann himself, in a letter quoted in the notes which follow this paper in his collected works, singles out two statements of the paper as not having been fully proved as yet, namely, the statement that the equation $\xi(\frac{1}{2} + i\alpha) = 0$ has approximately $(T/2\pi)\log(T/2\pi)$ *real* roots α in the range $0 < \alpha < T$ and the statement that the integral of Section 1.15 can be evaluated termwise. He expresses no doubt about the truth of these statements, however, and says that they follow from a new representation of the function ξ which he has not yet simplified sufficiently to publish. Nonetheless, as was stated in Section 1.9, the first of these two statements—at least if it is understood to mean that the relative error in the approximation approaches zero as $T \to \infty$—has never been proved. The second was proved by von Mangoldt in 1895, but by a method completely different from that suggested by Riemann, namely by proving first that Riemann's formula for $J(x)$ is valid and by concluding from this that the termwise value of the integral in Section 1.15 must be correct.

Riemann evidently believed that he had given a proof of the product formula for $\xi(s)$, but, at least from the reading of the paper given above, one cannot consider his proof to be complete, and, in particular, one must question Riemann's estimate of the number of roots ρ in the range $\{0 \leq \text{Im } \rho \leq T\}$ on which this proof is based. It was not until 1893 that Hadamard proved the product formula, and not until 1905 that von Mangoldt proved the estimate of the number of roots in $\{0 \leq \text{Im } \rho \leq T\}$.

Next, the original question of the validity of the approximation $\pi(x) \sim \int_2^x (dt/\log t)$ remained entirely unresolved by Riemann's paper. It can be shown that the relative error of this approximation approaches zero as $x \to \infty$ if and only if the same is true of the relative error in Riemann's approximation $J(x) \sim \text{Li}(x)$, so the original question is equivalent to the question of whether $\sum \text{Li}(x^\rho)/\text{Li}(x) \to 0$, but this unfortunately does not bring the problem any

nearer to a solution. It was not until 1896 that Hadamard and, independently, de la Vallée Poussin proved the prime number theorem to the effect that the relative error in $\pi(x) \sim \int_2^x (dt/\log t)$ does approach zero as $x \to \infty$.

Finally, the paper raised a question much greater than any question it answered, the question of the truth or falsity of the Riemann hypothesis.

The remainder of this book is devoted to the subsequent history of these six questions. In summary, they are as follows:

(a) Is Riemann's estimate of the number of roots ρ on the line segment from $\frac{1}{2}$ to $\frac{1}{2} + iT$ correct as $T \to \infty$? (Unknown.)

(b) Is termwise evaluation of the integral of Section 1.15 valid? (Yes, von Mangoldt, 1895.)

(c) Is the product formula for $\xi(s)$ valid? (Yes, Hadamard, 1893.)

(d) Is Riemann's estimate of the number of roots ρ in the strip $\{0 \leq \text{Im } \rho \leq T\}$ correct? (Yes, von Mangoldt, 1905.)

(e) Is the prime number theorem true? [Yes, Hadamard and de la Vallée Poussin (independently), 1896.]

(f) Is the Riemann hypothesis true? (Unknown.)

Chapter 2

The Product Formula for ξ

2.1 INTRODUCTION

In 1893 Hadamard published a paper [H1] in which he studied entire functions (functions of a complex variable which are defined and analytic at all points of the complex plane) and their representations as infinite products. One consequence of the general theory which he developed in this paper is the fact that the product formula

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right) \tag{1}$$

is valid; here ξ is the entire function defined in Section 1.8, ρ ranges over all roots ρ of $\xi(\rho) = 0$, and the infinite product is understood to be taken in an order which pairs each root ρ with the corresponding root $1 - \rho$. Hadamard's proof of the product formula for ξ was called by von Mangoldt [M1] "the first real progress in the field in 34 years," that is, the first since Riemann's paper.

This chapter is devoted to the proof of the product formula (1). Since only the specific function ξ is of interest here, Hadamard's methods for general entire functions can, of course, be considerably specialized and simplified† for this case, and in the end the proof which results is closer to the one outlined by Riemann than to Hadamard's proof. The first step of the proof is to make an estimate of the distribution of the roots ρ. This estimate, which is that the number of roots ρ in the disk $|\rho - \frac{1}{2}| < R$ is less than a constant times $R \log R$ as $R \longrightarrow \infty$, is based on Jensen's theorem and is much less exact than Riemann's estimate that the number of roots in the strip $\{0 < \text{Im}$

†A major simplification is the use of Jensen's theorem, which was not known at the time Hadamard was writing.

$\rho < T\}$ is $(T/2\pi)\log(T/2\pi) - (T/2\pi)$ with a relative error which is of the order of magnitude of T^{-1}. It is exact enough, however, to prove the convergence of the product (1). Once it has been shown that this product converges, the rest of the proof can be carried out more or less as Riemann suggests.

2.2 JENSEN'S THEOREM

Theorem Let $f(z)$ be a function which is defined and analytic throughout a disk $\{|z| \leq R\}$. Suppose that $f(z)$ has no zeros on the bounding circle $|z| = R$ and that inside the disk it has the zeros $z_1, z_2, \ldots, z_n$ (where a zero of order k is included k times in the list). Suppose, finally, that $f(0) \neq 0$. Then

$$\log\left|f(0)\cdot\frac{R}{z_1}\cdot\frac{R}{z_2}\cdot\ldots\cdot\frac{R}{z_n}\right| = \frac{1}{2\pi}\int_0^{2\pi}\log|f(Re^{i\theta})|\,d\theta. \tag{1}$$

***Proof*†** If $f(z)$ has no zeros inside the disk, then the equation is merely

$$\log|f(0)| = \frac{1}{2\pi}\int_0^{2\pi}\log|f(Re^{i\theta})|\,d\theta; \tag{2}$$

that is, the equation is the statement that the value of $\log|f(z)|$ at the center of the disk is equal to its average value on the bounding circle. This can be proved either by observing that $\log|f(z)|$ is the real part of the analytic function $\log f(z)$ and is therefore a harmonic function, or by taking the real part of the Cauchy integral formula

$$\log f(0) = \frac{1}{2\pi i}\int_{|z|=R}\frac{\log f(z)}{z}\,dz = \frac{1}{2\pi}\int_0^{2\pi}\log f(Re^{i\theta})\,d\theta,$$

where $\log f(z)$ is defined in the disk to be

$$\log|f(0)| + \int_0^z \frac{f'(t)}{f(t)}\,dt.$$

Applying this formula (2) to the function

$$F(z) = f(z)\frac{R^2 - \bar{z}_1 z}{R(z - z_1)}\cdot\frac{R^2 - \bar{z}_2 z}{R(z - z_2)}\cdots\frac{R^2 - \bar{z}_n z}{R(z - z_n)}$$

gives

$$\log|F(0)| = \frac{1}{2\pi}\int_0^{2\pi}\log|F(Re^{i\theta})|\,d\theta$$

because $F(z)$ is analytic and has no zeros in the disk. But this is the formula

†See Ahlfors [A3]. This method of proof of Jensen's threorem is to be found in Backlund's 1918 paper on the Lindelöf hypothesis [B3] (see Section 9.4).

of Jensen's theorem (1) because

$$\left|\frac{R^2 - \bar{z}_j \cdot 0}{R(0 - z_j)}\right| = \left|\frac{R}{z_j}\right|$$

and because by a basic formula in the theory of conformal mapping

$$\left|\frac{R^2 - \bar{z}_j z}{R(z - z_j)}\right| = 1 \qquad \text{when} \quad |z| = R.$$

(To prove this formula multiply the numerator by $\bar{z}/R$. This does not change the modulus if $|z| = R$ and it makes the numerator into the complex conjugate of the denominator.) This completes the proof of Jensen's theorem (1).

2.3 A SIMPLE ESTIMATE OF $|\xi(s)|$

Theorem For all sufficiently large values of R the estimate $|\xi(s)| \leq R^R$ holds throughout the disk $|s - \frac{1}{2}| \leq R$.

Proof It was shown in Section 1.8 that $\xi(s)$ can be expanded as a power series in $(s - \frac{1}{2})$:

$$\xi(s) = a_0 + a_2(s - \tfrac{1}{2})^2 + \cdots + a_{2n}(s - \tfrac{1}{2})^{2n} + \cdots,$$

where

$$a_{2n} = 4\int_1^\infty \frac{d}{dx}[x^{3/2}\psi'(x)]\, x^{-1/4}\frac{(\frac{1}{2}\log x)^{2n}}{(2n)!}\, dx.$$

The fact that the coefficients a_n are *positive* follows immediately from

$$\frac{d}{dx}[x^{3/2}\psi'(x)] = \frac{d}{dx}\left(-\sum_{n=1}^{\infty} x^{3/2} n^2 \pi e^{-n^2\pi x}\right)$$

$$= \sum_{n=1}^{\infty}\left(n^4\pi^2 x - \frac{3}{2}n^2\pi\right)x^{1/2}e^{-n^2\pi x}$$

because this shows that the integrand in the integral for a_{2n} is positive for $x \geq 1$. Thus the largest value of $\xi(s)$ on the disk $|s - \frac{1}{2}| \leq R$ occurs at the point $s = \frac{1}{2} + R$, and to prove the theorem it suffices to show that $\xi(\frac{1}{2} + R) \leq R^R$ for all sufficiently large R. Now

$$\xi(s) = \Pi(s/2)\pi^{-s/2}(s - 1)\zeta(s)$$

and $\zeta(s)$ decreases to 1 as $s \to +\infty$, so if R is given and if N is chosen so that $\frac{1}{2} + R \leq 2N < \frac{1}{2} + R + 2$, it follows that

$$\begin{aligned}\xi(\tfrac{1}{2} + R) \leq \xi(2N) &= (N!)\pi^{-N}(2N - 1)\zeta(2N)\\ &\leq N^N\pi^{-0}(2N)\zeta(2)\\ &= \text{const}\ N^{N+1}\\ &\leq \text{const}\ (\tfrac{1}{2}R + 2)^{(R/2)+3} < R^R\end{aligned}$$

for all sufficiently large R, which completes the proof of the theorem.

2.4 THE RESULTING ESTIMATE OF THE ROOTS ρ

Theorem Let $n(R)$ denote the number of roots ρ of $\xi(\rho) = 0$ which lie inside or on the circle $|s - \frac{1}{2}| = R$ (counted with multiplicities). Then $n(R) \leq 3R \log R$ for all sufficiently large R.

Proof Jensen's theorem applied to $\xi(s)$ on the disk $|s - \frac{1}{2}| \leq 2R$ gives

$$\log \xi\left(\frac{1}{2}\right) + \sum_{|\rho - 1/2| < 2R} \log \frac{2R}{|\rho - \frac{1}{2}|} \leq \log[(2R)^{2R}].$$

The terms of the sum over ρ are all positive and the terms corresponding to roots ρ inside the circle $|\rho - \frac{1}{2}| \leq R$ are all at least $\log 2$; hence,

$$n(R) \log 2 \leq 2R \log 2R - \log \xi(\tfrac{1}{2})$$

$$n(R) \leq \frac{2}{\log 2} R \log R + 2R - \frac{\log \xi(\frac{1}{2})}{\log 2}$$

$$\leq 3R \log R$$

for all sufficiently large R, as was to be shown. If there are roots ρ on the circle $|s - \frac{1}{2}| = 2R$, so that Jensen's theorem is not applicable, one can apply the above to the circle with radius $R + \epsilon$ and let $\epsilon \to 0$.

2.5 CONVERGENCE OF THE PRODUCT

As was noted in Section 1.10, in order to prove the convergence of the product

$$\prod \left(1 - \frac{s}{\rho}\right) = \prod_{\text{Im } \rho > 0} \left[1 - \frac{s(1-s)}{\rho(1-\rho)}\right] \tag{1}$$

for all s, it suffices to prove the convergence of the sum $\sum |\rho(1-\rho)|^{-1}$. Since all but a finite number of roots ρ satisfy the inequality

$$\frac{1}{|\rho(1-\rho)|} = \frac{1}{|(\rho - \frac{1}{2})^2 - \frac{1}{4}|} < \frac{1}{|\rho - \frac{1}{2}|^2},$$

it suffices therefore to prove the convergence of the sum $\sum |\rho - \frac{1}{2}|^{-2}$; here the sum can be considered either as a sum over roots ρ in the upper halfplane $\text{Im } \rho > 0$ or as a sum over all roots since the first of these is merely twice the second. The convergence of the product (1) is therefore a consequence of the case $\epsilon = 1$ of the following theorem.

Theorem For any given $\epsilon > 0$ the series

$$\sum \frac{1}{|\rho - \frac{1}{2}|^{1+\epsilon}}$$

converges, where ρ ranges over all roots ρ of $\xi(\rho) = 0$.

[Note that this theorem would follow immediately from Riemann's observation that the vertical density of the roots ρ is a constant times $(\log T)\,dT$ and from the fact that $\int^{\infty} T^{-1-\epsilon}(\log T)\,dT$ converges. This is Riemann's first step in his "proof" of the product formula for ξ.]

Proof Let the roots ρ be numbered $\rho_1, \rho_2, \rho_3, \ldots$ in order of increasing $|\rho - \frac{1}{2}|$. Furthermore, let $R_1, R_2, R_3, \ldots$ be the sequence of positive real numbers defined implicitly by the equation $4R_n \log R_n = n$. Then by the theorem of the preceding section there are at most $3n/4$ roots ρ inside the circle $|s - \frac{1}{2}| = R_n$; hence the nth root is not in this circle, that is, $|\rho_n - \frac{1}{2}| > R_n$. Thus

$$\sum \frac{1}{|\rho_n - \frac{1}{2}|^{1+\epsilon}} \leq \sum \frac{1}{R_n^{1+\epsilon}} = \sum \frac{(4 \log R_n)^{1+\epsilon}}{n^{1+\epsilon}}$$
$$= \sum \frac{1}{n^{1+(\epsilon/2)}} \cdot \frac{(4 \log R_n)^{1+\epsilon}}{n^{\epsilon/2}}.$$

Now $\log n = \log R_n + \log 4 + \log \log R_n > \log R_n$ for n large. Hence $(4 \log R_n)^{1+\epsilon} < 16 (\log n)^2 < n^{\epsilon/2}$ for all sufficiently large n and

$$\sum \frac{1}{|\rho - \frac{1}{2}|^{1+\epsilon}} < \text{const} + \sum \frac{1}{n^{1+(\epsilon/2)}} < \infty$$

as was to be shown.

2.6 RATE OF GROWTH OF THE QUOTIENT

Riemann states that $\log \xi(s) - \sum \log [1 - (s/\rho)]$ grows no faster than $s \log s$, from which he concludes, since it is an even function, that it must be a constant. In this section the weaker result that the growth of its real part is no faster than $|s|^{1+\epsilon}$ will be proved. This still permits one to conclude, as will be shown in the next section, that it is constant.

Theorem Let $\epsilon > 0$ be given. Then

$$\operatorname{Re} \log \frac{\xi(s)}{\prod_{\rho} \left(1 - \frac{s - \frac{1}{2}}{\rho - \frac{1}{2}}\right)} \leq \left|s - \frac{1}{2}\right|^{1+\epsilon}$$

for all sufficiently large $|s - \frac{1}{2}|$.

Proof Let R be given and let the function being estimated be written as a sum of two functions

$$\operatorname{Re} \log \frac{\xi(s)}{\prod \left(1 - \frac{s - \frac{1}{2}}{\rho - \frac{1}{2}}\right)} = u_R(s) + v_R(s),$$

where

$$u_R(s) = \operatorname{Re}\log\frac{\xi(s)}{\prod_{|\rho-1/2|\le 2R}\left(1-\frac{s-\frac12}{\rho-\frac12}\right)}$$

$$v_R(s) = \operatorname{Re}\log\frac{1}{\prod_{|\rho-1/2|>2R}\left(1-\frac{s-\frac12}{\rho-\frac12}\right)}.$$

These logarithms are defined only up to multiples of $2\pi i$, but their real parts are well defined except at the points $s=\rho$ for $|\rho-\frac12|>2R$ (at which points u_R is $-\infty$ and v_R is $+\infty$). It will suffice to show that for large R both $u_R(s)$ and $v_R(s)$ are at most $R^{1+\epsilon}$ on $|s-\frac12|=R$ since then, when ϵ is decreased slightly to ϵ', it follows that $u_R(s)+v_R(s)\le 2R^{1+\epsilon'}\le|s-\frac12|^{1+\epsilon}$ on $|s-\frac12|=R$ for all R large enough that $u_R\le R^{1+\epsilon'}$, $v_R\le R^{1+\epsilon'}$, and $2\le R^{\epsilon-\epsilon'}$.

First consider $u_R(s)$. On the circle $|s-\frac12|=4R$ the factors in the denominator are all at least 1; therefore

$$\begin{aligned} u_R(s) &\le \operatorname{Re}\log\xi(s) = \log|\xi(s)| \\ &\le \log[(4R)^{4R}] = 4R\log 4R \le R^{1+\epsilon} \end{aligned}$$

on the circle $|s-\frac12|=4R$, for large R (large enough that $4\log 4R<R^{\epsilon}$). Now u_R is a harmonic function on the disk $|s-\frac12|\le 4R$ except at the points $s=\rho$ in the range $2R<|s-\frac12|\le 4R$. But near these singular points $s=\rho$ the value of u_R is near $-\infty$, so the maximum value of the harmonic function u_R on the disk $|s-\frac12|\le 4R$ must occur on the outer boundary $|s-\frac12|=4R$. Thus the maximum of u_R on the disk, and in particular on the circle $|s-\frac12|=R$, is at most $R^{1+\epsilon}$ as was to be shown.

Now consider $v_R(s)$. For complex x in the disk $|x|\le\frac12$ the inequality

$$\begin{aligned} \operatorname{Re}\log\frac{1}{1-x} &= -\operatorname{Re}\log(1-x) = \operatorname{Re}\int_0^x\frac{dt}{1-t} \\ &\le\left|\int_0^x\frac{dt}{1-t}\right|\le|x|\max\frac{1}{|1-t|}=2|x| \end{aligned}$$

holds. Thus for $|s-\frac12|=R$ the inequality

$$\begin{aligned} v_R(s) &= \operatorname{Re}\log\frac{1}{\prod_{|\rho-1/2|>2R}\left(1-\frac{(s-\frac12)^2}{(\rho-\frac12)^2}\right)} \\ &\le 2\sum_{|\rho-1/2|>2R}\frac{R^2}{|\rho-\frac12|^2} \\ &= 2\sum\left(\frac{R}{|\rho-\frac12|}\right)^{1-\epsilon}\left(\frac{R}{|\rho-\frac12|}\right)^{1+\epsilon} \\ &\le 2\sum\left(\frac12\right)^{1-\epsilon}\frac{R^{1+\epsilon}}{|\rho-\frac12|^{1+\epsilon}} \\ &= 2^{\epsilon}R^{1+\epsilon}\sum_{|\rho-1/2|>2R}\frac{1}{|\rho-\frac12|^{1+\epsilon}} \end{aligned}$$

holds. Now the sum in this expression converges by the theorem of Section 2.5, and it decreases to zero as R increases. Thus $v_R(s) \leq R^{1+\epsilon}$ on $|s - \frac{1}{2}| = R$ for all sufficiently large R as was to be shown. This completes the proof.

2.7 RATE OF GROWTH OF EVEN ENTIRE FUNCTIONS

Theorem Let $f(s)$ be an analytic function, defined in the entire s-plane, which is even in the sense that $f(-s) \equiv f(s)$ and which grows more slowly than $|s|^2$ in the sense that for every $\epsilon > 0$ there is an R such that $\operatorname{Re} f(s) < \epsilon |s|^2$ at all points s satisfying $|s| \geq R$. Then f must be constant.

Proof The subtle point of the theorem is that only the *upward* growth of the *real part* of f is limited. The main step in the proof is the following lemma, which shows that this implies that the growth of the *modulus* of f is also limited.

Lemma Let $f(s)$ be an analytic function on the disk $\{|s| \leq r\}$, let $f(0) = 0$, and let M be the maximum value of $\operatorname{Re} f(s)$ on the bounding circle $|s| = r$ (and hence on the entire disk). Then for $r_1 < r$ the modulus of f on the smaller disk $\{|s| \leq r_1\}$ is bounded by

$$|f(s)| \leq 2r_1 M/(r - r_1) \qquad (|s| \leq r_1).$$

Proof of the Lemma Consider the function

$$\phi(s) = f(s)/s[2M - f(s)].$$

If $u(s)$ and $v(s)$ denote the real and imaginary parts, respectively, of f, then $|2M - u(s)| \geq M \geq u(s)$ on the circle $|s| = r$; so the modulus of ϕ on this circle is at most

$$|\phi(s)| = \frac{(u^2 + v^2)^{1/2}}{r[(2M - u)^2 + v^2]^{1/2}} \leq \frac{(u^2 + v^2)^{1/2}}{r(u^2 + v^2)^{1/2}} = \frac{1}{r}$$

which implies that $|\phi(s)| \leq r^{-1}$ throughout the disk $\{|s| \leq r\}$. But $f(s)$ can be expressed in terms of $\phi(s)$ as

$$\phi(s)s[2M - f(s)] = f(s), \qquad f(s) = \frac{2Ms\phi(s)}{1 + s\phi(s)}$$

which shows that for $|s| = r_1$ the modulus of $f(s)$ is at most

$$|f(s)| \leq 2Mr_1 r^{-1}/(1 - r_1 r^{-1}) = 2Mr_1/(r - r_1).$$

Hence the same inequality holds throughout the disk $\{|s| \leq r_1\}$ as was to be shown.

Now to prove the theorem let $f(s) = \sum_{n=0}^{\infty} a_n s^n$ be the power series expansion of a function $f(s)$ satisfying the conditions of the theorem. Note first

that it can be assumed without loss of generality that $a_0 = 0$ because $f(s)$ satisfies the growth condition of the theorem if and only if $f(s) - f(0)$ does and because $f(s)$ is constant if and only if $f(s) - f(0)$ is. Now Cauchy's integral formula for the coefficients is

$$a_n = \frac{1}{2\pi i}\int_{\partial D} \frac{f(s)\,ds}{s^{n+1}},$$

where D is any domain containing the origin. Let ϵ, R be as in the statement of the theorem and let D be the disk $\{|s| \leq \frac{1}{2}R\}$. Then the above formula gives

$$\begin{aligned} |a_n| &= \left|\frac{1}{2\pi i}\int_0^{2\pi} \frac{f(\frac{1}{2}Re^{i\theta})}{(\frac{1}{2}Re^{i\theta})^n} i\,d\theta\right| \\ &\leq \frac{1}{2\pi}\int_0^{2\pi} \frac{2^n\,|f(\frac{1}{2}Re^{i\theta})|}{R^n}\,d\theta. \end{aligned}$$

The right side is the average value of a function whose value is by the lemma at most

$$\frac{2^n}{R^n}\frac{2(\epsilon R^2)(\frac{1}{2}R)}{R - (\frac{1}{2}R)} = \frac{2^{n+1}\epsilon}{R^{n-2}}.$$

If $n \geq 2$, this is at most $2^{n+1}\epsilon$, and since ϵ is arbitrary, a_n must be zero for $n \geq 2$. Thus $f(s) = a_1 s$. However a_1 must be zero by the evenness condition $f(s) \equiv f(-s)$. Therefore $f(s) \equiv 0$ which is constant, as was to be shown.

2.8 THE PRODUCT FORMULA FOR ξ

The function $F(s) = \xi(s)/\prod_\rho [1 - (s - \frac{1}{2})/(\rho - \frac{1}{2})]$ is analytic in the entire s-plane and is an even function of $s - \frac{1}{2}$. Moreover, it has no zeros, so its logarithm is well defined up to an additive constant $2\pi ni$ (n an integer) by the formula $\log F(s) = \int_0^s F'(z)\,dz/F(z) + \log F(0)$, where $\log F(0)$ is determined to within an additive constant $2\pi ni$. The results of the preceding two sections then combine to give $\log F(s) = \text{const}$, and therefore upon exponentiation

$$\xi(s) = c\prod\left(1 - \frac{s - \frac{1}{2}}{\rho - \frac{1}{2}}\right),$$

where c is a constant. Dividing this by the particular value

$$\xi(0) = c\prod\left(1 - \frac{-\frac{1}{2}}{\rho - \frac{1}{2}}\right)$$

gives

$$\frac{\xi(s)}{\xi(0)} = \prod\left(1 - \frac{s - \frac{1}{2}}{\rho - \frac{1}{2}}\right)\left(1 - \frac{-\frac{1}{2}}{\rho - \frac{1}{2}}\right)^{-1}.$$

The factors on the right are linear functions of s which are 0 when $s = \rho$ and 1 when $s = 0$; hence they are $1 - (s/\rho)$ and the formula is the desired formula

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right),$$

where, as always, it is understood that the factors ρ and $1 - \rho$ are paired.†

†The same argument proves the validity of the product formula for the sine

$$\sin \pi s = \pi s \prod_{n=1}^{\infty} \left(1 - \frac{s^2}{n^2}\right)$$

mentioned in Section 1.3. The only other unproved statement in Section 1.3 which is not elementary is the equivalence of the two definitions (2) and (3) of $\Pi(s)$.

Chapter 3

Riemann's Main Formula

3.1 INTRODUCTION

Soon after Hadamard proved the product formula for $\xi(s)$, von Mangoldt [M1] proved Riemann's main formula

$$J(x) = \mathrm{Li}(x) - \sum_{\rho} \mathrm{Li}(x^{\rho}) - \log 2 + \int_{x}^{\infty} \frac{dt}{t(t^2-1)\log t} \qquad (x > 1). \tag{1}$$

Von Mangoldt also recast this formula in a simpler form which has virtually replaced Riemann's original statement (1) in the subsequent development of the theory. This simpler form of (1) can be derived as follows.

The essence of Riemann's derivation of (1) is the inversion of the relationship

$$\log \zeta(s) = \int_{0}^{\infty} x^{-s}\, dJ(x) \tag{2}$$

for $J(x)$ in terms of $\log \zeta(s)$ and then the use of

$$\Pi\left(\frac{s}{2}\right)\pi^{-s/2}(s-1)\zeta(s) = \frac{1}{2}\prod_{\rho}\left(1 - \frac{s}{\rho}\right) \tag{3}$$

to express $\log \zeta(s)$ in terms of elementary functions and in terms of the roots ρ. Now the function $\log \zeta(s)$ has logarithmic singularities at all the roots ρ, and as a function of a complex variable it is very awkward outside the half-plane $\mathrm{Re}\, s > 1$. On the other hand, its derivative $\zeta'(s)/\zeta(s)$ is analytic in the entire plane except for poles at the roots ρ, the pole 1, and the zeros $-2n$. This might well lead one to begin not with formula (2) but with its derivative

$$\frac{\zeta'(s)}{\zeta(s)} = -\int_{0}^{\infty} x^{-s}(\log x)\, dJ(x). \tag{4}$$

The measure $(\log x)\,dJ(x)$ is a point measure which assigns the weight $\log(p^n)\cdot(1/n)$ to prime powers p^n and the weight 0 to all other points. Thus it can be written as a Stieltjes measure $d\psi(x)$, where $\psi(x)$ is the step function which begins at 0 and has a jump of $\log(p^n)\cdot(1/n) = \log p$ at each prime power p^n. In other words

$$\psi(x) = \sum_{p^n<x} \log p$$

except when x is a prime power; at the jumps $x = p^n$ the value of ψ is defined, as usual, to be halfway between the new and old values $\psi(x) = \frac{1}{2}[\psi(x-\epsilon) + \psi(x+\epsilon)]$. This function $\psi(x)$ had already been considered by Chebyshev,† who named it $\psi(x)$ and who proved‡ among other things that the prime number theorem is essentially the same as the statement that $\psi(x) \sim x$ with a relative error which approaches zero as $x \longrightarrow \infty$. In terms of $\psi(x)$ formula (4) becomes

$$-\frac{\zeta'(s)}{\zeta(s)} = \int_0^\infty x^{-s}\,d\psi(x). \tag{5}$$

In other words, if J is replaced by ψ in the original formula (2), then the awkward function $\log\zeta(s)$ is replaced by the more tractable function $-\zeta'(s)/\zeta(s)$.

Now the argument by which Riemann went from formula (2) to the formula (1) for $J(x)$ can be applied equally well to go from formula (5) to a new formula for $\psi(x)$. The explicit computations of this argument are given in the next section. However, even without the explicit computations, one can guess the formula for $\psi(x)$ as follows. The simplest formulation of Riemann's result is his formula (see Section 1.18)

$$dJ = \left(\frac{1}{\log x} - \sum_\rho \frac{x^{\rho-1}}{\log x} - \frac{1}{x(x^2-1)\log x}\right)dx \qquad (x>1)$$

which gives

$$\begin{aligned} d\psi &= (\log x)\,dJ \\ &= \left(1 - \sum_\rho x^{\rho-1} - \sum_\rho x^{-2n-1}\right)dx \qquad (x>1) \end{aligned}$$

and leads to the guess

$$\psi(x) = x - \sum_\rho \frac{x^\rho}{\rho} + \sum_n \frac{x^{-2n}}{2n} + \text{const} \qquad (x>1). \tag{6}$$

This is von Mangoldt's reformulation of (1) referred to in the first paragraph. [The value of the constant is given in Section 3.2. It is assumed in von Mangoldt's formula (6), as it is in Riemann's formula (1), that the terms of the sum over ρ are taken in the order of increasing $|\operatorname{Im}\rho|$; these sums converge

†This work of Chebyshev [C3] in 1850 preceded Riemann's paper.

‡Actually Chebyshev does not state this result explicitly, but it follows trivially from the techniques he introduces for deducing estimates of π from estimates of ψ.

only conditionally—even when the terms ρ, $1-\rho$ are paired—so their order is essential.]

The main part of this chapter is devoted to von Mangoldt's proof of the formula (6) for $\psi(x)$. In Section 3.2 the formula is derived from the termwise evaluation of certain definite integrals, and in the following three sections the validity of these termwise evaluations is rigorously proved. Von Mangoldt's proof of Riemann's original formula (1) is outlined in the next two sections, and the last section deals with the numerical evaluation of the constant $\zeta'(0)/\zeta(0)$.

3.2 DERIVATION OF VON MANGOLDT'S FORMULA FOR $\psi(x)$

The technique of Section 1.12 applied to the formula $-\zeta'(s)/\zeta(s) = s\int_0^\infty \psi(x)x^{-s-1}\,dx$ instead of to the formula $\log\zeta(s) = s\int_0^\infty J(x)x^{-s-1}\,dx$ puts $\psi(x)$ in the form of a definite integral

$$\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\left[-\frac{\zeta'(s)}{\zeta(s)}\right]x^s\frac{ds}{s} \qquad (a>1). \tag{1}$$

Von Mangoldt's method of proving the formula for $\psi(x)$ is to evaluate this definite integral in two different ways, one of which gives the value $\psi(x)$ and the other $x - \sum(x^\rho/\rho) + \sum(x^{-2n}/2n) + \text{const.}$ (Neither of these evaluations uses Fourier's theorem, so the use of Fourier's theorem in Section 1.12 can be regarded as purely heuristic.)

The first method of evaluating the definite integral (1) is as follows. Beginning with the formula†

$$-\frac{\zeta'(s)}{\zeta(s)} = \int_0^\infty x^{-s}\,d\psi(x), \tag{2}$$

let $\Lambda(n)$ denote the weight assigned to the integer n by the measure $d\psi$—that is, $\Lambda(n)$ is zero unless n is a prime power, in which case $\Lambda(n)$ is the log of the prime of which n is a power—so that the integral (2) can be written as a sum

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^\infty \Lambda(n)n^{-s} \qquad (\operatorname{Re} s>1). \tag{3}$$

Substitute this formula in (1) and assume termwise integration is valid. This gives the value

$$\sum_{n=2}^\infty \Lambda(n)\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\left(\frac{x}{n}\right)^s\frac{ds}{s}$$

†This formula is essentially the Euler product formula (see the concluding remarks of Section 1.11). In fact, logarithmic differentiation of $\zeta(s) = \prod(1-p^{-s})^{-1}$ gives (3) immediately.

for (1). Now formula (2) of Section 1.14 (with $\beta = 0$) shows that the integral corresponding to n in this sum is 1 if $x/n > 1$ and 0 if $x/n < 1$; hence the sum is just

$$\sum_{n<x} \Lambda(n) = \psi(x)$$

as was to be shown.

To justify this sequence of steps leading to the value $\psi(x)$ for the definite integral (1), note first that the series (3) converges uniformly in any halfplane $\operatorname{Re} s \geq K > 1$ [by comparison with the convergent series $\sum (\log n) n^{-K}$]. This proves both that the termwise differentiation of $-\log \zeta(s) = \sum \log (1 - p^{-s})$ is valid (a series can be differentiated termwise if the result is uniformly convergent) and that the integral over any *finite* path can be computed termwise:

$$\frac{1}{2\pi i} \int_{a-ih}^{a+ih} \left[-\frac{\zeta'(s)}{\zeta(s)} \right] x^s \frac{ds}{s} = \sum_{n=1}^{\infty} \Lambda(n) \cdot \frac{1}{2\pi i} \int_{a-ih}^{a+ih} \left(\frac{x}{n} \right)^s \frac{ds}{s} \tag{4}$$

when $a > 1$. Now *if* it can be shown that *the limit as* $h \to \infty$ *of the sum on the right is equal to the sum of the limits*, then the fact that the integral (1) has the value $\psi(x)$ will follow immediately from the formula

$$\frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} y^s \frac{ds}{s} = \begin{cases} 0 & \text{if } 0 < y < 1 \\ \frac{1}{2} & \text{if } y = 1 \\ 1 & \text{if } y > 1 \end{cases} \qquad (a > 0). \tag{5}$$

This formula, which was deduced from Fourier's theorem in Section 1.14, will be proved directly in the next section. To summarize, then, the proof that the definite integral (1) is equal to $\psi(x)$ depends on the proof that the limit as $h \to \infty$ of the sum in (4) is the sum of the limits, and on the proof of the integral formula (5).

The second method of evaluating the integral (1) is as follows. Differentiate logarithmically the formula

$$\Pi\left(\frac{s}{2}\right) \pi^{-s/2} (s-1) \zeta(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right)$$

to find

$$\frac{d}{ds} \log \Pi\left(\frac{s}{2}\right) - \frac{1}{2} \log \pi + \frac{1}{s-1} + \frac{\zeta'(s)}{\zeta(s)} = \sum_{\rho} \frac{1}{1 - (s/\rho)} \cdot \left(-\frac{1}{\rho}\right).$$

Using the expression of $\Pi(x)$ as an infinite product [(4) of Section 1.3], and

differentiating termwise then gives

$$(6)\qquad -\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s-1} - \sum_\rho \frac{1}{s-\rho} + \sum_{n=1}^{\infty}\left[-\frac{1}{s+2n} + \frac{1}{2}\log\left(1+\frac{1}{n}\right)\right] - \frac{1}{2}\log\pi.$$

With $s = 0$ this gives

$$-\frac{\zeta'(0)}{\zeta(0)} = -1 - \sum_\rho\left(-\frac{1}{\rho}\right) + \sum_{n=1}^{\infty}\left[-\frac{1}{2n} + \frac{1}{2}\log\left(1+\frac{1}{n}\right)\right] - \frac{1}{2}\log\pi,$$

so subtraction gives

$$-\frac{\zeta'(s)}{\zeta(s)} + \frac{\zeta'(0)}{\zeta(0)} = \left[\frac{1}{s-1}+1\right] - \sum_\rho\left[\frac{1}{s-\rho}+\frac{1}{\rho}\right] - \sum_{n=1}^{\infty}\left[\frac{1}{s+2n}-\frac{1}{2n}\right]$$

and finally

$$(7)\qquad -\frac{\zeta'(s)}{\zeta(s)} = \frac{s}{s-1} - \sum_\rho \frac{s}{\rho(s-\rho)} + \sum_{n=1}^{\infty}\frac{s}{2n(s+2n)} - \frac{\zeta'(0)}{\zeta(0)}.$$

Substitute this formula in the definite integral (1) and assume that termwise integration is valid. This gives the value

$$\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} x^s\frac{ds}{s-1} - \sum_\rho \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} x^s\frac{ds}{\rho(s-\rho)} + \sum_n \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} x^s\frac{ds}{2n(s+2n)} + \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} x^s\left[-\frac{\zeta'(0)}{\zeta(0)}\right]\frac{ds}{s}$$

for (1). Now the change of variable $t = s - \beta$ in the previous integral formula (5) gives

$$(8)\qquad \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} x^s\frac{ds}{s-\beta} = \frac{1}{2\pi i}\int_{a-\beta-i\infty}^{a-\beta+i\infty} x^\beta x^t\frac{dt}{t} = x^\beta\frac{1}{2\pi i}\int_{\mathrm{Re}(a-\beta)-i\infty}^{\mathrm{Re}(a-\beta)+i\infty} x^t\frac{dt}{t} = x^\beta$$

provided $x > 1$ and $a > \mathrm{Re}\,\beta$. (The middle equation here, in which the limits of integration are switched from $a-\beta \pm i\infty$ to $\mathrm{Re}(a-\beta) \pm i\infty$, is not trivial because these two integrals when written as limits as $h \to \infty$ are different. However, the difference between them is two integrals of $x^t\,dt/t$ over intervals of the form $[a \pm ib, a \pm i(b+c)]$, where c is fixed and $b \to \infty$; thus the difference is less than a constant times $c/b \to 0$.) Thus the value of

(1) reduces, when $x > 1$, to

$$x - \sum_\rho \frac{x^\rho}{\rho} + \sum_n \frac{x^{-2n}}{2n} - \frac{\zeta'(0)}{\zeta(0)} \qquad (x > 1)$$

as desired.

To justify this sequence of steps leading to the value $x - \Sigma(x^\rho/\rho) + \Sigma(x^{-2n}/2n) - \zeta'(0)/\zeta(0)$ of the definite integral (1), note first that both of the infinite series in (6) converge uniformly in any disk $|s| \le K$. (The series in n converges uniformly because

$$\begin{aligned}
&|(s+2n)^{-1} - \tfrac{1}{2}\log(1+n^{-1})| \\
&\quad = |(s+2n)^{-1} - (2n)^{-1} + (2n)^{-1} \\
&\qquad - \tfrac{1}{2}(n^{-1} - \tfrac{1}{2}n^{-2} + \tfrac{1}{3}n^{-3} - \cdots)| \\
&\quad \le |s(s+2n)^{-1}(2n)^{-1}| + |\tfrac{1}{4}n^{-2} - \tfrac{1}{6}n^{-3} + \cdots| \\
&\quad \le K(2n)^{-2} + n^{-2} \le \text{const}/n^2
\end{aligned}$$

for all sufficiently large n, and the series in ρ converges uniformly because when the terms ρ and $1 - \rho$ are paired

$$\begin{aligned}
&|(s-\rho)^{-1} + [s-(1-\rho)]^{-1}| \\
&\quad = \left|\left[\left(s - \frac{1}{2}\right) - \left(\rho - \frac{1}{2}\right)\right]^{-1} + \left[\left(s - \frac{1}{2}\right) + \left(\rho - \frac{1}{2}\right)\right]^{-1}\right| \\
&\quad = \left|\frac{2(s-\frac{1}{2})}{(s-\frac{1}{2})^2 - (\rho - \frac{1}{2})^2}\right| \le \text{const}\left|\rho - \frac{1}{2}\right|^{-2}
\end{aligned}$$

for all sufficiently large ρ once K is fixed and because $\sum |\rho - \frac{1}{2}|^{-2}$ converges by the theorem of Section 2.5.) This proves that the termwise differentiation by which (6) was obtained is valid. Then it follows by an elementary theorem [$\sum (a_n + b_n) = \sum a_n + \sum b_n$ when $\sum a_n$, $\sum b_n$ both converge] that (7) is valid—except at the zeros and poles 1, ρ, $-2n$ of ζ—and that the series it contains both converge uniformly in $|s| \le K$. Thus it can be integrated termwise over *finite* intervals and the integral (1) is therefore equal to

$$\begin{aligned}
(9) \qquad &\lim_{h\to\infty} \frac{1}{2\pi i}\int_{a-ih}^{a+ih}\left[\frac{s}{s-1} - \sum_\rho \frac{s}{\rho(s-\rho)} + \sum_n \frac{s}{2n(s+2n)} - \frac{\zeta'(0)}{\zeta(0)}\right]x^s\,\frac{ds}{s} \\
&\quad = x - \lim_{h\to\infty}\sum_\rho \frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^s\,ds}{\rho(s-\rho)} \\
&\qquad + \lim_{h\to\infty}\sum_n \frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^s\,ds}{2n(s+2n)} - \frac{\zeta'(0)}{\zeta(0)}
\end{aligned}$$

[where use is made of the fact that the limit of a finite sum is the sum of the limits and the first and last terms are evaluated using (8) and (5)]. Now *if* the limits of these two sums are equal to the sums of the limits, then the rest of the argument is elementary and the value $x - \sum (x^\rho/\rho) + \sum (x^{-2n}/2n) - \zeta'(0)/\zeta(0)$ for (1) will be proved. Thus, in addition to the basic formula (5),

the rigorous proof that the definite integral (1) has this value depends on the validity of the termwise evaluation of the two limits in (9).

Thus von Mangoldt's formula

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} + \sum_{n} \frac{x^{-2n}}{2n} - \frac{\zeta'(0)}{\zeta(0)} \qquad (x > 1)$$

depends on the validity of the integral formula (5) and of the three interchanges of $\lim_{h\to\infty}$ with infinite sums in (4) and (9). The following three sections are devoted to proving that all are indeed valid. The numerical value of the constant $\zeta'(0)/\zeta(0) = \log 2\pi$ is found in Section 3.8.

3.3 THE BASIC INTEGRAL FORMULA

This section is devoted to the evaluation of

$$\lim_{h\to\infty} \frac{1}{2\pi i} \int_{a-ih}^{a+ih} \frac{x^s\, ds}{s} \qquad (x > 0, \quad a > 0). \tag{1}$$

Since the arguments of Section 3.2 deal with infinite sums of such limits, it will be necessary to find, in addition to the limit (1), the *rate* at which this limit is approached. For the case $0 < x < 1$ this is accomplished by the estimate

$$\left| \frac{1}{2\pi i} \int_{a-ih}^{a+ih} \frac{x^s\, ds}{s} \right| \leq \frac{x^a}{\pi h\, |\log x|} \qquad (a > 0, \quad 0 < x < 1), \tag{2}$$

which can be proved as follows. Because $a > 0$, the function x^s/s has no singularity in the rectangle $\{a \leq \operatorname{Re} s \leq K, -h \leq \operatorname{Im} s \leq h\}$, where K is a large constant. Hence by Cauchy's theorem the integral of $x^s\, ds/s$ around the boundary of this rectangle is zero, which gives

$$\begin{aligned} \frac{1}{2\pi i} \int_{a-ih}^{a+ih} \frac{x^s\, ds}{s} = &- \frac{1}{2\pi i} \int_{a+ih}^{K+ih} \frac{x^s\, ds}{s} \\ &+ \frac{1}{2\pi i} \int_{a-ih}^{K-ih} \frac{x^s\, ds}{s} + \frac{1}{2\pi i} \int_{K-ih}^{K+ih} \frac{x^s\, ds}{s}. \end{aligned}$$

The last integral has modulus at most $(2\pi)^{-1}(x^K/K)(2h)$. Each of the other two integrals on the right has modulus at most

$$\frac{1}{2\pi} \int_a^K \frac{x^{\sigma}\, d\sigma}{h} = \frac{1}{2\pi} \left[\frac{x^{\sigma}}{h \log x} \right]_{\sigma=a}^{\sigma=K}$$

which then gives

$$\left| \frac{1}{2\pi i} \int_{a-ih}^{a+ih} \frac{x^s\, ds}{s} \right| \leq \frac{1}{\pi} \frac{|x^K - x^a|}{h\, |\log x|} + \frac{1}{2\pi} \frac{x^K}{K} 2h.$$

When $0 < x < 1$, the limit of x^K as $K \to \infty$ is zero and (2) follows. Thus, in particular, the limit (1) is zero when $0 < x < 1$.

In the case $x = 1$ no estimate of the rate of approach to the limit will be needed and it suffices to note that (1) is

$$\lim_{h\to\infty} \frac{1}{2\pi i}\int_{-h}^{h} \frac{dt}{a+it} = \lim_{h\to\infty}\left(\frac{1}{2\pi}\int_{-h}^{h}\frac{a\,dt}{a^2+t^2} - i\frac{1}{2\pi}\int_{-h}^{h}\frac{t\,dt}{a^2+t^2}\right)$$
$$= \lim_{h\to\infty}\frac{1}{2\pi}\int_{-h/a}^{h/a}\frac{du}{1+u^2} = \frac{1}{2}$$

using the well-known† formula $\int_{-\infty}^{\infty}(1+u^2)^{-1}\,du = \pi$. Thus for $x = 1$ the limit (1) is $\frac{1}{2}$.

For $x > 1$ the estimate analogous to (2) is obtained by considering the integral of $x^s\,ds/(2\pi i s)$ around the boundary of a rectangle of the form $\{-K \le \operatorname{Re} s \le a,\ -h \le \operatorname{Im} s \le h\}$. Since x^s is analytic in this rectangle and $s = 0$ lies inside the rectangle, the Cauchy integral formula states that this integral is $x^0 = 1$, hence

$$\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^s\,ds}{s} + \frac{1}{2\pi i}\int_{a+ih}^{-K+ih}\frac{x^s\,ds}{s}$$
$$+\frac{1}{2\pi i}\int_{-K+ih}^{-K-ih}\frac{x^s\,ds}{s} + \frac{1}{2\pi i}\int_{-K-ih}^{a-ih}\frac{x^s\,ds}{s} = 1,$$

$$\left|\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^s\,ds}{s} - 1\right| \le \frac{1}{2\pi}\int_{-K}^{a}\frac{x^\sigma\,d\sigma}{h} + \frac{1}{2\pi}\frac{x^{-K}}{K}\cdot 2h + \frac{1}{2\pi}\int_{-K}^{a}\frac{x^\sigma\,d\sigma}{h}$$
$$= \frac{1}{\pi}\frac{x^a - x^{-K}}{h\log x} + \frac{1}{\pi}\frac{x^{-K}h}{K}.$$

Letting $K \longrightarrow \infty$ then gives

$$\left|\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^s\,ds}{s} - 1\right| \le \frac{x^a}{\pi h\log x} \qquad (a>0,\quad x>1) \tag{3}$$

which is the desired estimate. In particular, the limit (1) is 1 when $x > 1$.

One other estimate of integrals is used in the proof of von Mangoldt's formula, namely, von Mangoldt's estimate

$$\left|\frac{1}{2\pi i}\int_{a+ic}^{a+id}\frac{x^s\,ds}{s}\right| \le K\frac{x^a}{(a+c)\log x} \qquad (x>1,\quad a>0,\quad d>c\ge 0), \tag{4}$$

where K is a constant which may be taken to be $(4+\pi)/(2\pi\sqrt{2})$. To prove this formula, note that integration by parts

$$\int\frac{x^s\,ds}{s} = \frac{x^s}{s\log x} + \int\frac{x^s\,ds}{s^2\log x}$$

gives

$$\left|\int_{a+ic}^{a+id}\frac{x^s\,ds}{s}\right| \le \left|\frac{x^{a+id}}{(a+id)\log x}\right| + \left|\frac{x^{a+ic}}{(a+ic)\log x}\right|$$
$$+\frac{x^a}{\log x}\left|\int_c^d\frac{x^{it}\,dt}{(a+it)^2}\right|.$$

†See Edwards [E1, p. 65].

Since $|x^{it}| = 1$ and $|a + id| \geq |a + ic| \geq (a + c)/\sqrt{2}$, the first two terms are each less then $\sqrt{2}\, x^a/(a + c) \log x$ and the integral in the third term is less than

$$\int_c^\infty \frac{dt}{a^2 + t^2} = \int_0^\infty \frac{dt}{a^2 + (c + t)^2} \leq \int_0^\infty \frac{dt}{a^2 + c^2 + t^2}$$
$$= \int_0^\infty \frac{(a^2 + c^2)^{1/2}\, du}{a^2 + c^2 + [u(a^2 + c^2)^{1/2}]^2} = \frac{1}{(a^2 + c^2)^{1/2}} \int_0^\infty \frac{du}{1 + u^2}$$
$$\leq \frac{\sqrt{2}}{a + c} \cdot \frac{\pi}{2}.$$

Thus

$$\left| \frac{1}{2\pi i} \int_{a+ic}^{a+id} \frac{x^s\, ds}{s} \right| \leq \frac{1}{2\pi} \left(2\sqrt{2} + \frac{\pi}{\sqrt{2}} \right) \frac{x^a}{(a + c) \log x}$$

which is the desired result.

3.4 THE DENSITY OF THE ROOTS

This section is devoted to the proof of the following theorem:

Theorem The vertical density of the roots ρ of $\xi(\rho) = 0$ is less than $2 \log T$ for large T. More specifically, there is an H such that for $T \geq H$ the number of roots ρ with imaginary parts in the range $T \leq \operatorname{Im} \rho \leq T + 1$ is less than $2 \log T$.

Proof Von Mangoldt's proof of this fact is based on Hadamard's proof of the product formula and on a strong version of Stirling's formula which was published by Stieltjes in 1889.

Hadamard's theorem that the series $\sum |\rho - \frac{1}{2}|^{-2}$ converges implies (see Sect. 3.2) that the termwise integration of the series $\xi'(s)/\xi(s) = \sum (s - \rho)^{-1}$ is valid over any finite segment. Hence, in particular,

$$\int_{2+iT}^{2+i(T+1)} \frac{\xi'(s)}{\xi(s)}\, ds = \sum_\rho \int_{2+iT}^{2+i(T+1)} \frac{ds}{s - \rho}$$

for any T. Now for any fixed ρ the imaginary part of the integral on the right

$$\operatorname{Im}\left(\int_{2+iT}^{2+i(T+1)} \frac{ds}{s - \rho} \right)$$

is equal (because $dz/z = d \log r + i\, d\theta$) to the angle subtended by the segment $[2 + iT, 2 + i(T + 1)]$ at the point ρ. Thus it is always positive—because the roots ρ are to the left of $\operatorname{Re} s = 2$—and if ρ lies in the range $T \leq \operatorname{Im} \rho \leq T + 1$, then it is at least the angle subtended by the segment $[2 + iT, 2 + i(T + 1)]$ at the point iT, which is $\operatorname{Arctan} \frac{1}{2}$. Thus if n denotes the number

of roots ρ in $T \leq \operatorname{Im} \rho \leq T+1$, it follows that

$$n \cdot \operatorname{Arctan} \frac{1}{2} \leq \operatorname{Im} \int_{2+iT}^{2+i(T+1)} \frac{\xi'(s)}{\xi(s)}\, ds. \tag{1}$$

Thus an upper bound on n will result from an upper bound on the integral on the right.

On the other hand, the integral of $\xi'(s)\, ds/\xi(s)$ over the interval from $2+iT$ to $2+i(T+1)$ is, by the fundamental theorem of calculus, equal to the amount by which the function

$$\log \xi(s) = \log \Pi(s/2) - (s/2) \log \pi + \log (s-1) + \log \zeta(s)$$

changes between these two points. It is in the estimation of $\log \Pi(s/2)$ that von Mangoldt uses Stieltjes' version of Stirling's formula. Specifically, he uses the fact that the modulus of the error in the approximation

$$\log \Pi(z) \sim (z + \tfrac{1}{2}) \log z - z + \tfrac{1}{2} \log 2\pi$$

is at most $(6|z|)^{-1}$ for z in the halfplane $\operatorname{Re} z \geq 0$. (For a complete statement and proof of Stieltjes' result see Section 6.3.) Thus

$$\begin{aligned}
\log \xi(s) &\sim \left(\frac{s}{2} + \frac{1}{2}\right) \log s - \left(\frac{s}{2} + \frac{1}{2}\right) \log 2 - \frac{s}{2} + \frac{1}{2} \log 2\pi \\
&\quad - \frac{s}{2} \log \pi + \log(s-1) + \log \zeta(s) \\
&= \frac{s+1}{2} \log s - \frac{s}{2} \log 2\pi - \frac{s}{2} \\
&\quad + \log (s-1) + \log \zeta(s) + \text{const}
\end{aligned}$$

with an error of at most $(6|s|)^{-1}$. As was noted in Section 1.13, formula (3), the modulus of $\log \zeta(s)$ is at most $\log \zeta(2)$ on the line $\operatorname{Re} s = 2$, so neglecting this term introduces an error of at most $\log \zeta(2) = \log (\pi^2/6) < 1$. Thus the change in $\log \xi(s)$ between $2 + i(T+1)$ and $2+iT$ is approximately

$$\begin{aligned}
&\frac{3 + i(T+1)}{2} \log[2 + i(T+1)] - \frac{3+iT}{2} \log(2+iT) \\
&\quad - \frac{i}{2} \log 2\pi - \frac{i}{2} + \log \frac{1 + i(T+1)}{1+iT} \\
&= \frac{i}{2} \log[2 + i(T+1)] + \frac{3+iT}{2} \log\left(1 + \frac{i}{2+iT}\right) \\
&\quad + \text{const} + \log\left(1 + \frac{1}{1+iT}\right)
\end{aligned}$$

and the error has modulus at most $2(6T)^{-1} + 2$. Neglecting terms which are bounded for large T puts this estimate in the form

$$\frac{i}{2} \log(iT) + \frac{3+iT}{2} \cdot \frac{i}{2+iT} \sim \frac{i}{2} \log T.$$

Thus the error in the approximation

$$\int_{2+iT}^{2+i(T+1)} \frac{\zeta'(s)}{\zeta(s)}\,ds \sim \frac{i}{2}\log T$$

remains bounded in modulus as $T \to \infty$—say by K—hence (1) gives

$$n \cdot \operatorname{Arctan}\tfrac{1}{2} \leq \tfrac{1}{2}\log T + K,$$

$$n \leq \frac{\frac{1}{2}\log T + K}{\pi/8} < 2\log T$$

for all sufficiently large T, as was to be shown.

3.5 PROOF OF VON MANGOLDT'S FORMULA FOR $\psi(x)$

The derivation of von Mangoldt's formula

$$\psi(x) = x - \sum_{\rho}\frac{x^{\rho}}{\rho} + \sum_{n}\frac{x^{-2n}}{2n} - \frac{\zeta'(0)}{\zeta(0)} \qquad (x > 1) \tag{1}$$

which was given in Section 3.2 depended on the integral formula which was proved in Section 3.3 and on three termwise integrations. This section is devoted to proving that these termwise integrations are valid. In all three cases, the series converge uniformly on any finite segment $[a - ih, a + ih]$ $(a > 1)$, so the integral can be computed termwise on finite segments and the problem is to show that the limit of their sum as $h \to \infty$ is equal to the sum of their limits.

Consider first the limit

$$\lim_{h\to\infty}\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\left(\frac{x}{n}\right)^{s}\frac{ds}{s}. \tag{2}$$

The limit of a finite sum is the sum of the limits; hence one can disregard the finite number of terms in which $n \leq x$ and consider only the terms $n > x$. For these, the estimate (2) of Section 3.3 gives

$$\left|\Lambda(n)\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\left(\frac{x}{n}\right)^{s}\frac{ds}{s}\right| \leq \log n\,\frac{x^{a}}{n^{a}\pi h(\log n - \log x)}$$

$$\leq \text{const}\,\frac{1}{n^{a}h}.$$

Hence their sum over $n > x$ is at most a constant times h^{-1} and therefore approaches zero as $h \to \infty$. This shows that the limit (2) can be evaluated termwise.

Consider next the limit

$$\lim_{h\to\infty}\sum_{n=1}^{\infty}\frac{x^{-2n}}{2n}\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^{s+2n}\,ds}{s+2n} \qquad (x > 1). \tag{3}$$

The limit of the nth term of this series is $x^{-2n}/2n$ by (3) of Section 3.3; hence the sum of the limits converges. Now by (4) of Section 3.3 the nth term is at most

$$\frac{x^{-2n}}{2n} \cdot 2\left|\frac{1}{2\pi i}\int_{a+2n}^{a+2n+ih} \frac{x^t\,dt}{t}\right| \leq \frac{x^{-2n}}{n} \cdot K\frac{x^{a+2n}}{(a+2n)\log x}$$
$$\leq \frac{\text{const}}{n^2}$$

for all h. Thus the series (3) converges uniformly in h and one can pass to the limit $h \longrightarrow \infty$ termwise, as was to be shown. [Given $\epsilon > 0$, choose N large enough that the sum (3) differs by at most ϵ from the sum of the first N terms for all h. By enlarging N if necessary, one can also assume that the sum of the limits $x^{-2n}/2n$ differs by at most ϵ from the sum of the first N of them. Now choose H large enough that each of the first N terms of (3) differs by at most ϵ/N from its limit when $h \geq H$. Then the sum (3) differs by at most 3ϵ from the sum of the limits provided only that $h \geq H$. Since ϵ is arbitrary this proves the desired result.]

Consider finally the limit

$$\lim_{h\to\infty} \sum_{\rho} \frac{x^\rho}{\rho} \frac{1}{2\pi i}\int_{a-ih}^{a+ih} \frac{x^{s-\rho}}{s-\rho}\,ds. \tag{4}$$

This limit has now been shown to *exist* because it has been shown that the limit (9) of Section 3.2 exists [and is equal to $\psi(x)$] and it has been shown that the limit of the sum over n in (9) also exists. It has also been shown that the individual terms of (4) approach limits x^ρ/ρ as $h \longrightarrow \infty$, but it has *not* been shown that the sum of these limits converges; in fact, the proof that $\sum x^\rho/\rho$ converges when summed in the order of increasing $|\operatorname{Im}\rho|$ is the major difficulty in the proof of von Mangoldt's formula. Broadly speaking, von Mangoldt overcomes this difficulty by approaching the limit (4) "diagonally," that is, by considering the limit

$$\lim_{h\to\infty} \sum_{|\operatorname{Im}\rho|\leq h} \frac{x^\rho}{\rho} \frac{1}{2\pi i}\int_{a-ih}^{a+ih} \frac{x^{s-\rho}}{s-\rho}\,ds \tag{5}$$

as an intermediate step between the limit (4) which is known to exist and the sum of the limits

$$\lim_{h\to\infty} \sum_{|\operatorname{Im}\rho|\leq h} \frac{x^\rho}{\rho} \tag{6}$$

which is to be shown to exist and be equal to (4).

Specifically, consider for each h the differences

$$\sum_{\rho} \frac{x^\rho}{\rho} \frac{1}{2\pi i}\int_{a-ih}^{a+ih} \frac{x^{s-\rho}}{s-\rho}\,ds - \sum_{|\operatorname{Im}\rho|\leq h} \frac{x^\rho}{\rho} \frac{1}{2\pi i}\int_{a-ih}^{a+ih} \frac{x^{s-\rho}}{s-\rho}\,ds \tag{7}$$

and

$$\text{(8)} \qquad \sum_{|\operatorname{Im}\rho|\le h} \frac{x^\rho}{\rho}\,\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^{s-\rho}}{s-\rho}\,ds - \sum_{|\operatorname{Im}\rho|\le h}\frac{x^\rho}{\rho}.$$

It will be shown that both these differences approach zero as $h \to \infty$. Then, since the limit (4) exists, it follows first that the "diagonal" limit (5) exists and is equal to it [because (7) goes to zero] and hence that the limit (6) exists and is equal to it [because (8) goes to zero hence (6) equals (5)] as desired.

Consider first the estimate of (7). Let $\rho = \beta + i\gamma$ denote a typical root. Then by (4) of Section 3.3 the modulus of (7) is at most

$$\begin{aligned}
&\sum_{|\gamma|>h}\left|\frac{x^\rho}{\rho}\right|\left|\frac{1}{2\pi i}\int_{a-ih}^{a+ih}\frac{x^{s-\rho}}{s-\rho}\,ds\right| \\
&\quad\le 2\sum_{\gamma>h}\frac{x^\beta}{\gamma}\left|\frac{1}{2\pi i}\int_{a-\beta+i(\gamma-h)}^{a-\beta+i(\gamma+h)}\frac{x^t\,dt}{t}\right| \\
&\quad\le 2\sum_{\gamma>h}\frac{x^\beta}{\gamma}\cdot K\frac{x^{a-\beta}}{(a-\beta+\gamma-h)\log x} \\
&\quad\le 2K\frac{x^a}{\log x}\sum_{\gamma>h}\frac{1}{\gamma(\gamma-h+c)}
\end{aligned}$$

where $c = a - 1 > 0$ so that $c \le a - \beta$ for all roots ρ. Now if the roots γ beyond h are grouped in intervals $h < \gamma \le h+1, h+1 < \gamma \le h+2$, $h+2 < \gamma \le h+3, \ldots$, then (assuming h is large enough that the estimate of Section 3.4 applies beyond h) the interval $h + j < \gamma \le h + j + 1$ contains at most $2\log(h+j)$ of the γ's and the modulus of (7) is at most a constant times

$$\sum_{j=0}^{\infty}\frac{\log(h+j)}{(h+j)(j+c)},$$

and it remains only to show that this sum approaches zero as $h \to \infty$. One can do this by choosing h large enough that $\log(h+j) < (h+j)^{1/2}$ for all $j \ge 0$; then the summand is at most one over $(h+j)^{1/4}(c+j)^{1/4}\cdot(c+j)$, so the sum is at most a constant times $h^{-1/4} \to 0$ as was to be shown.

Consider now the estimate of (8). The fact that the terms corresponding to ρ and $\bar\rho$ give equal contributions and the estimates (3) and (4) of Section 3.3 show that the modulus of (8) is at most

$$\begin{aligned}
&2\sum_{0<\gamma\le h}\left|\frac{x^\rho}{\rho}\right|\left|\frac{1}{2\pi i}\int_{a-\beta-i\gamma-ih}^{a-\beta-i\gamma+ih}\frac{x^t\,dt}{t} - 1\right| \\
&\quad\le 2\sum_{0<\gamma\le h}\frac{x^\beta}{\gamma}\left|\frac{1}{2\pi i}\int_{a-\beta-i(h+\gamma)}^{a-\beta+i(h+\gamma)}\frac{x^t\,dt}{t} - 1\right| \\
&\qquad + 2\sum_{0<\gamma\le h}\frac{x^\beta}{\gamma}\left|\frac{1}{2\pi i}\int_{a-\beta+i(h-\gamma)}^{a-\beta+i(h+\gamma)}\frac{x^t\,dt}{t}\right|
\end{aligned}$$

$$\leq 2 \sum_{0<\gamma\leq h} \frac{x^\beta}{\gamma} \frac{x^{a-\beta}}{\pi(h+\gamma)\log x}$$
$$+ 2 \sum_{0<\gamma\leq h} \frac{x^\beta}{\gamma} K \frac{x^{a-\beta}}{(a-\beta+h-\gamma)\log x}$$
$$\leq \frac{2x^a}{\pi \log x} \sum_{0<\gamma\leq h} \frac{1}{\gamma(h+\gamma)}$$
$$+ \frac{2Kx^a}{\log x} \sum_{0<\gamma\leq h} \frac{1}{\gamma(c+h-\gamma)},$$

where $c = a - 1 > 0$ as before. Thus it suffices to prove that these two sums over γ approach zero as $h \longrightarrow \infty$. Let H be a large integer such that the estimate of Section 3.4 applies beyond H and let the roots be grouped in intervals $H \leq \gamma < H+1$, $H+1 \leq \gamma < H+2, \ldots$. Then the interval $H+j \leq \gamma \leq H+j+1$ contains at most $2\log(H+j)$ of the γ's and

$$\sum_{0<\gamma\leq h} \frac{1}{\gamma(h+\gamma)} \leq \sum_{0<\gamma\leq H} \frac{1}{\gamma(h+\gamma)} + \sum_{0\leq j\leq h-H} \frac{2\log(H+j)}{(H+j)(h+H+j)}.$$

The first sum has a fixed finite number of terms and therefore clearly has the limit zero as $h \longrightarrow \infty$. The second sum is at most

$$2 \sum_{0\leq j\leq h-H} (\log h)\left[\frac{1}{h}\left(\frac{1}{H+j} - \frac{1}{h+H+j}\right)\right]$$
$$\leq \frac{\log h}{h} 2 \sum_{0\leq j\leq h-H} \frac{1}{H+j}$$
$$\leq 2\frac{\log h}{h} \int_{H-1}^{h} \frac{dt}{t} \leq 2\frac{(\log h)^2}{h}$$

which approaches zero as $h \longrightarrow \infty$. A similar estimate shows that

$$\sum_{H\leq\gamma\leq h} \frac{1}{\gamma(c+h-\gamma)}$$

approaches zero as $h \longrightarrow \infty$ and completes the proof that (6) equals (4).

Thus the limits (2), (3), and (4) can be evaluated termwise—provided the sum of the limits of (4) is defined in the sense of (6)—and the derivation of Section 3.2 proves von Mangoldt's formula (1).

3.6 RIEMANN'S MAIN FORMULA

There are at least two reasons why Riemann's formula for $J(x)$ [Section 3.1, formula (1)] is generally neglected today. First, it contains essentially the same information as von Mangoldt's formula for $\psi(x)$ but is less "natural" than this formula in the sense that it is harder to prove and harder to generalize. Secondly, Riemann's reason for establishing the formula in the first

place—to show an explicit analytic connection between the arithmetic function $\pi(x)$ and the empirically derived approximation $\mathrm{Li}(x)$—was rendered superfluous by Chebyshev's observation that the prime number theorem $\pi(x) \sim \mathrm{Li}(x)$ can be deduced from the more natural theorem $\psi(x) \sim x$. Thus, in all respects, the formula for $\psi(x)$ is preferable to that for $J(x)$.

Nonetheless, it is the formula for $J(x)$ that was stated† by Riemann and for this reason, if for no other, it is of great interest to know whether or not the formula is valid. Von Mangoldt proved that it is. Von Mangoldt did not, however, follow Riemann's method of proving the formula for $J(x)$; once the product formula for $\xi(s)$ was established by Hadamard, the only real difficulty which remained in Riemann's derivation of the formula for $J(x)$ was the proof that the termwise integration of the sum over ρ (Section 1.15) is valid, but von Mangoldt does not justify this termwise integration directly and proves the formula for $J(x)$ by a quite different method. It may be that he did this simply as a matter of convenience or it may be that he was in fact unable to derive estimates which would justify the termwise integration of Section 1.15 directly. In any event, Landau [L2] in 1908 proved in a more or less direct manner that termwise integration is valid.

3.7 VON MANGOLDT'S PROOF OF RIEMANN'S MAIN FORMULA

For $r > 0$ consider the definite integral

$$\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\left[-\frac{\zeta'(s+r)}{\zeta(s+r)}\right]x^s\,ds \qquad (a>1, \quad x>1). \tag{1}$$

When the formula $-\zeta'(s+r)/\zeta(s+r) = \sum \Lambda(n)n^{-s-r}$ is substituted in this equation and integration is carried out termwise, one obtains the value $\sum_{n<x}\Lambda(n)n^{-r}$ where, as usual, at any point $x = p^n$ where the value jumps, it is defined to split the difference. Von Mangoldt denotes this function by $\psi(x, r)$. An equivalent definition of $\psi(x, r)$ is

$$\psi(x,r) = \int_0^x x^{-r}\,d\psi(x). \tag{2}$$

On the other hand, the derivation of the formula

$$-\frac{\zeta'(s)}{\zeta(s)} = \frac{s}{s-1} - \sum_\rho \frac{s}{\rho(s-\rho)} + \sum_n \frac{s}{2n(s+2n)} - \frac{\zeta'(0)}{\zeta(0)}$$

†Landau [L3] began a tradition of referring to this and other statements of Riemann as "conjectures," which gives the very mistaken impression that Riemann had some doubts about them. The only conjecture which Riemann makes in the paper is that $\mathrm{Re}\,\rho = \frac{1}{2}$.

of Section 3.2 is easily modified to give the more general formula

$$-\frac{\zeta'(s+r)}{\zeta(s+r)} = -\frac{s}{(r-1)(s+r-1)} + \sum_\rho \frac{s}{(r-\rho)(s+r-\rho)} + \sum_n \frac{s}{(r+2n)(s+r+2n)} - \frac{\zeta'(r)}{\zeta(r)}$$

which is valid for $r > 0$ except at $r = 1$. When this expression is substituted in (1) and the integration is carried out termwise, the result is

$$\frac{x^{1-r}}{1-r} - \sum_\rho \frac{x^{\rho-r}}{\rho-r} + \sum_n \frac{x^{-2n-r}}{2n+r} - \frac{\zeta'(r)}{\zeta(r)}. \tag{3}$$

The estimates of Section 3.5 can be modified to show that for fixed $r > 0$, $x > 1$, $r \neq 1$ the termwise integrations are valid and hence that

$$\int_0^x x^{-r}\, d\psi(x) = \frac{x^{1-r}}{1-r} - \sum_\rho \frac{x^{\rho-r}}{\rho-r} + \sum_n \frac{x^{-2n-r}}{2n+r} - \frac{\zeta'(r)}{\zeta(r)}.$$

The first and last terms on the right have poles at $r = 1$ which cancel each other, so the entire right side defines a continuous function of r as can be seen by rewriting it in the form

$$\int_0^x x^{-r}\, d\psi(x) = \left[\frac{x^{1-r}}{1-r} - \frac{\zeta'(r)}{\zeta(r)}\right] - x^{-r}\sum_\rho \frac{x^\rho}{\rho} - x^{-r}\sum_\rho \frac{r x^\rho}{\rho(\rho-r)} + \sum_n \frac{x^{-2n-r}}{2n+r} \qquad (x > 1) \tag{4}$$

and noting that near any value of r these series converge uniformly in r and hence define continuous functions of r. Thus the formula is valid for $r = 1$ as well.

Now integrate both sides dr from $r = 0$ *to* $r = \infty$. On the left one obtains

$$\int_0^\infty \int_0^x x^{-r}\, d\psi(x)\, dr = \int_0^x \left(\int_0^\infty x^{-r}\, dr\right) d\psi(x) = \int_0^x \frac{1}{\log x}\, d\psi(x) = \int_0^x dJ(x) = J(x).$$

(Or, less elegantly,

$$\int_0^\infty \sum_{n<x} \Lambda(n) n^{-r}\, dr = \sum_{n<x} \frac{\Lambda(n)}{\log n} = J(x)$$

with the usual adjustment if x is a prime power.) The sum over n on the right can be integrated termwise by the Lebesgue dominated convergence theorem because it is dominated by $x^{-r}\sum_n x^{-2n}$. The result is

$$\sum_n \int_0^\infty \frac{x^{-2n-r}\, dr}{2n+r} = \sum_n \int_{2n}^\infty \frac{x^{-u}\, du}{u} = \sum_n \int_{2n}^\infty \int_x^\infty t^{-u-1}\, dt\, du = \sum_n \int_x^\infty \frac{t^{-2n-1}}{\log t}\, dt = \int_x^\infty \frac{dt}{t(t^2-1)\log t}.$$

The first sum over ρ does not involve r and can therefore be integrated termwise. The second sum over ρ can be integrated termwise by the Lebesgue dominated convergence theorem because it is dominated by $rx^{1-r}\sum|\rho|^{-2}$. The result of these two termwise integrations is

$$-\sum_\rho\left[\int_0^\infty x^{-r}\frac{x^\rho}{\rho}\,dr+\int_0^\infty x^{-r}\frac{rx^\rho}{\rho(\rho-r)}\,dr\right]$$
$$=-\sum_\rho\int_0^\infty\frac{x^{\rho-r}}{\rho-r}\,dr=-\sum_{\operatorname{Im}\rho>0}[\operatorname{Li}(x^\rho)+\operatorname{Li}(x^{1-\rho})]$$

as will now be shown. [Here, as in (4), the sum over ρ is to be taken in the order of increasing $|\operatorname{Im}\rho|$.]

The formula needed above is

$$\int_0^\infty\frac{x^{\rho-r}}{\rho-r}\,dr=\operatorname{Li}(x^\rho)\mp i\pi, \tag{5}$$

where $x>1$, $\operatorname{Re}\rho>0$, and† where the sign of $i\pi$ is opposite to that of $\operatorname{Im}\rho$. This can be proved by setting $t=(\rho-r)\log x$, $dt=-(\log x)\,dr$ to put the integral in the form

$$-\int_\infty^0\frac{x^{\rho-r}(\log x)\,dr}{(\rho-r)\log x}=\int_{\rho\log x-\infty\log x}^{\rho\log x}\frac{e^t\,dt}{t}.$$

Since the integrand vanishes very rapidly near $-\infty$, the lower limit of integration can be taken as $-\infty$; hence

$$\int_0^\infty\frac{x^{\rho-r}\,dr}{\rho-r}=\int_{-\infty}^{\rho\log x}\frac{e^t\,dt}{t},$$

where the path of integration passes above the singularity at $t=0$ if $\operatorname{Im}\rho>0$, below if $\operatorname{Im}\rho<0$. Now, if it is stipulated that the path of integration in the integral

$$\int_{-\infty}^{\beta\log x}\frac{e^t\,dt}{t}$$

must enter the halfplane $\operatorname{Re}t>0$ by crossing the *positive* imaginary axis, then this integral defines an analytic function of β in $\operatorname{Re}\beta>0$ which for real β is equal to

$$\int_0^{x^\beta}\frac{du}{\log u}=\operatorname{Li}(x^\beta)-i\pi$$

(see Section 1.15). Hence the same is true by analytic continuation for all

†Thus, as with Riemann, the possibility $\operatorname{Re}\rho=0$ is not included. The extension of the formula to cover this case is trivial, but, as Hadamard showed (see Section 4.2), none of the ρ's lies on the imaginary axis.

β in $\operatorname{Re}\beta > 0$, and

$$\int_0^\infty \frac{x^{\rho-r}}{\rho - r}\,dr = \operatorname{Li}(x^\rho) - i\pi$$

for $\operatorname{Im}\rho > 0$ follows. The case $\operatorname{Im}\rho < 0$ is analogous and (5) is proved.

It remains only to show that the remaining pair of terms in (4) when integrated dr from 0 to ∞ give the remaining terms in Riemann's formula, that is, to show that

$$\int_0^\infty \left[\frac{x^{1-r}}{1-r} - \frac{\zeta'(r)}{\zeta(r)}\right] dr = \operatorname{Li}(x) + \log\xi(0).$$

This can be proved as follows.

The two terms can be integrated separately

$$\int_0^\infty \left[\frac{x^{1-r}}{1-r} - \frac{\zeta'(r)}{\zeta(r)}\right] dr = \int_0^\infty \frac{x^{1-r}}{1-r}\,dr - \int_0^\infty \frac{\zeta'(r)}{\zeta(r)}\,dr$$

provided the path from 0 to ∞ is perturbed slightly to avoid the singularity at $r = 1$, say by passing slightly above it. The first integral is then the limiting case $\rho = 1 - i\epsilon$ of formula (5) and is therefore $\operatorname{Li}(x) + i\pi$. The second integral can be evaluated by integrating

$$\frac{d}{dr}\log\xi(r) = \frac{d}{dr}\log\left[\pi^{-r/2}\,\Pi\left(\frac{r}{2}\right)\right] + \frac{1}{r-1} + \frac{\zeta'(r)}{\zeta(r)}$$

from 0 to K passing above $r = 1$ to obtain

$$\log\xi(K) - \log\xi(0) = \log\pi^{-K/2}\,\Pi\left(\frac{K}{2}\right) + \log(K-1) - i\pi + \int_0^K \frac{\zeta'(r)}{\zeta(r)}\,dr,$$

$$-\int_0^K \frac{\zeta'(r)}{\zeta(r)}\,dr = \log\xi(0) - i\pi + \log\zeta(K).$$

The limit as $K \longrightarrow \infty$ is thus $\log\xi(0) - i\pi$ and the desired formula follows.

In summary, then, when (4) is integrated dr from 0 to ∞, the result is Riemann's main formula

$$J(x) = \operatorname{Li}(x) - \sum_\rho \operatorname{Li}(x^\rho) + \int_x^\infty \frac{dt}{t(t^2-1)\log t} + \log\xi(0) \qquad (x > 1)$$

which is thereby proved.

3.8 NUMERICAL EVALUATION OF THE CONSTANT

Von Mangoldt found that the numerical value of the constant in the formula for $\psi(x)$ is

$$\zeta'(0)/\zeta(0) = \log 2\pi. \tag{1}$$

The series $\Sigma(x^{-2n}/2n)$ can also be summed using the series expansion

$$\log[1/(1-x)] = x + \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 + \cdots$$

to put the formula for $\psi(x)$ in the form

$$\psi(x) = x - \sum_\rho \frac{x^\rho}{\rho} + \frac{1}{2}\log\left(\frac{x^2}{x^2-1}\right) - \log 2\pi \qquad (x > 1),$$

where the sum over ρ is in the order of increasing $|\operatorname{Im}\rho|$.

The value (1) of the constant can be obtained as follows.† Consider the function

$$\frac{\zeta(s)}{\Pi(-s)} = \frac{1}{2\pi i}\int_{+\infty}^{+\infty} \frac{(-x)^s}{e^x - 1}\cdot\frac{dx}{x}, \tag{2}$$

where the path of integration is as in Section 1.4. It will first be shown that the derivative of this function is zero at $s = 1$, that is, it will be shown that

$$\frac{1}{2\pi i}\int_{+\infty}^{+\infty} \frac{(-x)\log(-x)}{e^x - 1}\frac{dx}{x} = 0. \tag{3}$$

Let the path of integration be written as a sum of three parts as in Section 1.4 so that this integral becomes

$$\begin{aligned}
&\frac{1}{2\pi i}\int_\infty^\epsilon \frac{(-x)(\log x - i\pi)}{e^x - 1}\frac{dx}{x} \\
&\qquad + \frac{1}{2\pi i}\int_{|x|=\epsilon} \frac{(-x)(\log\epsilon + i\theta - i\pi)}{e^x - 1}\frac{dx}{x} \\
&\qquad + \frac{1}{2\pi i}\int_\epsilon^\infty \frac{(-x)(\log x + i\pi)}{e^x - 1}\frac{dx}{x} \\
&= -\int_\epsilon^\infty \frac{dx}{e^x - 1} - \frac{\log\epsilon}{2\pi i}\int_{|x|=\epsilon}\frac{x}{e^x - 1}\cdot\frac{dx}{x} \\
&\qquad - \frac{1}{2\pi i}\int_{-\pi}^{\pi}\frac{x}{e^x - 1}\phi\, d\phi,
\end{aligned}$$

where $x = \epsilon e^{i(\phi+\pi)}$ in the last integral. Since $x(e^x - 1)^{-1}$ is 1 at $x = 0$, the middle integral is $(-\log\epsilon)$ by the Cauchy integral formula and the last integral approaches zero as $\epsilon \downarrow 0$.

†For an alternative proof see Section 6.8.

The first integral can be evaluated directly

$$-\int_\epsilon^\infty \frac{dx}{e^x-1} = -\int_\epsilon^\infty \left(\sum_{n=1}^\infty e^{-nx}\right) dx = -\left[\sum_{n=1}^\infty \frac{e^{-nx}}{-n}\right]\Bigg|_\epsilon^\infty = -\sum \frac{(e^{-\epsilon})^n}{n}$$

$$= \log(1-e^{-\epsilon}) = \log\left(\epsilon - \frac{\epsilon^2}{2} + \frac{\epsilon^3}{6} - \cdots\right)$$

$$= \log\epsilon + \log\left(1 - \frac{\epsilon}{2} + \cdots\right).$$

Thus the $\log\epsilon$'s cancel and the limit as $\epsilon \downarrow 0$ of the remaining terms is zero, which proves (3). Now by the functional equation the function (2) can also be written in the form

$$\zeta(s)/\Pi(-s) = (2\pi)^{s-1}\zeta(1-s)2\sin(\pi s/2).$$

Since its derivative is zero at $s = 1$, its logarithmic derivative

$$\log(2\pi) - \frac{\zeta'(1-s)}{\zeta(1-s)} + \frac{\pi}{2}\frac{\cos(\pi s/2)}{\sin(\pi s/2)}$$

must also be zero at $s = 1$, which gives (1).

Since the logarithmic derivative of $\xi(s)$ is on the one hand

$$\sum_\rho \frac{d}{ds}\log\left(1 - \frac{s}{\rho}\right) = \sum_\rho \frac{1}{s-\rho}$$

and on the other hand

$$\frac{d}{ds}\log\Pi\left(\frac{s}{2}\right) - \frac{1}{2}\log\pi + \frac{1}{s-1} + \frac{\zeta'(s)}{\zeta(s)},$$

the sum of the series $\sum(1/\rho)$ is the value of

$$-\frac{1}{2}\frac{\Pi'(s/2)}{\Pi(s/2)} + \frac{1}{2}\log\pi + \frac{1}{1-s} - \frac{\zeta'(s)}{\zeta(s)}$$

at $s = 0$. Now logarithmic differentiation of the product formula for $\Pi(x)$ [(4) of Section 1.3] gives

$$\frac{\Pi'(s)}{\Pi(s)} = \sum_{n=1}^\infty \left[-\frac{1}{s+n} - \log n + \log(n+1)\right]$$

$$-\frac{\Pi'(0)}{\Pi(0)} = \lim_{n\to\infty}\left[1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log(n+1)\right].$$

The number on the right side of this equation is by definition *Euler's constant*, and is traditionally denoted γ. Thus

$$\sum_\rho \frac{1}{\rho} = \frac{1}{2}\gamma + \frac{1}{2}\log\pi + 1 - \log 2\pi. \tag{4}$$

This formula was known to Riemann, who used it in his computations of the roots ρ (see Section 7.6 below). From this it is clear that the formula (1) was also known to Riemann.

Chapter 4

The Prime Number Theorem

4.1 INTRODUCTION

The prime number theorem is the statement that the relative error in the approximation $\pi(x) \sim \mathrm{Li}(x)$ approaches zero as $x \to \infty$. Following the work of Chebyshev it was well known that the prime number theorem could be deduced from the theorem that the relative error in the approximation $\psi(x) \sim x$ approaches zero as $x \to \infty$. But von Mangoldt's formula for $\psi(x)$ shows that $\psi(x) \sim x$ has this property if and only if

$$\lim_{x\to\infty} \frac{-\sum_{\rho} (x^{\rho}/\rho) + \sum_{n} (x^{-2n}/2n) + \text{const}}{x} = 0$$

or, what is the same, if and only if

$$\lim_{x\to\infty} \sum_{\rho} \frac{x^{\rho-1}}{\rho} = 0. \tag{1}$$

If the limit of this sum could be taken termwise, then it would suffice to prove that $x^{\rho-1} \to 0$ for all ρ or, what is the same, that $\operatorname{Re} \rho < 1$ for all ρ. Since $\operatorname{Re} \rho \leq 1$ for all ρ (by the Euler product formula—see Section 1.9), this amounts to proving that there are no roots ρ on the line $\operatorname{Re} s = 1$. Thus, given von Mangoldt's 1894 formula for $\psi(x)$, the proof of the prime number theorem can be reduced to proving that there are no roots ρ on the line $\operatorname{Re} s = 1$ and to proving that the above limit can be evaluated termwise.

Both Hadamard [H2] and de la Vallée Poussin [V1] succeeded in 1896 in filling in these remaining steps in the proof of the prime number theorem. They both circumvented proving that the limit (1) can be evaluated termwise

and instead they each derived a variation of von Mangoldt's formula, namely,

$$\int_0^x t^{-2}\psi(t)\,dt = \log x - \sum_\rho \frac{x^{\rho-1}}{\rho(\rho-1)} - \sum_n \frac{x^{-2n-1}}{2n(2n+1)} + \frac{1}{x}\frac{\zeta'(0)}{\zeta(0)} + \text{const} \qquad (x > 1) \tag{2}$$

in de la Vallée Poussin's case† and

$$\int_0^x t^{-1}\psi(t)\,dt = x - \sum_\rho \frac{x^\rho}{\rho^2} - \sum_n \frac{x^{-2n}}{(2n)^2} - \frac{\zeta'(0)}{\zeta(0)}\log x + \text{const} \qquad (x > 1) \tag{3}$$

in Hadamard's case. Either of these formulas is quite easy to prove using von Mangoldt's methods—easier‡ in fact than von Mangoldt's formula for $\psi(x)$—and if it is known that $\operatorname{Re}\rho < 1$ for all roots ρ, then either of them can be used to conclude by straightforward estimates that $\psi(x) \sim x$. Thus, although it certainly required insight to see that a formula such as (2) or (3) could be used, the substantial step beyond von Mangoldt's work which was required for the proof of the prime number theorem was the proof that there are no roots ρ on the line $\operatorname{Re} s = 1$.

Hadamard's proof that there are no roots ρ on $\operatorname{Re} s = 1$ is given in Section 4.2. De la Vallée Poussin admitted that Hadamard's proof was the simpler of the two, and although simpler proofs have since been found (see Section 5.2), Hadamard's is perhaps still the most straightforward and natural proof of this fact. Section 4.3 is devoted to a proof that $\psi(x) \sim x$. This proof follows the same general line of argument as was followed by both Hadamard and de la Vallée Poussin, but it is somewhat simpler in that it is based on the formula

$$\int_0^x \psi(t)\,dt = \frac{x^2}{2} - \sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)} - \sum_n \frac{x^{-2n+1}}{2n(2n-1)} - x\frac{\zeta'(0)}{\zeta(0)} + \text{const} \qquad (x > 1) \tag{4}$$

rather than on the analogous, but somewhat more complicated, formulas (2) or (3). Finally, Section 4.4 gives the very simple deduction of $\pi(x) \sim \operatorname{Li}(x)$ from $\psi(x) \sim x$.

†For the value of the constant, which is $-1 - \gamma$, see footnote †, p. 94.

‡At the time, de la Vallee Poussin was under the mistaken impression that von Mangoldt's proof was fallacious, so of course he gave his own proof. Hadamard too preferred to avoid making appeal to von Mangoldt's more difficult estimates and produced his own proof.

4.2 HADAMARD'S PROOF THAT Re $\rho < 1$ FOR ALL ρ

The representation

$$\log\zeta(s) = \int_0^\infty x^{-s}\,dJ(x) \tag{1}$$
$$= \sum_p \frac{1}{p^s} + \frac{1}{2}\sum_p \frac{1}{p^{2s}} + \frac{1}{3}\sum_p \frac{1}{p^{3s}} + \cdots$$

is valid throughout the halfplane Re $s > 1$. The presence of zeros ρ of $\zeta(s)$ on the line Re $s = 1$ would imply the presence of points $s = \sigma + it$ slightly to the right of Re $s = 1$ where Re $\log\zeta(s)$ was near $-\infty$. The series in (1) has the property that the sum of the terms after the first is bounded by the number

$$B = \frac{1}{2}\sum_p \frac{1}{p^2} + \frac{1}{3}\sum_p \frac{1}{p^3} + \frac{1}{4}\sum_p \frac{1}{p^4} + \cdots$$

for the entire halfplane Re $s \geq 1$ including Re $s = 1$; hence

$$\text{Re}\log\zeta(\sigma + it) \geq \sum_p \frac{\cos(t\log p)}{p^\sigma} - B$$

can approach $-\infty$ as $\sigma \downarrow 1$ only if the first term approaches $-\infty$. In short, if $1 + it$ were a zero ρ of $\zeta(s)$, then it would follow that

$$\lim_{\sigma\downarrow 1} \sum_p \frac{\cos(t\log p)}{p^\sigma} = -\infty \tag{2}$$

for this value of t. The objective is to show that this is impossible.

Now the fact that

$$\lim_{s\to 1}(s-1)\zeta(s) = \frac{\xi(1)}{\Pi(\frac{1}{2})\pi^{-1/2}} = \frac{(\frac{1}{2})}{\frac{1}{2}\pi^{1/2}\pi^{-1/2}} = 1$$

implies that

$$\lim_{\sigma\downarrow 1}[\text{Re}\log\zeta(\sigma) + \log(\sigma - 1)] = 0,$$

$$\lim_{\sigma\downarrow 1}\left[\sum_p \frac{1}{p^\sigma} + \text{bounded} + \log(\sigma - 1)\right] = 0,$$

which means that for σ slightly larger than one

$$\sum_p \frac{1}{p^\sigma} \sim -\log(\sigma - 1) \sim +\infty. \tag{3}$$

On the other hand, the derivation of (2) is easily strengthened to give

$$\lim_{\sigma\downarrow 1}\left[\sum_p \frac{\cos(t\log p)}{p^\sigma} + \text{bounded} - \text{Re}\log(\sigma - 1)\right] = 0,$$

$$\sum_p \frac{\cos(t\log p)}{p^\sigma} \sim \log(\sigma - 1) \sim -\infty. \tag{4}$$

If this were true then, because of (3), one would expect that the number $\cos(t \log p)$ would have to be nearly -1 for the overwhelming majority of primes p. This would imply a surprising regularity in the distribution of the numbers $\log p$, namely, that most of them lie near the points of the arithmetic progression $(2n + 1)t^{-1}\pi$ for this particular t. However, such a regularity cannot exist because it would imply that $\cos(2t \log p)$ was nearly $+1$ for the overwhelming majority of primes p, hence

$$\sum_p \frac{\cos(2t \log p)}{p^\sigma} \sim +\infty,$$

$$\operatorname{Re} \log \zeta(\sigma + 2it) \sim +\infty$$

which would imply the existence of a pole of $\zeta(s)$ at $s = 1 + 2it$. Since $\zeta(s)$ has no poles other than $s = 1$, this line of argument might be expected to yield a proof of the impossibility of (4) as desired. The actual proof requires little more than a quantitative description of the "overwhelming majority" of the primes p for which $t \log p \sim (2n + 1)\pi$ would have to be true.

Assume that $1 + it$ is a zero of $\zeta(s)$. Then $\zeta(s)/(s - 1 - it)$ would be analytic near $s = 1 + it$ so the real part of its log would be bounded above, say by K; hence with $s = \sigma + it$

$$\sum_p \frac{\cos(t \log p)}{p^\sigma} + \frac{1}{2}\sum_p \frac{\cos(2t \log p)}{p^{2\sigma}} + \frac{1}{3}\sum_p \frac{\cos(3t \log p)}{p^{3\sigma}} + \cdots - \operatorname{Re} \log(\sigma - 1) < K,$$

$$\sum_p \frac{\cos(t \log p)}{p^\sigma} < \log(\sigma - 1) + K + B \tag{5}$$

for all $\sigma > 1$ near 1. This is a quantitative version of (4); the objective is to show that no t has this property.

Let δ be a small positive number (for the sake of definiteness, $\delta = \pi/8$ will work in the following proof) and let the terms of the sum on the left side of (5) be divided into those terms corresponding to primes p for which there is an n such that

$$|(2n + 1)\pi - t \log p| < \delta \tag{6}$$

and those terms for which p does not satisfy this condition. In the terms of the second group $\cos(t \log p) = \cos(\pi - \alpha) = -\cos \alpha$ where $\delta \leq |\alpha| \leq \pi$, hence $\cos(t \log p) > -\cos \delta$ for these terms and (5) implies

$$-S' - (\cos \delta)S'' < \log(\sigma - 1) + K + B$$

where S' denotes the sum of $p^{-\sigma}$ over all primes p which satisfy (6) and S'' denotes the sum of $p^{-\sigma}$ over all primes p which do not. Now (3) says that $S' + S'' \sim -\log(\sigma - 1)$, and the derivation of (3) shows easily that there is a K' such that $S' + S'' < -\log(\sigma - 1) + K'$ for all $\sigma > 1$ near 1. Thus (5)

implies

$$-S' - (\cos\delta)S'' < -S' - S'' + K' + K + B,$$
$$(1 - \cos\delta)S'' < \text{const}$$

for all $\sigma > 1$ near 1. Since $S' + S''$ becomes infinite as $\sigma \downarrow 1$, this shows that (5) implies

$$\lim_{\sigma\downarrow 1} \frac{S''}{S' + S''} = 0, \qquad \lim_{\sigma\downarrow 1} \frac{S'}{S' + S''} = 1 \tag{7}$$

which is a specific sense in which the "overwhelming majority" of primes must satisfy (6) if (5) is true.

However, since $1 + 2it$ is not a pole of $\zeta(s)$, the real part of $\log\zeta(s)$ is bounded above near $s = 1 + 2it$, say by K'', hence

$$\sum_p \frac{\cos(2t\log p)}{p^\sigma} < K'' + B. \tag{8}$$

When the terms in this sum are split into those terms for which p satisfies (6) and those terms for which p does not, the terms in the first group have $\cos(2t\log p) = \cos(2\alpha)$, where $|\alpha| < 2\delta$; hence $\cos(2t\log p) > \cos 2\delta > 0$, so (8) implies

$$S'\cos 2\delta - S'' < K'' + B,$$
$$\frac{S''}{S' + S''} > \cos 2\delta \frac{S'}{S' + S''} - \frac{\text{const}}{S' + S''}$$

which contradicts (7). Thus (5) is impossible and the proof is complete.

4.3 PROOF THAT $\psi(x) \sim x$

Since von Mangoldt's formula

$$\psi(x) = x - \sum \frac{x^\rho}{\rho} + \sum \frac{x^{-2n}}{2n} - \frac{\zeta'(0)}{\zeta(0)} \qquad (x > 1)$$

is obtained by evaluating the definite integral

$$\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\left[-\frac{\zeta'(s)}{\zeta(s)}\right]\frac{x^s\,ds}{s} \qquad (x > 1, \quad a > 1)$$

in two different ways, it is natural to expect that the antiderivative of von Mangoldt's formula, namely,

$$\int_0^x \psi(t)\,dt = \frac{x^2}{2} - \sum \frac{x^{\rho+1}}{\rho(\rho+1)} - \sum \frac{x^{-2n+1}}{2n(2n-1)}$$
$$- \frac{\zeta'(0)}{\zeta(0)}x + \text{const} \qquad (x > 1)$$

could be obtained by evaluating the antiderivative of this definite integral, namely,

$$(1) \qquad \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\left[-\frac{\zeta'(s)}{\zeta(s)}\right]\frac{x^{s+1}\,ds}{s(s+1)} \qquad (x>1, \quad a>1)$$

in two different ways.

The first way of evaluating (1) is to use the expansion $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)n^{-s}$. If termwise integration of this expansion is valid, then (1) is

$$(2) \qquad \sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\frac{x^{s+1}\,ds}{n^s s(s+1)}.$$

The partial fractions expansion

$$(3) \qquad \frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

gives

$$\frac{x^{s+1}}{n^s s(s+1)} = \frac{x}{s}\left(\frac{x}{n}\right)^s - \frac{n}{s+1}\left(\frac{x}{n}\right)^{s+1}$$

$$\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\frac{x^{s+1}\,ds}{n^s s(s+1)} = \frac{x}{2\pi i}\int_{a-i\infty}^{a+i\infty}\left(\frac{x}{n}\right)^s\frac{ds}{s} - \frac{n}{2\pi i}\int_{a+1-i\infty}^{a+1+i\infty}\left(\frac{x}{n}\right)^u\frac{du}{u}$$

$$= \begin{cases} x-n & \text{if } n \le x, \\ 0 & \text{if } n \ge x, \end{cases}$$

so that if termwise integration is valid, then (1) is equal to†

$$\sum_{n\le x}\Lambda(n)(x-n) = \int_0^x (x-t)\,d\psi(t) = \int_0^x \psi(t)\,dt.$$

The proof that termwise integration is valid, and hence that (1) is equal to $\int_0^x \psi(t)\,dt$, is easily accomplished by writing the integrand of (1) in the form

$$-\frac{\zeta'(s)}{\zeta(s)}\frac{x^{s+1}}{(s+1)s} = \left[x\sum_{n=1}^{\infty}\Lambda(n)\left(\frac{x}{n}\right)^s\frac{1}{s}\right] - \left[\sum_{n=1}^{\infty}\Lambda(n)\left(\frac{x}{n}\right)^{s+1}\frac{n}{s+1}\right]$$

to express (1) as the difference of two integrals of infinite sums over n, and by then observing that von Mangoldt's method shows immediately that each of these two integrals can be evaluated termwise.

The second way of evaluating (1) is to use the expansion

$$(4) \qquad -\frac{\zeta'(s)}{\zeta(s)} = \frac{s}{s-1} - \sum_{\rho}\frac{s}{\rho(s-\rho)} + \sum_{n=1}^{\infty}\frac{s}{2n(s+2n)} - \frac{\zeta'(0)}{\zeta(0)}$$

of Section 3.2, (7). When this is used in conjunction with (3) in expressing the integrand of (1), the term $1/s$ divides evenly, but the term $1/(s+1)$ does not.

†Note that the case $x = n$ presents no difficulties because then $x - n = 0$. This occurs because the antiderivative of a function with jump discontinuities has no jump discontinuities.

To simplify the resulting expression note that

$$-\frac{\zeta'(s)}{\zeta(s)}+\frac{\zeta'(-1)}{\zeta(-1)}=\frac{s}{s-1}-\frac{-1}{-1-1}-\sum_{\rho}\left[\frac{s}{\rho(s-\rho)}-\frac{-1}{\rho(-1-\rho)}\right]$$
$$+\sum_{n}\left[\frac{s}{2n(s+2n)}-\frac{-1}{2n(-1+2n)}\right]$$
$$-\frac{\zeta'(s)}{\zeta(s)}=\frac{s+1}{2(s-1)}-\sum_{\rho}\frac{s+1}{(\rho+1)(s-\rho)}$$
$$+\sum_{n}\frac{s+1}{(2n-1)(s+2n)}-\frac{\zeta'(-1)}{\zeta(-1)}$$

so that the integrand in (1) is equal to

$$-\frac{\zeta'(s)}{\zeta(s)}x^{s+1}\left(\frac{1}{s}-\frac{1}{s+1}\right)=\frac{x^{s+1}}{s-1}-\sum_{\rho}\frac{x^{s+1}}{\rho(s-\rho)}+\sum_{n}\frac{x^{s+1}}{2n(s+2n)}$$
$$-\frac{\zeta'(0)}{\zeta(0)}\frac{x^{s+1}}{s}-\frac{x^{s+1}}{2(s-1)}+\sum_{\rho}\frac{x^{s+1}}{(\rho+1)(s-\rho)}$$
$$-\sum_{n}\frac{x^{s+1}}{(2n-1)(s+2n)}+\frac{\zeta'(-1)}{\zeta(-1)}\frac{x^{s+1}}{s+1}.$$

Von Mangoldt's estimates (Section 3.5) with only slight modifications show that both of the sums over ρ and both of the sums over n can be integrated termwise. This gives for (1) the value

$$x^2-\sum_{\rho}\frac{x^{\rho+1}}{\rho}+\sum_{n}\frac{x^{1-2n}}{2n}-\frac{\zeta'(0)}{\zeta(0)}x-\frac{x^2}{2}+\sum_{\rho}\frac{x^{\rho+1}}{\rho+1}-\sum_{n}\frac{x^{1-2n}}{2n-1}+\frac{\zeta'(-1)}{\zeta(-1)}$$
$$=\frac{x^2}{2}-\sum_{\rho}\frac{x^{\rho+1}}{\rho(\rho+1)}-\sum_{n}\frac{x^{1-2n}}{2n(2n-1)}-\frac{\zeta'(0)}{\zeta(0)}x+\frac{\zeta'(-1)}{\zeta(-1)}$$

which by the above is also equal to $\int_0^x \psi(t)\,dt$. This completes the proof of the formula†

$$\int_0^x \psi(t)\,dt=\frac{x^2}{2}-\sum_{\rho}\frac{x^{\rho+1}}{\rho(\rho+1)}-\sum_{n}\frac{x^{-2n+1}}{2n(2n-1)}-\frac{\zeta'(0)}{\zeta(0)}x+\frac{\zeta'(-1)}{\zeta(-1)}$$

which holds for all $x>1$.

†This is a special case ($u=0, v=-1$) of the formula

$$y^u\int_0^y t^{-u}\,d\psi(t)-y^v\int_0^y t^{-v}\,d\psi(t)=-\left[y^u\frac{\zeta'(u)}{\zeta(u)}-y^v\frac{\zeta'(v)}{\zeta(v)}\right]+\frac{u-v}{(u-1)(v-1)}y$$
$$-(u-v)\sum_{\rho}\frac{y^{\rho}}{(u-\rho)(v-\rho)}$$
$$-(u-v)\sum_{n}\frac{y^{-2n}}{(u+2n)(v+2n)}\qquad(y>1)$$

which de la Vallée Poussin proved in 1896 by using the above method in conjunction with elementary arguments—not those of von Mangoldt—to justify the termwise integrations. However, he used the case $u=1, v=0$, which gives (2) of Section 4.1, in his proof of the prime number theorem. Note that the constant in formula (4) of Section 4.1 has been shown to be $\zeta'(-1)/\zeta(-1)$.

Using this formula for $\int_0^x \psi(t)\,dt$ it is quite easy to show that

$$\int_0^x \psi(t)\,dt \sim \frac{x^2}{2}, \tag{5}$$

where, as usual, the symbol $\sim$ means that the relative error in the approximation approaches zero as $x \to \infty$. One need only note that $\int_0^x \psi(t)\,dt - (x^2/2)$ divided by $x^2/2$ is, by the formula, equal to $2\Sigma x^{\rho-1}/\rho(\rho+1)$ plus terms which go to zero as $x \to \infty$. Since $\Sigma\rho^{-1}(\rho+1)^{-1}$ converges absolutely and since $|x^{\rho-1}| \leq 1$ (because $\operatorname{Re}\rho \leq 1$), it follows that the series $\Sigma x^{\rho-1}/\rho(\rho+1)$ converges uniformly in x and hence that the limit as $x \to \infty$ can be evaluated termwise; since each term goes to zero (because $\operatorname{Re}\rho < 1$), it follows that this limit is zero and hence that the relative error in (5) approaches zero as $x \to \infty$.

To deduce $\psi(x) \sim x$, let $\epsilon > 0$ be given and let X be such that

$$(1-\epsilon)\frac{x^2}{2} < \int_0^x \psi(t)\,dt < (1+\epsilon)\frac{x^2}{2}$$

for all $x \geq X$. Then for $y > x \geq X$ it follows that

$$\int_0^y \psi(t)\,dt - \int_0^x \psi(t)\,dt \tag{6}$$

is at least

$$(1-\epsilon)\frac{y^2}{2} - (1+\epsilon)\frac{x^2}{2} = (1-\epsilon)\frac{y^2-x^2}{2} - 2\epsilon\frac{x^2}{2}$$

and at most

$$(1+\epsilon)\frac{y^2}{2} - (1-\epsilon)\frac{x^2}{2} = (1+\epsilon)\frac{y^2-x^2}{2} + 2\epsilon\frac{x^2}{2}.$$

On the other hand, ψ is an increasing function, so (6) is at least

$$\int_x^y \psi(t)\,dt \geq (y-x)\psi(x)$$

and at most

$$\int_x^y \psi(t)\,dt \leq (y-x)\psi(y).$$

Combining these inequalities gives

$$(y-x)\psi(x) \leq (6) \leq (1+\epsilon)\frac{y^2-x^2}{2} + 2\epsilon\frac{x^2}{2},$$

$$(y-x)\psi(y) \geq (6) \geq (1-\epsilon)\frac{y^2-x^2}{2} - 2\epsilon\frac{x^2}{2},$$

for all $y > x \geq X$. With $y = \beta x$ these inequalities give

$$\frac{\psi(x)}{x} \leq (1+\epsilon)\frac{\beta+1}{2} + \frac{\epsilon}{\beta-1},$$

$$\frac{\psi(y)}{y} \geq (1-\epsilon)\frac{\beta+1}{2\beta} - \frac{\epsilon}{\beta(\beta-1)}.$$

Given any $\beta > 1, \epsilon > 0$, these inequalities are satisfied for all sufficiently large x, y. Now the quantity on the right side of the first inequality can be made less than any number greater than 1 by first choosing $\beta > 1$ near enough to 1 that the first term is very near 1 and by then choosing ϵ so small that the second term is still small; this shows that $\psi(x)/x$ is less than any number greater than 1 for x sufficiently large. Similarly the second inequality can be used to show that $\psi(y)/y$ is greater than any number less than 1 for y sufficiently large. This completes the proof that $\psi(x) \sim x$.

4.4 PROOF OF THE PRIME NUMBER THEOREM

Once the estimate $\psi(x) \sim x$ has been proved, the prime number theorem $\pi(x) \sim \mathrm{Li}(x)$ is easily deduced. The technique used below is essentially the technique used by Chebyshev in 1850 [C3] to deduce his estimate of $\pi(x)$ from his estimate of $\psi(x)$.

Let $\theta(x)$ denote† the sum of the logarithms of all the primes p less than x, with the usual understanding that if x itself is a prime, then $\theta(x) = \frac{1}{2}[\theta(x + \epsilon) + \theta(x - \epsilon)]$. Then the relationship of θ and ψ is analogous to the relationship of π and J, and in analogy to the formula (1) of Section 1.17 relating π and J there is the formula

$$\psi(x) = \theta(x) + \theta(x^{1/2}) + \theta(x^{1/3}) + \theta(x^{1/4}) + \cdots$$

relating θ and ψ. The series on the right has the property that each term is larger than the following term and that all terms $\theta(x^{1/n})$ in which $x^{1/n} < 2$ are zero. Thus there are at most $\log x/\log 2$ nonzero terms and

$$\theta(x) < \psi(x) < \theta(x) + \theta(x^{1/2}) \log x/\log 2$$

which gives

$$\psi(x) - \theta(x^{1/2}) \log x/\log 2 < \theta(x) < \psi(x). \tag{1}$$

The inequality on the right together with $\psi(x) \sim x$ shows that $\theta(x)/x^{1+\epsilon} \to 0$ as $x \to \infty$. Hence $\theta(x^{1/2}) \log x/x = [\theta(x^{1/2})/(x^{1/2})^{1+\epsilon}][\log x/(x^{1/2})^{1-\epsilon}]$ goes to zero as $x \to \infty$ and $\theta(x) \sim x$ follows from (1) and $\psi(x) \sim x$.

Now let $\epsilon > 0$ be given and let X be such that $(1 - \epsilon)x \leq \theta(x) \leq (1 + \epsilon)x$ whenever $x \geq X$. Then for $y > x \geq X$ it follows that

$$\begin{aligned} \pi(y) - \pi(x) &= \int_x^y \frac{d\theta(t)}{\log t} \\ &= \left[\frac{\theta(t)}{\log t}\right]_x^y + \int_x^y \frac{\theta(t)\,dt}{(\log t)^2 t} \end{aligned}$$

†This notation, introduced by Chebyshev, is now standard.

is at most

$$\begin{aligned}\frac{(1+\epsilon)y}{\log y} - \frac{(1-\epsilon)x}{\log x} + \int_x^y \frac{(1+\epsilon)t\,dt}{(\log t)^2 t} &= 2\epsilon\frac{x}{\log x} + (1+\epsilon)\left\{\left[\frac{t}{\log t}\right]_x^y + \int_x^y \frac{t\,dt}{(\log t)^2 t}\right\} \\ &= 2\epsilon\frac{x}{\log x} + (1+\epsilon)\left\{\int_x^y \frac{dt}{\log t}\right\} \\ &= 2\epsilon\frac{x}{\log x} + (1+\epsilon)[\mathrm{Li}(y) - \mathrm{Li}(x)]\end{aligned}$$

and is at least

$$-2\epsilon\frac{x}{\log x} + (1-\epsilon)[\mathrm{Li}(y) - \mathrm{Li}(x)].$$

Thus for fixed x it follows that $\pi(y)/\mathrm{Li}(y)$ is at most

$$1 + \epsilon + \frac{\pi(x) + 2\epsilon x(\log x)^{-1} - (1+\epsilon)\,\mathrm{Li}(x)}{\mathrm{Li}(y)} \leq 1 + 2\epsilon$$

for all sufficiently large y and similarly that it is at least $1 - 2\epsilon$ for all sufficiently large y. Since ϵ was arbitrary, this proves the prime number theorem $\mathrm{Li}(y) \sim \pi(y)$.

Chapter 5

De la Vallée Poussin's Theorem

5.1 INTRODUCTION

After it was proved that the relative error in the approximation $\pi(x) \sim \mathrm{Li}(x)$ approaches zero as x approaches infinity, the next step was an estimate of the *rate* at which it approaches zero. De la Vallée Poussin [V2] proved in 1899 that there is a $c > 0$ such that the relative error approaches zero at least as fast as $\exp[-(c \log x)^{1/2}]$ does; that is,

$$\left|\frac{\pi(x) - \mathrm{Li}(x)}{\mathrm{Li}(x)}\right| < e^{-\sqrt{c \log x}}$$

for all sufficiently large x. The next two sections are devoted to the proof of this fact. Section 5.4 contains an application of this theorem to the question of comparing $\mathrm{Li}(x)$ to other possible approximations; it is shown, in essence, that Dirichlet, Gauss, Chebyshev, and Riemann were correct in preferring $\mathrm{Li}(x)$ to another approximation suggested earlier by Legendre. The next section, 5.5, shows that de la Vallée Poussin's theorem can be improved considerably if the Riemann hypothesis is true and that in fact the Riemann hypothesis is *equivalent* to the statement that the relative error in $\pi(x) \sim \mathrm{Li}(x)$ goes to zero faster than $x^{-(1/2)+\epsilon}$ as $x \to \infty$ (for all $\epsilon > 0$). Finally, the last section is devoted to the very simple proof of von Mangoldt's theorem that Euler's formula $\sum \mu(n)/n = 0$ [$\mu(n)$ is the Möbius function] is true in the strong sense that the series $\sum \mu(n)/n$ is *convergent* to the sum zero; this proof makes very effective use of de la Vallée Poussin's theorem that the relative error in the prime number theorem approaches zero at least as fast as $\exp[-(c \log x)^{1/2}]$.

5.2 AN IMPROVEMENT OF Re $\rho < 1$

De la Vallée Poussin's estimate of the error in the prime number theorem is based on a strengthened version of the theorem that the roots $\rho = \beta + i\gamma$ satisfy $\beta < 1$, namely, the theorem that there exist constants† $c > 0$, $K > 1$, such that

$$\beta < 1 - \frac{c}{\log \gamma} \tag{1}$$

for all roots $\rho = \beta + i\gamma$ in the range $\gamma > K$. Since $\log \gamma > \log K > 0$, the inequality (1) is stronger than $\beta < 1$, but the amount by which it is stronger decreases as γ increases and (1) does not preclude the possibility that there are roots ρ arbitrarily near to the line Re $s = 1$; although (1) has been improved upon somewhat since 1899, no one has yet been able to prove that Re ρ has any upper bound less than 1.

De la Vallée Poussin's proof of (1) makes use of a technique by which Mertens [M6] simplified the proof that Re $\rho < 1$. This technique is based on the elementary inequality

$$\begin{aligned} 4 &\geq 2(1 - \cos\theta), \\ 4(1 + \cos\theta) &\geq 2(1 - \cos^2\theta) = 1 - \cos 2\theta, \\ 3 + 4\cos\theta + \cos 2\theta &\geq 0, \end{aligned}$$

which holds for all θ. Combining this with the formula‡ $-\zeta'(s)/\zeta(s) = \int_0^\infty x^{-s}\, d\psi(x)$ for Re $s > 1$ gives

$$\begin{aligned} &\mathrm{Re}\left\{-3\frac{\zeta'(\sigma)}{\zeta(\sigma)} - 4\frac{\zeta'(\sigma + it)}{\zeta(\sigma + it)} - \frac{\zeta'(\sigma + 2it)}{\zeta(\sigma + 2it)}\right\} \\ &\quad = \int_0^\infty x^{-\sigma}[3 + 4\cos(t \log x) + \cos(2t \log x)]\, d\psi(x) \geq 0. \end{aligned}$$

Hence

$$\mathrm{Re}\left\{3\frac{\zeta'(\sigma)}{\zeta(\sigma)} + 4\frac{\zeta'(\sigma + it)}{\zeta(\sigma + it)} + \frac{\zeta'(\sigma + 2it)}{\zeta(\sigma + 2it)}\right\} \leq 0 \tag{2}$$

for all $\sigma > 1$ and for all real t.

This inequality can be used to prove $\beta < 1$ as follows. Assume there is a real value of t such that $\zeta(1 + it) = 0$ and let $f(s)$ be the function $[\zeta(s)]^3 \cdot [\zeta(s + it)]^4 \cdot [\zeta(s + 2it)]$. Then the first factor of $f(s)$ has a pole of order 3 at

†Specifically, de la Vallée Poussin proved that $\beta < 1 - c(\log \gamma - \log n)^{-1}$ for $\gamma \geq 705$, where $c = 0.034$ and $n = 47.8$. This requires, of course, much more careful estimates than those given here.

‡Mertens uses $\log \zeta(s) = \int_0^\infty x^{-s}\, dJ(x)$ instead.

$s = 1$, the second has a zero of order at least 4, and the third has no pole; hence $f(s)$ has a zero of order at least 1 at $s = 1$, say $f(s) = (s-1)^n g(s)$, where $n \geq 1$ and where $g(s)$ has no zero and no pole at $s = 1$. Then the logarithmic derivative of g approaches a limit as $s \to 1$, so the logarithmic derivative of $f(s)$ differs from $n(s-1)^{-1}$ by a bounded amount as $s \to 1$. Thus the logarithmic derivative of $f(s)$ has large positive real part when $s = \sigma$ is real and greater than 1. But (2) shows that this is impossible and hence that no t satisfies $\zeta(1 + it) = 0$.

De la Vallée Poussin used the inequality (2) to prove (1) by using the formula

$$\frac{\zeta'(s)}{\zeta(s)} = -\frac{1}{s-1} + \sum_\rho \frac{1}{s-\rho} - \frac{1}{2}\frac{\Pi'(s/2)}{\Pi(s/2)} + \frac{1}{2}\log \pi$$

[see (6) of Section 3.2] to estimate the terms of (2). Specifically, this formula and the formula $\Pi'(x)/\Pi(x) \sim \log x$ of Section 6.3 show that for $1 \leq \sigma \leq 2$ and for t large, the real part of $\zeta'(s)/\zeta(s)$ is approximately

$$\sum_\rho \operatorname{Re}\frac{1}{s-\rho} - \frac{1}{2}\log\left|\frac{t}{2}\right| + \frac{1}{2}\log \pi$$

[because $(s-1)^{-1}$ is small] and in particular that one can choose $K > 1$ such that

$$\operatorname{Re}\frac{\zeta'(\sigma + it)}{\zeta(\sigma + it)} \geq \sum_\rho \operatorname{Re}\frac{1}{\sigma + it - \rho} - \log t + \frac{1}{2}\log \pi$$

holds throughout the region $1 \leq \sigma \leq 2$, $t \geq K$. The terms of the sum over ρ are positive (because $\sigma \geq 1 > \operatorname{Re} \rho$), so the same inequality holds *a fortiori* if some or all terms of the sum over ρ are omitted. Now $(\sigma - 1)\zeta(\sigma)$ is nonsingular at $\sigma = 1$, so the logarithmic derivative of $\zeta(\sigma)$ differs from $-(\sigma - 1)^{-1}$ by a bounded amount and (2) gives

$$0 \geq -\frac{3}{\sigma - 1} + \frac{4}{\sigma + it - \rho} - 4\log t - \log(2t) + \text{const}$$

when all but one term of the sums over ρ are omitted. Thus for all roots ρ and $1 < \sigma \leq 2$, $t \geq K$

$$\frac{3}{4}\frac{1}{\sigma - 1} + C\log t \geq \frac{1}{\sigma + it - \rho},$$

where $C > 0$ is independent of σ, t, and ρ. If $\rho = \beta + i\gamma$, where $\gamma \geq K$, then one can set $t = \gamma$ to find

$$\frac{3}{4}\frac{1}{\sigma - 1} + C\log \gamma \geq \frac{1}{\sigma - \beta}$$

for all σ in the range $1 < \sigma \leq 2$. Thus

$$\left[\frac{3}{4(\sigma-1)} + C\log\gamma\right]^{-1} \leq \sigma - \beta = (\sigma-1) - (\beta-1),$$

$$\beta - 1 \leq (\sigma-1) - \frac{4(\sigma-1)}{3 + 4(\sigma-1)C\log\gamma}$$

$$= \frac{-(\sigma-1) + 4(\sigma-1)^2 C\log\gamma}{3 + 4(\sigma-1)C\log\gamma},$$

$$\beta \leq 1 - \frac{y - 4y^2}{3 + 4y} \cdot \frac{1}{C\log\gamma},$$

where $y = (\sigma-1)C\log\gamma$. Thus y can have any value between 0 and $C\log\gamma \geq C\log K$. Since $y - 4y^2 > 0$ for small values of y, one need only fix a small positive value of y such that $y < C\log K$, $y - 4y^2 > 0$ in order to draw the desired conclusion (1) for all roots $\rho = \beta + i\gamma$ which satisfy $\gamma \geq K$.

5.3 DE LA VALLÉE POUSSIN'S ESTIMATE OF THE ERROR

Since the main step in the proof of the prime number theorem is to use the estimate $\beta < 1$ to prove that $\sum x^{\rho-1}/\rho(\rho+1)$ approaches zero as $x \to \infty$, it is natural to try to use the improved estimate $\beta < 1 - c(\log\gamma)^{-1}$ (for $\gamma \geq K$) of the last section to prove an improved estimate of $\sum x^{\rho-1}/\rho(\rho+1)$. De la Vallée Poussin accomplished this by the following very simple argument. Note first that

$$\left|\sum_{\rho} \frac{x^{\rho-1}}{\rho(\rho+1)}\right| \leq \sum_{\rho} \frac{x^{\beta-1}}{\gamma^2} = \sum_{|\gamma|<K} \frac{x^{\beta-1}}{\gamma^2} + 2\sum_{\gamma\geq K} \frac{x^{\beta-1}}{\gamma^{1-\delta}} \cdot \frac{1}{\gamma^{1+\delta}}.$$

The first term on the right is the sum of a finite number of terms each of which is a constant times a negative power of x (namely, $x^{\beta-1}$); hence there are positive constants C, ϵ such that this term is less than $Cx^{-\epsilon}$ for all $x > 1$. If δ is any positive constant, then $2\sum \gamma^{-1-\delta}$ converges by the theorem of Section 2.5; so the second term on the right is less than a constant times the maximum value of $x^{\beta-1}/\gamma^{1-\delta}$ if this expression does have a maximum value for $\gamma \geq K$. But

$$\frac{x^{\beta-1}}{\gamma^{1-\delta}} \leq \frac{x^{-c/(\log\gamma)}}{\gamma^{1-\delta}}$$

and the logarithmic derivative of the quantity on the right with respect to γ (considered for the moment as a continuous variable) is

$$\frac{c\log x}{(\log\gamma)^2} \cdot \frac{1}{\gamma} - \frac{1-\delta}{\gamma}$$

which is positive, negative, or zero according to whether $c\log x$ is greater

than, less than, or equal to $(1-\delta)(\log \gamma)^2$. Thus if x is large enough that $c \log x > (1-\delta)(\log K)^2$ and if $1-\delta > 0$, then there is a maximum for $\gamma \geq K$ at the point where $c \log x = (1-\delta)(\log \gamma)^2$. At this point

$$\frac{c \log x}{\log \gamma} = (1-\delta) \log \gamma = [c(1-\delta) \log x]^{1/2};$$

hence

$$\frac{x^{\beta-1}}{\gamma^{1-\delta}} \leq \frac{x^{-c/(\log \gamma)}}{\gamma^{1-\delta}} \leq \frac{\exp\{-[c(1-\delta) \log x]^{1/2}\}}{\exp\{[c(1-\delta) \log x]^{1/2}\}} = \exp\{-2[c(1-\delta) \log x]^{1/2}\}.$$

Set $\delta = \frac{3}{4}$ and let C_1 denote $2 \sum \gamma^{-1-\delta}$. Then the above estimates combine to give

$$\left| \sum \frac{x^{\rho-1}}{\rho(\rho+1)} \right| < Cx^{-\epsilon} + C_1 \exp[-(c \log x)^{1/2}]$$

for all sufficiently large x. Finally, since $x^{-\epsilon}$ goes to zero much faster than $\exp[-(c \log x)^{1/2}]$, since the constants C, C_1 can be absorbed by decreasing c slightly, and since $2 \sum x^{\rho-1}/\rho(\rho+1)$ is the relative error in the approximation $\int_0^x \psi(t)\, dt \sim x^2/2$ except for terms which are like x^{-1} as $x \to \infty$ [see (4) of Section 4.1] this proves that *there is a constant $c > 0$ such that the relative error in the approximation $\int_0^x \psi(t)\, dt \sim x^2/2$ is less than* $\exp[-(c \log x)^{1/2}]$ *for all sufficiently large x.*

Now, by essentially the same arguments which were used to deduce the prime number theorem $\pi(x) \sim \mathrm{Li}(x)$ from $\int_0^x \psi(t)\, dt \sim x^2/2$, one can deduce an estimate of the relative error in the prime number theorem from the above estimate of the relative error in $\int_0^x \psi(t)\, dt \sim x^2/2$.

The first step is to derive an estimate of the relative error in $\psi(x) \sim x$. Let $\varepsilon(x) = \exp[-(c \log x)^{1/2}]$, where c is as above so that

$$[1-\varepsilon(x)]\frac{x^2}{2} \leq \int_0^x \psi(t)\, dt \leq [1+\varepsilon(x)]\frac{x^2}{2}$$

for all sufficiently large x. Then, as before, $\int_0^y \psi(t)\, dt - \int_0^x \psi(t)\, dt$ for $y > x$ is on the one hand at most

$$[1+\varepsilon(y)]\frac{y^2}{2} - [1-\varepsilon(x)]\frac{x^2}{2} \leq [1+\varepsilon(x)]\left(\frac{y^2-x^2}{2}\right) + 2\varepsilon(x)\frac{x^2}{2}$$

and at least

$$[1-\varepsilon(x)]\left(\frac{y^2-x^2}{2}\right) - 2\varepsilon(x)\frac{x^2}{2}$$

and on the other hand is at most

$$\int_x^y \psi(y)\, dt = (y-x)\psi(y)$$

and at least

$$\int_x^y \psi(x)\, dt = (y-x)\psi(x).$$

Thus

$$(y-x)\psi(x) \le [1+\varepsilon(x)]\left(\frac{y^2-x^2}{2}\right) + 2\varepsilon(x)\frac{x^2}{2},$$

$$(y-x)\psi(y) \ge [1-\varepsilon(x)]\left(\frac{y^2-x^2}{2}\right) - 2\varepsilon(x)\frac{x^2}{2},$$

and, therefore,

$$\psi(x) - x \le \frac{y-x}{2} + \varepsilon(x)\frac{y+x}{2} + \varepsilon(x)\frac{x^2}{y-x},$$

$$\psi(y) - y \ge -\frac{y-x}{2} - \varepsilon(x)\frac{y+x}{2} - \varepsilon(x)\frac{x^2}{y-x}.$$

In the first inequality set $y = \{1 + [\varepsilon(x)]^{1/2}\}x$ to obtain

$$\begin{aligned}\frac{\psi(x)-x}{x} &\le \frac{[\varepsilon(x)]^{1/2}}{2} + \varepsilon(x)\left[1 + \frac{1}{2}\varepsilon(x)\right] + [\varepsilon(x)]^{1/2} \\ &\le \text{const}\,[\varepsilon(x)]^{1/2},\end{aligned}$$

and in the second inequality set $x = \{1 - [\varepsilon(y)]^{1/2}\}y$ to obtain

$$\begin{aligned}\frac{\psi(y)-y}{y} &\ge -\frac{[\varepsilon(y)]^{1/2}}{2} - \varepsilon(x)\left\{1 - \frac{1}{2}[\varepsilon(y)]^{1/2}\right\} - \varepsilon(x)\frac{1}{[\varepsilon(y)]^{1/2}} \\ &\ge -\frac{[\varepsilon(x)]^{1/2}}{2} - \varepsilon(x) - \frac{\varepsilon(x)}{[\varepsilon(y)]^{1/2}} \ge \frac{-2\varepsilon(x)}{[\varepsilon(y)]^{1/2}} \\ &= -2\exp\{-[c\log y + c\log[1-(\varepsilon(y))^{1/2}]]^{1/2} + \tfrac{1}{2}(c\log y)^{1/2}\} \\ &\ge -2\exp[-(c\log y)^{1/2}[(1-\text{small})^{1/2} - \tfrac{1}{2}]].\end{aligned}$$

Thus *there is a constant* $c_0 > 0$ *such that the relative error in the approximation* $\psi(x) \sim x$ *is less than* $\exp[-(c_0 \log x)^{1/2}]$ *for all sufficiently large* x.

The next step is to consider the approximation $\theta(x) \sim x$. But since, as before,

$$\psi(x) - \theta(x^{1/2})(\log x)/\log 2 < \theta(x) < \psi(x)$$

and since $\theta(x^{1/2})\log x \sim x^{1/2}\log x$ grows much more slowly than $x\exp[-(c_0 \log x)^{1/2}]$, it follows immediately that the relative error in this approximation is also less than $\exp[-(c_0 \log x)^{1/2}]$ for all sufficiently large x.

The final step is to consider the approximation $\pi(x) \sim \text{Li}(x)$. Here the formula

$$\pi(y) - \pi(x) = \int_x^y \frac{d\theta(t)}{\log t} = \frac{\theta(y)}{\log y} - \frac{\theta(x)}{\log x} + \int_x^y \frac{\theta(t)\,dt}{(\log t)^2 t}$$

shows that if $y > x$ are in the range where the relative error in $\theta(x) \sim x$ is less than $\varepsilon(x) = \exp[-(c_0 \log x)^{1/2}]$, then $\pi(y)$ is at most

$$\begin{aligned}&\pi(x) + \frac{y[1+\varepsilon(y)]}{\log y} - \frac{x[1-\varepsilon(x)]}{\log x} + \int_x^y \frac{t[1+\varepsilon(t)]}{(\log t)^2 t}\,dt \\ &\quad = \pi(x) + \frac{y\varepsilon(y)}{\log y} + \frac{x\varepsilon(x)}{\log x} + \int_x^y \frac{dt}{\log t} + \int_x^y \frac{\varepsilon(t)\,dt}{(\log t)^2}.\end{aligned}$$

Let x be fixed, let y be larger than x^2, and let the final integral be divided into an integral from x to $y^{1/2}$ and an integral from $y^{1/2}$ to y. Then

$$\begin{aligned}\pi(y) &\leq \text{const} + \frac{y\varepsilon(y)}{\log y} + \text{Li}(y) \\ &\quad + (y^{1/2} - x)\frac{\varepsilon(x)}{(\log x)^2} + (y - y^{1/2})\frac{4\varepsilon(y^{1/2})}{(\log y)^2} \\ &\leq \text{const} + \text{Li}(y) \\ &\quad + \frac{y}{\log y}\left[\varepsilon(y) + \frac{\text{const} \log y}{y^{1/2}} + \frac{4\varepsilon(y^{1/2})}{\log y}\right]\end{aligned}$$

and similarly

$$\pi(y) \geq \text{const} + \text{Li}(y) - \frac{y}{\log y}\left[\varepsilon(y) + \frac{\text{const} \log y}{y^{1/2}} + \frac{4\varepsilon(y^{1/2})}{\log y}\right].$$

Since the quantity in square brackets is less than $\varepsilon(y^{1/2})$ for all sufficiently large y, it will suffice to prove that $y/\log y$ divided by $\text{Li}(y)$ is bounded as $y \to \infty$ in order to prove that the relative error in $\pi(y) \sim \text{Li}(y)$ is less than $\varepsilon(y^{1/2})$ for y sufficiently large. It will be shown in the next section that $\text{Li}(y) \sim y/\log y$, and this will complete the proof that *there is a constant* $c_1 > 0$ *such that the relative error in the approximation* $\pi(y) \sim \text{Li}(y)$ *is less than* $\exp[-(c_1 \log y)^{1/2}]$ *for all sufficiently large values of* y. This is de la Vallée Poussin's estimate† of the error in the prime number theorem.

5.4 OTHER FORMULAS FOR $\pi(x)$

The approximate formula for $\pi(x)$ which appears in Legendre's *Theorie des Nombres* [L4] is

$$\pi(x) \sim \frac{x}{\log x - A}, \tag{1}$$

where A is a constant whose value Legendre gives as 1.08366, apparently on empirical grounds. Legendre does not specify the sense in which the approximation (1) is to be understood, but if it is interpreted in the usual sense of "the relative error approaches zero as $x \to \infty$," then the value of A is irrelevant because

$$\frac{x}{\log x - A} \sim \frac{x}{\log x - B}$$

†De la Vallée Poussin wrote the estimate in the form $(c_1 \log y)^{1/2} \exp[-(c_1 \log y)^{1/2}]$ and gave the explicit value 0.032 of c_1. He did not, however, give any explicit estimate of how large y must be in order for this estimate of the relative error in $\pi(y) \sim \text{Li}(y)$ to be valid.

for any two numbers A, B (because the ratio is $\log x - A$ over $\log x - B$, which approaches 1) and hence if (1) is true for one value of A, it must be true for all values of A (because "$\sim$" is transitive). Therefore if Legendre's value $A = 1.08366$ has any significance, it must lie in some other interpretation of the approximation (1).

The prime number theorem $\pi(x) \sim \mathrm{Li}(x)$ shows that (1) is true if and only if $\mathrm{Li}(x) \sim x/(\log x - A)$ for some—and hence all—values of A. But by integration by parts

$$\begin{aligned}\mathrm{Li}(x) &= \mathrm{Li}(2) + \int_2^x \frac{dt}{\log t} \\ &= \mathrm{Li}(2) + \frac{x}{\log x} - \frac{2}{\log 2} + \int_2^x \frac{dt}{(\log t)^2}, \\ \mathrm{Li}(x) - \frac{x}{\log x} &= \text{const} + \int_2^x \frac{dt}{(\log t)^2},\end{aligned}$$

and it suffices to show that the integral on the right divided by $x/\log x$ approaches zero as $x \to \infty$ in order to conclude that (1) is true with $A = 0$. [Intuitively this is the obvious statement that the average value of $(\log t)^{-2}$ for $2 \le t \le x$ is much less than $(\log x)^{-1}$.] This is easily accomplished by dividing the interval of integration at $x^{1/2}$ to find

$$\begin{aligned}\frac{\log x}{x}\int_2^x \frac{dt}{(\log t)^2} &= \frac{\log x}{x}\int_2^{x^{1/2}} \frac{dt}{(\log t)^2} + \frac{\log x}{x}\int_{x^{1/2}}^x \frac{dt}{(\log t)^2} \\ &\le \frac{\log x}{x}\cdot\frac{x^{1/2}-2}{(\log 2)^2} + \frac{\log x}{x}\cdot\frac{x - x^{1/2}}{(\log x^{1/2})^2} \\ &\le \frac{\log x}{(\log 2)^2 x^{1/2}} + \frac{4}{\log x} \to 0.\end{aligned}$$

Thus the prime number theorem implies (1) but it implies no particular value of A.

Chebyshev [C2] was able to show that if any value of A is any better than any other, then this value must be $A = 1$. The special property of the value $A = 1$ which is needed is the fact that the approximation

$$\mathrm{Li}(x) \sim \frac{x}{\log x - A} \tag{2}$$

is best when $A = 1$. To see this, note that

$$\begin{aligned}\frac{x}{\log x - A} &= \frac{x}{\log x}\left[1 + \left(\frac{A}{\log x}\right) + \left(\frac{A}{\log x}\right)^2 + \cdots\right], \\ \frac{x}{\log x - A} &\sim \frac{x}{\log x} + \frac{Ax}{(\log x)^2}, \qquad (3)\end{aligned}$$

where $\sim$ means that the error is much smaller than the last term $Ax(\log x)^{-2}$

in the sense that their ratio approaches zero as $x \longrightarrow \infty$, while

$$\begin{aligned}\mathrm{Li}(x) &= \frac{x}{\log x} + \text{const} + \int_2^x \frac{dt}{(\log t)^2} \\ &= \frac{x}{\log x} + \text{const} + \frac{x}{(\log x)^2} - \frac{2}{(\log 2)^2} + 2\int_2^x \frac{dt}{(\log t)^3}.\end{aligned}$$

Hence

$$\mathrm{Li}(x) \sim \frac{x}{\log x} + \frac{x}{(\log x)^2}, \tag{4}$$

where the error is much smaller than the last term because

$$\frac{(\log x)^2}{x}\int_2^x \frac{dt}{(\log t)^3} \leq \frac{(\log x)^2}{x}\frac{x^{1/2}-2}{(\log 2)^3} + \frac{(\log x)^2}{x}\frac{x - x^{1/2}}{(\log x^{1/2})^3} \longrightarrow 0.$$

The combination of (3) and (4) shows that the error in (2) divided by $x(\log x)^{-2}$ is $A - 1$ plus a quantity which approaches zero as $x \longrightarrow \infty$. This gives a precise sense in which $A = 1$ is the best value of A in (2). Chebyshev was able to prove enough about the approximation $\pi(x) \sim \mathrm{Li}(x)$ to prove that if A has a best value in (1), then that value must also be $A = 1$.

However, even the prime number theorem is not enough to prove that $A = 1$ is best in (1). To prove this, one would have to show that the approximation $\pi(x) \sim \mathrm{Li}(x)$, like the approximations (3) and (4), has the property that its error grows much less rapidly than $x(\log x)^{-2}$, and the prime number theorem says only that it grows much less rapidly than $\mathrm{Li}(x)$, which in turn grows like $x(\log x)^{-1}$ by (2). Thus a stronger estimate of the error in $\pi(x) \sim \mathrm{Li}(x)$ is needed.

Now since $\exp[(c \log x)^{1/2}]$ grows more rapidly than any power of $(c \log x)^{1/2}$, it follows from de la Vallée Poussin's estimate of the error in $\pi(x) \sim \mathrm{Li}(x)$ that

$$\left|\frac{\pi(x) - \mathrm{Li}(x)}{x(\log x)^{-2}}\right| \leq \frac{\mathrm{Li}(x)}{x} \cdot \frac{(\log x)^2}{\exp[(c \log x)^{1/2}]} \longrightarrow 0$$

and hence that the error in (1) divided by $x(\log x)^{-2}$ is $A - 1$ plus a quantity which approaches zero as $x \longrightarrow \infty$. Thus de la Vallée Poussin's estimate proves that the value $A = 1$ in (1) is better than any other value.

More generally, successive integration by parts shows that (4) can be generalized to

$$\begin{aligned}\mathrm{Li}(x) \sim \frac{x}{\log x} &+ \frac{x}{(\log x)^2} + 2\frac{x}{(\log x)^3} \\ &+ 6\frac{x}{(\log x)^4} + \cdots + (n-1)!\frac{x}{(\log x)^n},\end{aligned}$$

where the error (for any fixed n) grows much less rapidly than the last term $x(\log x)^{-n}$ as $x \longrightarrow \infty$. De la Vallée Poussin's estimate shows that the error in

$\pi(x) \sim \mathrm{Li}(x)$ also grows less rapidly than $x(\log x)^{-n}$ and hence proves that the approximation

$$\pi(x) \sim \frac{x}{\log x} + \frac{x}{(\log x)^2} + \cdots + (n-1)!\frac{x}{(\log x)^n} \tag{5}$$

is valid in the sense that (for any fixed n) the error in the approximation divided by the last term on the right approaches zero as $x \longrightarrow \infty$. The case $n = 1$ is essentially the prime number theorem and the case $n = 2$ is essentially the theorem that $A = 1$ is best in (1).

Thus de la Vallée Poussin's estimate of the error proves that the approximation $\pi(x) \sim \mathrm{Li}(x)$ is better than the approximation (5) for any value of n. This is the principal consequence which de la Vallée Poussin himself drew from his estimate of the error.

Formula (5) is an example of an *asymptotic expansion*, which is an expansion such as $\pi(x) = \sum (n-1)!\, x(\log x)^{-n}$ in which the error resulting from taking a finite number of terms is of a lower order of magnitude than the last term used. Another more familiar example of an asymptotic expansion is the Taylor series of an infinitely differentiable function $f(x) = \sum f^{(n)}(a)(x-a)^n/n!$. The fact that this is an asymptotic expansion—that is, the fact that the error resulting from using just n terms decreases more rapidly than $(x-a)^n$ as $x \longrightarrow a$—is Taylor's theorem. This in no way implies, of course, that for fixed $x \neq a$ the error approaches zero as $n \longrightarrow \infty$. For any fixed x formula (5) for $\pi(x)$ becomes worthless as $n \longrightarrow \infty$ because $(n-1)!$ grows much faster than $(\log x)^n$. Another example of an asymptotic expansion is Stirling's series (3) of Section 6.3. Although pure mathematicians shun asymptotic expansions which do not converge as $n \longrightarrow \infty$, mathematicians who engage in computation are well aware that asymptotic expansions (for example, Stirling's series) are often more practical than convergent expansions [for example, the product formula (3) of Section 1.3 for Π].

Recall that Riemann's approximate formula for $\pi(x)$ was

$$\pi(x) \sim \mathrm{Li}(x) + \sum_{n=2}^{N} \frac{\mu(n)}{n} \mathrm{Li}(x^{1/n}), \tag{6}$$

where $N > \log x/\log 2$, and that on empirical grounds this formula appeared to be much better than $\pi(x) \sim \mathrm{Li}(x)$. The second term in this formula is $-\frac{1}{2}\mathrm{Li}(x^{1/2}) \sim -\frac{1}{2}x^{1/2}(\log x^{1/2})^{-1} = -x^{1/2}(\log x)^{-1}$, whereas de la Vallée Poussin's estimate shows that the error in $\pi(x) \sim \mathrm{Li}(x)$ is less than $\mathrm{Li}(x) \cdot \exp[-(c \log x)^{1/2}] \sim x(\log x)^{-1} \exp[-(c \log x)]^{1/2}$ so that $-\frac{1}{2}\mathrm{Li}(x^{1/2})$ divided by the error estimate is about $-x^{-1/2} \exp[+(c \log x)^{1/2}] = -\exp[-\frac{1}{2} \log x + (c \log x)^{1/2}] \longrightarrow 0$. Thus de la Vallée Poussin's estimate is not strong enough to prove that even the second term of (6) has any validity, much less the remaining terms. It was, in fact, proved by Littlewood [L13] that Riemann's formula (6) is *not* better as $x \longrightarrow \infty$ than the simpler formula $\pi(x) \sim \mathrm{Li}(x)$. In other words, Littlewood proved that in the long run as $x \longrightarrow \infty$ the

"periodic" terms $\mathrm{Li}(x^\rho)$ in the formula for $\pi(x)$ (see Section 1.17) are as significant as the monotone increasing term $-\frac{1}{2}\mathrm{Li}(x^{1/2})$ and *a fortiori* as significant as the following terms $\mathrm{Li}(x^{1/3})$, $\mathrm{Li}(x^{1/4})$, ... of (6).

5.5 ERROR ESTIMATES AND THE RIEMANN HYPOTHESIS

In view of the strong relationship between de la Vallée Poussin's estimate of the error in the prime number theorem and his estimate $\beta < 1 - c(\log \gamma)^{-1}$ of $\beta = \mathrm{Re}\,\rho$, it is not surprising that the Riemann hypothesis $\mathrm{Re}\,\rho = \frac{1}{2}$ should imply much stronger estimates of the error. The best such estimate that has been found up to now is the estimate proved by von Koch [K1] in 1901, namely, that if the Riemann hypothesis is true then the relative errors in $\psi(x) \sim x$ and $\pi(x) \sim \mathrm{Li}(x)$ are both less than a constant times $(\log x)^2 x^{-1/2}$ for all sufficiently large x. This estimate implies that the relative errors are eventually less than $x^{-(1/2)+\epsilon}$ for all $\epsilon > 0$, whereas de la Vallée Poussin's estimate $\exp[-(c \log x)^{1/2}]$ fails to show that they are less than $x^{-\epsilon}$ for any $\epsilon > 0$.

If the Riemann hypothesis is true, then the magnitude of the relative error in the approximation $\int_0^x \psi(t)\,dt \sim \frac{1}{2}x^2$ is less than

$$\begin{aligned} &2\left|\sum_\rho \frac{x^{\rho-1}}{\rho(\rho+1)}\right| + \text{const}\, x^{-2} + x^{-1}\zeta'(0)/\zeta(0) \\ &\quad \le 2 \cdot x^{-1/2} \cdot \sum_\rho \left|\frac{1}{\rho(\rho+1)}\right| + \text{const}\, x^{-1} \\ &\quad \le \text{const}\, x^{-1/2} \end{aligned}$$

for all sufficiently large x. However, the previous method of deducing from this an estimate of the relative error in $\psi(x) \sim x$ involves taking a square root and hence yields only the estimate $x^{-1/4}$ and not von Koch's estimate $x^{-1/2}(\log x)^2$. Therefore some other method of estimating the error in $\psi(x) \sim x$ is necessary. The following method is due to Holmgren [H10]. In the estimate (see Section 4.3)

$$\begin{aligned} \psi(x) &\le \int_x^{x+1} \psi(t)\,dt = \int_0^{x+1} \psi(t)\,dt - \int_0^x \psi(t)\,dt \\ &= \left|\frac{(x+1)^2}{2} - \frac{x^2}{2} + \sum_\rho \frac{(x+1)^{\rho+1} - x^{\rho+1}}{\rho(\rho+1)} + \text{bounded}\right| \\ &\le x + \sum_\rho \left|\frac{(x+1)^{\rho+1} - x^{\rho+1}}{\rho(\rho+1)}\right| + \text{const} \end{aligned}$$

let the terms corresponding to roots $\rho = \frac{1}{2} + i\gamma$ (assuming the Riemann

hypothesis) for which $|\gamma| \le x$ be estimated using

$$\left|\frac{(x+1)^{\rho+1} - x^{\rho+1}}{\rho(\rho+1)}\right| = \frac{1}{|\rho|}\left|\int_x^{x+1} t^\rho\, dt\right| \le \frac{|x+1|^{1/2}}{|\rho|} \le \frac{(2x)^{1/2}}{|\gamma|}$$

and let those corresponding to roots for which $|\gamma| > x$ be estimated using

$$\left|\frac{(x+1)^{\rho+1} - x^{\rho+1}}{\rho(\rho+1)}\right| \le \frac{2|x+1|^{\operatorname{Re}\rho+1}}{|\rho(\rho+1)|} \le \frac{2(2x)^{3/2}}{\gamma^2}.$$

Let H be as in the theorem of Section 3.4 so that the number of roots $\rho = \frac{1}{2} + i\gamma$ in the interval $t \le \gamma \le t+1$ is less than $2\log t$ for $t \ge H$. Then the above estimates give (when $x > H$)

$$\begin{aligned}
\psi(x) &\le x + x^{1/2} \sum_{|\gamma|<H} \frac{\sqrt{2}}{|\gamma|} + 2x^{1/2} \sum_{H<\gamma<x} \frac{\sqrt{2}}{\gamma} + 2x^{3/2} \sum_{x<\gamma} \frac{2^{5/2}}{\gamma^2} \\
&\le x + \text{const}\, x^{1/2} + \text{const}\, x^{1/2} \int_H^x \frac{\log t\, dt}{t} + \text{const}\, x^{3/2} \int_x^\infty \frac{\log t\, dt}{t^2} \\
&\le x + \text{const}\, x^{1/2} + \text{const}\, x^{1/2} \frac{(\log t)^2}{2}\Big|_H^x \\
&\quad + \text{const}\, x^{3/2} \left\{ -\frac{\log t}{t}\Big|_x^\infty + \int_x^\infty \frac{dt}{t^2} \right\} \\
&\le x + \text{const}\, x^{1/2} + \text{const}\, x^{1/2} (\log x)^2 + \text{const}\, x^{3/2} \left\{ \frac{\log x}{x} + \frac{1}{x} \right\} \\
&\le x + \text{const}\, x^{1/2} (\log x)^2.
\end{aligned}$$

The same technique applied to the estimate

$$\psi(x) \ge \int_{x-1}^x \psi(t)\, dt = \left| \frac{x^2}{2} - \frac{(x-1)^2}{2} + \sum_\rho \frac{x^{\rho+1} - (x-1)^{\rho+1}}{\rho(\rho+1)} + \text{bounded} \right|$$

gives

$$\begin{aligned}
\psi(x) &\ge x - \sum_{|\gamma|<H} \frac{x^{1/2}}{|\gamma|} - 2 \sum_{H<\gamma<x} \frac{x^{1/2}}{\gamma} - 2 \sum_{x>\gamma} \frac{2x^{3/2}}{\gamma^2} \\
&\ge x - \text{const}\, x^{1/2} - \text{const}\, x^{1/2} \int_H^x \frac{\log t\, dt}{t} - \text{const}\, x^{3/2} \int_x^\infty \frac{\log t\, dt}{t^2} \\
&\ge x - \text{const}\, x^{1/2} (\log x)^2
\end{aligned}$$

which completes the proof that the relative error in $\psi(x) \sim x$ is less than a constant times $(\log x)^2 x^{-1/2}$ (assuming the Riemann hypothesis). Since

$$\psi(x) - \theta(x^{1/2}) \log x / \log 2 \le \theta(x) \le \psi(x)$$

and since $\theta(x^{1/2}) \log x \sim x^{1/2} \log x$ is much smaller than $x^{1/2}(\log x)^2$, the same is true of $\theta(x) \sim x$. Now

$$\begin{aligned}
\pi(x) - \operatorname{Li}(x) &= \int_c^x \frac{d[\theta(t) - t]}{\log t} + \text{const} \\
&= \frac{\theta(x) - x}{\log x} + \int_c^x \frac{\theta(t) - t}{t(\log t)^2}\, dt + \text{const}
\end{aligned}$$

for any constant $c > 1$; hence if $|\theta(t) - t| < Kt^{1/2}(\log t)^2$ for $t \geq c$, then $x > c$ implies

$$|\pi(x) - \mathrm{Li}(x)| \leq \frac{Kx^{1/2}(\log x)^2}{\log x} + \int_c^x \frac{K\,dt}{t^{1/2}} + \text{const}$$
$$\leq K\frac{x}{\log x}\frac{(\log x)^2}{x^{1/2}} + Kx^{1/2} + \text{const},$$

which proves, since $\mathrm{Li}(x) \sim (x/\log x)$ as $x \longrightarrow \infty$, that the relative error in $\pi(x) - \mathrm{Li}(x)$ is eventually less than a constant times $(\log x)^2 x^{-1/2}$ if the Riemann hypothesis is true, as was to be shown.

On the other hand, if the Riemann hypothesis is false, then there is a root ρ with $\mathrm{Re}\,\rho > \frac{1}{2}$ and hence (see below) a "periodic" term in Riemann's formula for $\pi(x)$ which grows more rapidly in magnitude than $x^{1/2}$, so it is reasonable to assume that the error in the prime number theorem $\pi(x) \sim \mathrm{Li}(x)$ would not in that case grow less rapidly than $x^{(1/2)+\epsilon}$.

The rate of growth of $\mathrm{Li}(x^\rho)$ for ρ in the first quadrant $\mathrm{Re}\,\rho > 0$, $\mathrm{Im}\,\rho > 0$ is easily estimated using the formula $\mathrm{Li}(x^\rho) = \int_{C^+} (\log t)^{-1} t^{\rho-1}\,dt + i\pi$ of Section 1.15 and integration by parts as in Section 5.4 to find

$$\mathrm{Li}(x^\rho) = \int_{C^+} \frac{t^{\rho-1}\,dt}{\log t} + i\pi$$
$$= \int_{C^+} \frac{d}{dt}\left\{\frac{t^\rho}{\rho \log t}\right\} dt - \int_{C^+} \frac{t^\rho}{\rho}\frac{(-1)}{(\log t)^2}\frac{dt}{t} + i\pi$$
$$= \frac{x^\rho}{\rho \log x} + \int_{C^+} \frac{t^{\rho-1}\,dt}{\rho(\log t)^2} + i\pi$$
$$= \frac{x^\rho}{\rho \log x} + \int_2^x \frac{t^{\rho-1}\,dt}{\rho(\log t)^2} + \text{const}.$$

The first term, which has modulus $x^{\mathrm{Re}\,\rho}\,|\rho|^{-1}(\log x)^{-1}$, dominates as $x \longrightarrow \infty$ because

$$\int_2^{x^{1/2}} \frac{t^{\rho-1}\,dt}{(\log t)^2} + \int_{x^{1/2}}^x \frac{t^{\rho-1}\,dt}{(\log t)^2}$$
$$\leq \int_2^{x^{1/2}} \frac{t^{\mathrm{Re}\,\rho-1}\,dt}{(\log 2)^2} + \int_{x^{1/2}}^x \frac{t^{\mathrm{Re}\,\rho-1}\,dt}{(\log x^{1/2})^2}$$
$$\leq \frac{x^{(1/2)\,\mathrm{Re}\,\rho}}{\mathrm{Re}\,\rho(\log 2)^2} + \frac{4x^{\mathrm{Re}\,\rho}}{\mathrm{Re}\,\rho(\log x)^2}$$

has modulus much less than a constant times $x^{\mathrm{Re}\,\rho}(\log x)^{-1}$.

Theorem The Riemann hypothesis is equivalent to the statement that for every $\epsilon > 0$ the relative error in the prime number theorem $\pi(x) \sim \mathrm{Li}(x)$ is less than $x^{-(1/2)+\epsilon}$ for all sufficiently large x. [If they are true, then the relative error in the prime number theorem is in fact less than a constant times $x^{-1/2}(\log x)^2$.]

Proof It remains only to show that if the relative error in $\pi(x) \sim \mathrm{Li}(x)$ is less than $x^{-(1/2)+\epsilon}$, then the Riemann hypothesis must be true. Assume therefore that for every $\epsilon > 0$ the relative error in $\pi(x) \sim \mathrm{Li}(x)$ is eventually

less than $x^{-(1/2)+\epsilon}$. Since $\mathrm{Li}(x) \sim (x/\log x)$, this implies that the absolute error is eventually less than $x^{(1/2)+\epsilon}$, say for $x \geq c$. Then since

$$\theta(x) - x = \int_c^x \log t \, d[\pi(t) - \mathrm{Li}(t)] + \text{const}$$
$$= (\log x)[\pi(x) - \mathrm{Li}(x)] - \int_c^x \frac{\pi(t) - \mathrm{Li}(t)}{t} \, dt + \text{const},$$

it follows that

$$|\theta(x) - x| \leq (\log x)x^{(1/2)+\epsilon} + \int_c^x t^{-(1/2)+\epsilon} \, dt + \text{const} \leq x^{(1/2)+2\epsilon}$$

for all sufficiently large x. Since

$$\theta(x) \leq \psi(x) \leq \theta(x) + \frac{\log x}{\log 2}\theta(x^{1/2}) \sim \theta(x) + \text{const}\, x^{1/2} \log x,$$

this implies immediately that $|\psi(x) - x| \leq x^{(1/2)+2\epsilon}$ for all sufficiently large x. But the formulas

$$-\frac{\zeta'(s)}{\zeta(s)} = \int_0^\infty x^{-s} \, d\psi(x) = s\int_1^\infty x^{-s-1}\psi(x) \, dx,$$

$$-\frac{1}{(1-s)} = \int_1^\infty x^{-s} \, dx = -1 + s\int_1^\infty x^{-s-1} \cdot x \, dx,$$

which hold for $\mathrm{Re}\, s > 1$, combine to give

$$-\frac{d}{ds}\log[(s-1)\zeta(s)] = 1 + s\int_1^\infty x^{-s-1}[\psi(x) - x] \, dx$$

for $\mathrm{Re}\, s > 1$. If $|\psi(x) - x| < x^{(1/2)+2\epsilon}$ for all large x, then the integral on the right converges throughout the halfplane $\mathrm{Re}\, s > \frac{1}{2} + 2\epsilon$. By analytic continuation the right side (which is analytic by differentiation under the integral sign) must be equal to the left side throughout the halfplane $\mathrm{Re}\, s > \frac{1}{2} + 2\epsilon$, which shows that $\zeta(s)$ could have no zeros in this halfplane. Thus if the relative error in $\pi(x) \sim \mathrm{Li}(x)$ is less than $x^{-(1/2)+\epsilon}$ for all $\epsilon > 0$, the Riemann hypothesis must be true.

5.6 A POSTSCRIPT TO DE LA VALLÉE POUSSIN'S PROOF

The Euler product formula for $\zeta(s)$ implies that for $\mathrm{Re}\, s > 1$

$$\text{(1)} \qquad \frac{1}{\zeta(s)} = \prod_p \left(1 - \frac{1}{p^s}\right) = 1 - \frac{1}{2^s} - \frac{1}{3^s} - \frac{1}{5^s} + \frac{1}{6^s} - \cdots$$
$$= \sum_{n=1}^\infty \frac{\mu(n)}{n^s},$$

where, as in the Möbius inversion formula, $\mu(n)$ is 0 if n is divisible by a square, -1 if n is a product of an odd number of distinct prime factors, and

$+1$ if n is a product of an even number of distinct prime factors [$\mu(1) = 1$]. Since $\zeta(s)$ has a pole at $s = 1$, $[\zeta(s)]^{-1}$ has a zero at $s = 1$; so if (1) were valid for $s = 1$, it would say

$$(2)\quad 0 = 1 - \tfrac{1}{2} - \tfrac{1}{3} - \tfrac{1}{5} + \tfrac{1}{6} - \tfrac{1}{7} + \tfrac{1}{10} - \tfrac{1}{11} - \tfrac{1}{13} + \tfrac{1}{14} + \tfrac{1}{15} - \cdots.$$

This equation was stated by Euler [E5], but Euler quite often dealt with divergent series and his statement of this equation should not necessarily be understood to imply *convergence* of the series on the right but only *summability*. At any rate Euler did not give any proof that the series (2) was convergent, and the first proof of this fact was given by von Mangoldt [M2] in 1897. Von Mangoldt's proof is rather difficult and a much simpler proof of the formula (2), together with an estimate of the *rate* of convergence, was given by de la Vallée Poussin in 1899 in connection with his proof of the improved error estimate $\exp[-(c \log x)^{1/2}]$ in the prime number theorem.

Specifically, de la Vallée Poussin proved that there is a constant K such that

$$(3)\qquad \left|\sum_{n<x} \frac{\mu(n)}{n}\right| < \frac{K}{\log x}$$

for all sufficiently large x. As $x \to \infty$ this of course implies (2). De la Vallée Poussin's proof is based on two observations concerning the function $P(x)$ defined by

$$P(x) = \int_0^x t^{-1}\, d\psi(t) = \sum_{n<x} \frac{\Lambda(n)}{n},$$

namely, the observation that $P(x)$ is related to the series (2) by the formula

$$(4)\qquad \sum_{n<x} \frac{\mu(n)}{n}\left[P\left(\frac{x}{n}\right) + \log n\right] \equiv 0$$

and the observation that an estimate of $P(x)$ can be derived from the estimate of $\psi(x)$ obtained in Section 5.3.

Consider first the proof of (4). The essence of this identity is the identity

$$\frac{d}{ds}\left[\frac{1}{\zeta(s)}\right] = -\left[\frac{\zeta'(s)}{\zeta(s)}\right]\left[\frac{1}{\zeta(s)}\right]$$

which, since $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)n^{-s}$ and $[\zeta(s)]^{-1} = \sum \mu(n)n^{-s}$ for $\operatorname{Re} s > 1$, gives

$$(5)\qquad \sum_{n=1}^{\infty} \frac{-\mu(n)\log n}{n^s} = \left[\sum_{j=1}^{\infty} \frac{\Lambda(j)}{j^s}\right]\left[\sum_{k=1}^{\infty} \frac{\mu(k)}{k^s}\right]$$
$$= \sum_{n=1}^{\infty} \frac{1}{n^s}\left[\sum_{jk=n} \Lambda(j)\mu(k)\right]$$

for $\operatorname{Re} s > 1$. It is natural to suppose that since these series have the same sum

for all s they must be identical, and hence for all x

$$\sum_{n<x} \frac{-\mu(n)\log n}{n^s} \equiv \sum_{n<x} \frac{1}{n^s}\Big[\sum_{jk=n} \Lambda(j)\mu(k)\Big] \tag{6}$$

which for $s = 1$ gives

$$-\sum_{n<x} \frac{\mu(n)\log n}{n} = \sum_{n<x} \sum_{jk=n} \frac{\Lambda(j)}{j} \cdot \frac{\mu(k)}{k}. \tag{7}$$

Now for every fixed $k \le n < x$ the sum on the right is over all integers j such that $jk < x$, that is, it is over all integers $j < x/k$, hence

$$-\sum_{n<x} \frac{\mu(n)\log n}{n} = \sum_{k<x} \frac{\mu(k)}{k} P\Big(\frac{x}{k}\Big)$$

from which (4) follows. The one step of the argument which requires further justification is the truncation (6) of the series (5). This is essentially a question of recovering the coefficients of a series $\sum A_n n^{-s}$ from a knowledge of its sum, which can be done by a technique similar to that used in Section 3.2 to recover $\Lambda(n)$ from $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)n^{-s}$. Specifically, one can go directly from (5) to the desired equation (7) by using the identity

$$\begin{aligned}
\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\Big[\sum_{n=1}^{\infty}\frac{A_n}{n^s}\Big]\frac{x^s\,ds}{s-1} &= x\cdot\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\Big[\sum_{n=1}^{\infty}\frac{A_n}{n}\Big(\frac{x}{n}\Big)^{s-1}\Big]\frac{ds}{s-1}\\
&= x\cdot\frac{1}{2\pi i}\int_{a-1-i\infty}^{a-1+i\infty}\Big[\sum_{n=1}^{\infty}\frac{A_n}{n}\Big(\frac{x}{n}\Big)^{t}\Big]\frac{dt}{t}\\
&= x\cdot\sum_{n<x}\frac{A_n}{n},
\end{aligned}$$

where the termwise integration in the last step is justified, by the same argument as before, whenever the A_n are such that $\sum (|A_n|/n)n^{-(a-1)} < \infty$, which is certainly the case for both series in (5) for large a. This completes the proof of (4).

Consider now the estimation of $P(x)$. The desired estimate is easily found heuristically by using

$$d\psi(x) = (1 - \sum x^{\rho-1} - \sum x^{-2n-1})\,dx$$

to find

$$dP(x) = x^{-1}\,d\psi(x) = (x^{-1} - \sum x^{\rho-2} - \sum x^{-2n-2})\,dx,$$

$$P(x) = \log x - \sum \frac{x^{\rho-1}}{\rho-1} + \sum \frac{x^{-2n-1}}{2n+1} + \text{const},$$

which indicates that $P(x) \sim \log x + \text{const}$ with an error which goes to zero faster than

$$\begin{aligned}
\Big|\sum \frac{x^{\rho-1}}{\rho-1}\Big| &\le \Big|\sum \frac{x^{\rho-1}}{\rho}\Big| + \Big|\sum \frac{x^{\rho-1}}{\rho(\rho-1)}\Big|\\
&\le \exp[-(c\log x)^{1/2}] + \exp[-(c\log x)^{1/2}].
\end{aligned}$$

In order to prove this it suffices to write

$$\begin{aligned} P(x) &= \int_0^x t^{-1}\, d\psi(t) = t^{-1}\psi(t)\Big|_0^x + \int_0^x t^{-2}\psi(t)\, dt \\ &= \frac{\psi(x)}{x} + \int_1^x t^{-2}\psi(t)\, dt \\ &= \frac{\psi(x)-x}{x} + \int_1^x t^{-1}\, dt + \int_1^x t^{-1}\frac{\psi(t)-t}{t}\, dt + 1 \\ &= \log x + \frac{\psi(x)-x}{x} + \int_1^x t^{-1}\frac{\psi(t)-t}{t}\, dt + 1. \end{aligned}$$

The integral in the last expression converges as $x \longrightarrow \infty$ because if x increases to x', it increases for large x by less than

$$\begin{aligned} &\int_x^{x'} t^{-1}\exp[-(c\log t)^{1/2}]\, dt = \frac{1}{c}\int_{c\log x}^{c\log x'} \exp(-u^{1/2})\, du \\ &= \frac{2}{c}\int_{(c\log x)^{1/2}}^{(c\log x')^{1/2}} e^{-v} v\, dv = \frac{2}{c}(-e^{-v}v - e^{-v})\Big|_{(c\log x)^{1/2}}^{(c\log x')^{1/2}} \\ &\leq \frac{2}{c}(c\log x)^{1/2}\exp[-(c\log x)^{1/2}] + \frac{2}{c}\exp[-(c\log x)^{1/2}]. \end{aligned}$$

Thus the above formula can be rewritten

$$P(x) = \log x + 1 + \int_1^\infty \frac{\psi(t)-t}{t^2}\, dt + \frac{\psi(x)-x}{x} - \int_x^\infty \frac{\psi(t)-t}{t^2}\, dt,$$

$$P(x) = \log x + C + \eta(x), \tag{8}$$

where C is the constant† $1 + \int_1^\infty t^{-2}[\psi(t) - t]\, dt$ and where the error $\eta(x)$ is less than

$$\left(1 + \frac{2}{c}\right)\exp[-(c\log x)^{1/2}] + \frac{2}{c}(c\log x)^{1/2}\exp[-(c\log x)^{1/2}]$$

for all x large enough that $[\psi(t) - t]/t \leq \exp[-(c\log t)^{1/2}]$ for $t \geq x$. Thus it is possible to choose $K' > 0$, $c > 0$ such that the error $\eta(x)$ is less than

†The constant C is in fact the negative of Euler's constant (see Section 3.8). To derive this fact, note first that C is $1 + \lim_{s\to 1}\int_1^\infty t^{1-s}[\psi(t) - t]\, dt$ and then integrate by parts to find that it is the limit as $s \longrightarrow 1$ of $-s^{-1}\{[\zeta'(s)/\zeta(s)] + 1 - s + (s-1)^{-1}\}$ which is the limit as $s \longrightarrow 1$ of $-(d/ds)\log[(s-1)\zeta(s)]$. But the functional equation gives $(s-1)\zeta(s) = -\Pi(1-s)(2\pi)^{s-1}\zeta(1-s)2\sin(\pi s/2)$. Using the fact that the logarithmic derivative of ζ at 0 is $\log 2\pi$ while that of Π is $-\gamma$ (see Section 3.8), the result $C = -\gamma$ then follows. Since the left side of formula (2) of Section 4.1 is $P(x) - x^{-1}\psi(x) = \log x + C - 1 + \cdots$, this implies that the constant on the right is $-\gamma - 1$. The same fact can also be derived by setting $v = 0$ in the formula of the note of Section 4.3 and letting $u \longrightarrow 1$. Yet another proof that $C = -\gamma$ is given in Section 12.10.

$K' \exp[-(c \log x)^{1/2}]$ for all $x \geq 0$. Then (4) and (8) combine to give

$$\sum_{n<x} \frac{\mu(n)}{n}\left[\log\left(\frac{x}{n}\right) + C - \eta\left(\frac{x}{n}\right) + \log n\right] \equiv 0,$$

$$(\log x + C) \sum_{n<x} \frac{\mu(n)}{n} \equiv \sum_{n<x} \frac{\mu(n)}{n} \eta\left(\frac{x}{n}\right),$$

$$\left|\sum_{n<x} \frac{\mu(n)}{n}\right| \leq \frac{\sum_{n<x} (1/n)K' \exp[-(c \log (x/n))^{1/2}]}{\log x + C},$$

and the desired inequality (3) will follow if it can be shown that

$$\sum_{n<x} \frac{1}{n} \exp\left[-\left(c \log \frac{x}{n}\right)^{1/2}\right]$$

remains bounded as $x \longrightarrow \infty$. Now the logarithmic derivative of the summand with respect to n (considered for the moment as a continuous variable) is

$$-\frac{1}{n} - \frac{1}{2}\left(c \log \frac{x}{n}\right)^{-1/2} \frac{cn}{x} \cdot \left(-\frac{x}{n^2}\right) = \frac{1}{n}\left[-1 + \frac{c}{2}\left(c \log \frac{x}{n}\right)^{-1/2}\right]$$

which is negative until

$$c/2 = [c \log (x/n)]^{1/2}, \qquad \log (x/n) = c/4, \qquad n = xe^{-c/4},$$

after which it is positive. Let N be the last integer before the sign change. Then

$$\begin{aligned}
&\sum_{n<x} \frac{1}{n} \exp\left[-\left(c \log \frac{x}{n}\right)^{1/2}\right] \\
&\quad \leq \exp[-(c \log x)^{1/2}] + \sum_{n=2}^{N} \frac{1}{n} \exp\left[-\left(c \log \frac{x}{n}\right)^{1/2}\right] \\
&\qquad + \sum_{N+1 \leq n < x-1} \frac{1}{n} \exp\left[-\left(c \log \frac{x}{n}\right)^{1/2}\right] \\
&\qquad + \frac{1}{x} \exp\left[-\left(c \log \frac{x}{x}\right)^{1/2}\right] \\
&\quad \leq 1 + \int_1^N \frac{1}{t} \exp\left[-\left(c \log \frac{x}{t}\right)^{1/2}\right] dt \\
&\qquad + \int_{N+1}^{x} \frac{1}{t} \exp\left[-\left(c \log \frac{x}{t}\right)^{1/2}\right] dt + \frac{1}{x} \\
&\quad \leq 1 + \int_1^x \frac{1}{t} \exp\left[-\left(c \log \frac{x}{t}\right)^{1/2}\right] dt + \frac{1}{x}
\end{aligned}$$

and it suffices to show that this integral remains bounded as $x \longrightarrow \infty$. But $u = x/t$ gives

$$\int_1^x \frac{1}{t} \exp\left[-\left(c \log \frac{x}{t}\right)^{1/2}\right] dt = \int_1^x \frac{1}{u} \exp[-(c \log u)^{1/2}]\, du$$

and it was shown above that this integral converges as $x \longrightarrow \infty$, so the proof of (3), and hence of (2), is complete.

Chapter 6

Numerical Analysis of the Roots by Euler–Maclaurin Summation

6.1 INTRODUCTION

The first substantial numerical information on the roots ρ was given by Gram [G5], who in 1903 published a list of 15 roots on the line $\operatorname{Re} s = \frac{1}{2}$. Gram computed the first 10 of these roots to about 6 decimal places and the remaining 5 to about 1 place. Specifically, the values he gave were $\rho = \frac{1}{2} + i\alpha$, where

$$\begin{aligned}
\alpha_1 &= 14.134\,725, & \alpha_6 &= 37.586\,176, & \alpha_{11} &= 52.8,\\
\alpha_2 &= 21.022\,040, & \alpha_7 &= 40.918\,720, & \alpha_{12} &= 56.4,\\
\alpha_3 &= 25.010\,856, & \alpha_8 &= 43.327\,073, & \alpha_{13} &= 59.4,\\
\alpha_4 &= 30.424\,878, & \alpha_9 &= 48.005\,150, & \alpha_{14} &= 61.0,\\
\alpha_5 &= 32.935\,057, & \alpha_{10} &= 49.773\,832, & \alpha_{15} &= 65.0.
\end{aligned}$$

Subsequent calculations have confirmed that these values are correct except, as Gram stated, for slight errors in the last place given. (For the correct values to 6 places see Haselgrove's tables [H8].) Gram was also able to prove that this list includes *all* of the roots ρ in the range $0 \leq \operatorname{Im} s \leq 50$ and thus to prove that the Riemann hypothesis is true in this range.

The basis of Gram's calculations was the straightforward method of Euler–Maclaurin summation to evaluate both the function ζ and the factorial function Π, and consequently to evaluate $\xi(s) = \Pi(\frac{1}{2}s)\pi^{-s/2}(s-1)\zeta(s)$. It is interesting from the point of view of the psychology of mathematical discovery to note that Gram had initially attempted more original and more complicated techniques but had met with very limited success. It was several years before he tried using the classical method of Euler–Maclaurin summa-

tion, and when he did he was surprised at the ease with which he was able to compute the numbers he had been searching for for so long.

Euler–Maclaurin summation is a technique for the numerical evaluation of sums which was developed in the early part of the eighteenth century. The original impetus came from Bernoulli's success in generalizing $\sum_{n=1}^{N} n = N(N+1)/2$ to find an analogous formula for $\sum_{n=1}^{N} n^k$ which involved the "Bernoulli numbers." Forms of the technique were used by Stirling and De Moivre as early as 1730, but the definitive statement of the method together with a proof of sorts did not come until around 1740 when it was published by Euler and, independently (see Cantor [C1]), by Maclaurin. Euler in his well-known calculus book [E6] included examples of the use of Euler–Maclaurin summation to compute $\zeta(s)$ for $s = 2, 3, \ldots, 15, 16$ and to compute $\Pi(s)$ for large s (Stirling's series), so Gram's computations are direct descendants of Euler's.

Gram's work was carried farther by Backlund [B3, B4] around 1912–1915. Backlund's major contribution was a method of computing, for certain values of T, the number of roots in the range $0 \leq \operatorname{Im} s \leq T$. This method enabled him to show that the Riemann hypothesis was true up to the level $T = 200$, that is, to prove that all of the roots in the range $0 \leq \operatorname{Im} s \leq 200$ lie on the line $\operatorname{Re} s = \frac{1}{2}$. It also enabled him to prove that Riemann's estimate [see (d) of Section 1.19] of the number of roots in $0 \leq \operatorname{Im} s \leq T$ for large T was correct, a fact which von Mangoldt had already proved in 1905 by a method which was more complicated. Backlund's proof of this fact is included in this chapter, even though it does not contribute to the numerical analysis of the roots, because it is a natural outgrowth of techniques which Backlund developed for the numerical analysis of the roots.

Some ten years later the Riemann hypothesis was verified up to the level $T = 300$ by Hutchinson [H11], who contributed some improvements of Gram's and Backlund's methods. As far as the distribution of the roots on the line $\operatorname{Re} s = \frac{1}{2}$ is concerned, Hutchinson showed that it is usually true in the range $0 \leq \operatorname{Im} s \leq 300$ that there is exactly one root between two consecutive Gram points (see Section 6.5)—in other words the set of roots and the set of Gram points usually separate each other—but that there are two slight exceptions to this rule in this range, the first near $\operatorname{Im} s = 282.5$ and the second near $\operatorname{Im} s = 295.5$.

Broadly speaking, the computations of Gram, Backlund, and Hutchinson contributed substantially to the plausibility of the Riemann hypothesis, but they gave no insight into the question of why it might be true or into the question of why Riemann might have been led to make such a hypothesis.

6.2 EULER–MACLAURIN SUMMATION

Consider the problem of finding the numerical value of the sum S defined by

$$S = \left(\frac{1}{10}\right)^2 + \left(\frac{1}{11}\right)^2 + \left(\frac{1}{12}\right)^2 + \cdots + \left(\frac{1}{100}\right)^2. \tag{1}$$

As a first approximation to S one might note that if half the first term and half the last term are omitted from S, then the sum which remains is an approximation to $\int_{10}^{100} x^{-2}\,dx$; specifically, the trapezoidal rule

$$\int_a^b f(x)\,dx \sim \sum_{i=1}^{n} \frac{f(x_i) + f(x_{i-1})}{2}(x_i - x_{i-1})$$

$$(a = x_0 < x_1 < \cdots < x_n = b)$$

in the case $x_0 = 10, x_1 = 11, \ldots, x_n = 100, f(x) = x^{-2}$ gives

$$\int_{10}^{100} \frac{dx}{x^2} \sim \frac{1}{2}\left[\left(\frac{1}{10}\right)^2 + \left(\frac{1}{11}\right)^2\right] \cdot 1 + \frac{1}{2}\left[\left(\frac{1}{11}\right)^2 + \left(\frac{1}{12}\right)^2\right] \cdot 1 + \cdots + \frac{1}{2}\left[\left(\frac{1}{99}\right)^2 + \left(\frac{1}{100}\right)^2\right] \cdot 1.$$

Hence

$$S \sim \int_{10}^{100} \frac{dx}{x^2} + \frac{1}{2}\left(\frac{1}{10}\right)^2 + \frac{1}{2}\left(\frac{1}{100}\right)^2$$

$$= -\frac{1}{x}\Big|_{10}^{100} + 0.005 + 0.00005 = 0.09505.$$

Euler–Maclaurin summation is a method of computing the error in this approximation and in the analogous approximation

$$\sum_{n=M}^{N} f(n) \sim \int_M^N f(x)\,dx + \tfrac{1}{2}[f(M) + f(N)] \tag{2}$$

for other sums.

The first step is to develop a closed formula for the error. Let $[x]$ denote, as usual, the largest integer less than or equal to x. Then the function $[x]$ is a step function which has jumps of one at integers, so the Stieltjes measure $d([x])$ assigns the weight one to integers and is zero elsewhere. Hence

$$\int_M^N f(x)\,d([x]) = \tfrac{1}{2}f(M) + f(M+1) + f(M+2) + \cdots + f(N-1) + \tfrac{1}{2}f(N),$$

where the usual convention of counting half the weight at an end point is followed. Thus to make the approximation (2) correct, the right side should

be increased by

$$-\int_M^N f(x)\,dx + \int_M^N f(x)\,d([x]) = \int_M^N f(x)\,d([x]-x).$$

It is more natural to describe the measure $d([x]-x)$ as $d([x]-x+\frac{1}{2})$ because the function $[x]-x+\frac{1}{2}$ is positive half the time and negative half the time, and because it is zero when x is an integer (by the usual convention that at discontinuities the value is the average of the left-hand limit and the right-hand limit). Integration by parts then expresses the error in the form

$$\begin{aligned}\int_M^N f(x)\,d([x]-x+\tfrac{1}{2}) &= -\int_M^N ([x]-x+\tfrac{1}{2})\,df(x)\\ &= \int_M^N (x-[x]-\tfrac{1}{2})f'(x)\,dx.\end{aligned}$$

Having arrived at this formula by sketchy arguments based on Stieltjes integration, one can easily justify it using ordinary Riemann integration to find

$$\begin{aligned}\int_M^N (x-[x]-\tfrac{1}{2})f'(x)\,dx &= \sum_{n=M}^{N-1}\int_n^{n+1}(x-[x]-\tfrac{1}{2})f'(x)\,dx\\ &= \sum_{n=M}^{N-1}\int_0^1 (t-\tfrac{1}{2})f'(n+t)\,dt\\ &= \sum_{n=M}^{N-1}\Big[(t-\tfrac{1}{2})f(n+t)\Big|_0^1 - \int_0^1 f(n+t)\,dt\Big]\\ &= \sum_{n=M}^{N-1}[\tfrac{1}{2}f(n+1)+\tfrac{1}{2}f(n)] - \sum_{n=M}^{N-1}\int_0^1 f(n+t)\,dt\\ &= \tfrac{1}{2}f(M) + f(M+1) + f(M+2) + \cdots\\ &\quad + f(N-1) + \tfrac{1}{2}f(N) - \int_M^N f(x)\,dx\end{aligned}$$

which proves the desired formula

$$\begin{aligned}\sum_{n=M}^{N} f(n) &= \int_M^N f(x)\,dx + \tfrac{1}{2}[f(M)+f(N)]\\ &\quad + \int_M^N (x-[x]-\tfrac{1}{2})f'(x)\,dx\end{aligned} \tag{3}$$

for continuously differentiable functions f on $[M, N]$.

In the case of the sum S in (1) this formula shows that its value is 0.09505 plus

$$\int_{10}^{100} \frac{(x-[x]-\tfrac{1}{2})(-2)}{x^3}\,dx. \tag{4}$$

The integrand in this integral is positive from 10 to $10\frac{1}{2}$, negative from $10\frac{1}{2}$ to 11, positive from 11 to $11\frac{1}{2}$, etc. Since x^{-3} decreases, the integral can thus be written as an alternating series of terms which decrease in absolute value,

so its value is positive but less than

$$\int_{10}^{10.5} \frac{(x - [x] - \frac{1}{2})(-2)}{x^3}\,dx = \int_0^{1/2} \frac{1 - 2t}{(10 + t)^3}\,dt$$
$$\leq \frac{1}{10^3}\int_0^{1/2} (1 - 2t)\,dt = 0.00025.$$

Thus S lies between 0.09505 and 0.09530, which gives its value to three places.

This much is quite elementary. The real substance of Euler–Maclaurin summation is *repeated integration by parts of the last term of* (3) which puts this term in a form that can in many cases be evaluated numerically with great accuracy. This integration by parts requires the use of *Bernoulli polynomials*. The nth Bernoulli polynomial is by definition the unique polynomial of degree n with the property that

$$\int_x^{x+1} B_n(t)\,dt = x^n. \tag{5}$$

Thus, for example, $B_3(x) = ax^3 + bx^2 + cx + d$ is determined by the equation

$$\begin{aligned} x^3 &= \int_x^{x+1} (at^3 + bt^2 + ct + d)\,dt \\ &= a\frac{(x+1)^4 - x^4}{4} + b\frac{(x+1)^3 - x^3}{3} + c\frac{(x+1)^2 - x^2}{2} \\ &\quad + d\frac{x + 1 - x}{1} \\ &= ax^3 + \left(\frac{3}{2}a + b\right)x^2 + (a + b + c)x \\ &\quad + \left(\frac{a}{4} + \frac{b}{3} + \frac{c}{2} + \frac{d}{1}\right) \end{aligned}$$

from which one finds successively $a = 1$, $b = -3/2$, $c = 1/2$, $d = 0$. It is easy to see that this process always yields a polynomial $B_n(x)$ satisfying (5) and, since a polynomial $p(x)$ which satisfies $\int_x^{x+1} p(t)\,dt \equiv 0$ must be identically zero, this suffices to prove that condition (5) defines a unique polynomial $B_n(x)$. Differentiation of (5) gives

$$B_n(x + 1) - B_n(x) = nx^{n-1}, \tag{6}$$
$$\int_x^{x+1} \frac{1}{n} B_n'(t)\,dt = x^{n-1},$$

which shows that $B_n'(x)/n$ satisfies the definition of $B_{n-1}(x)$ and hence that

$$B_n'(x) = nB_{n-1}(x). \tag{7}$$

Thus, starting with $B_3(x)$, which was computed above, one has immediately

$$B_3(x) = x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x, \qquad B_2(x) = x^2 - x + \tfrac{1}{6},$$
$$B_1(x) = x - \tfrac{1}{2}, \qquad B_0(x) = 1.$$

These polynomials can be used to integrate the last term of (3) by parts to put it in the form

$$\begin{aligned}
&\int_M^N \left(x - [x] - \frac{1}{2}\right) f'(x)\,dx \\
&\quad = \sum_{n=M}^{N-1} \int_0^1 \left(t - \frac{1}{2}\right) f'(n+t)\,dt = \sum_{n=M}^{N-1} \int_0^1 B_1(t) f'(n+t)\,dt \\
&\quad = \sum_{n=M}^{N-1} \left[\frac{B_2(t)}{2} f'(n+t)\Big|_0^1 - \frac{1}{2}\int_0^1 B_2(t) f''(n+t)\,dt\right] \\
&\quad = -\frac{B_2(0)}{2} f'(M) + \frac{B_2(1)}{2} f'(M+1) - \frac{B_2(0)}{2} f'(M+1) \\
&\qquad + \frac{B_2(1)}{2} f'(M+2) - \frac{B_2(0)}{2} f'(M+2) + \cdots + \frac{B_2(1)}{2} f'(N) \\
&\qquad - \frac{1}{2}\int_M^N B_2(x - [x]) f''(x)\,dx.
\end{aligned}$$

The long sum telescopes because (6) with $n = 2$, $x = 0$ gives $B_2(1) = B_2(0)$. Thus if $\bar{B}_2(x)$ is used to denote the periodic function $B_2(x - [x])$, the last term of (3) can be written in the form

$$\frac{B_2(0)}{2} f'(x)\Big|_M^N - \frac{1}{2}\int_M^N \bar{B}_2(x) f''(x)\,dx.$$

The second term in this formula can be integrated by parts by exactly the same sequence of steps to put the last term of (3) in the form

$$\frac{B_2(0)}{2} f'(x)\Big|_M^N - \frac{B_3(0)}{2\cdot 3} f''(x)\Big|_M^N + \frac{1}{2\cdot 3}\int_M^N \bar{B}_3(x) f'''(x)\,dx,$$

where, of course, $\bar{B}_3(x)$ denotes the periodic function $B_3(x - [x])$.

Applying this formula to the evaluation of the integral (4) gives

$$\begin{aligned}
&\frac{1}{2}\frac{-2}{x^3}\Big|_{10}^{100} - 0 + \frac{1}{2\cdot 3}\int_{10}^{100} \bar{B}_3(x) f'''(x)\,dx \\
&\quad = \frac{1}{6}\left[\frac{1}{10^3} - \frac{1}{100^3}\right] + \frac{(-2)(-3)(-4)}{2\cdot 3}\int_{10}^{100} \frac{\bar{B}_3(x)\,dx}{x^5} \\
&\quad = \frac{999}{6\cdot 10^6} - 4\int_{10}^{100} \frac{\bar{B}_3(x)\,dx}{x^5}.
\end{aligned}$$

The first term is 1.665×10^{-4} and the second term is much smaller, as can be seen by an alternating series technique similar to the one used to estimate (4). Note first that $B_3(x)$ is a polynomial of degree three which is zero at $x = 0$ and $x = \frac{1}{2}$ [by direct evaluation] and at $x = 1$ [because (6) shows that $B_n(1) = B_n(0)$ for $n \geq 2$]. This accounts for all its zeros and proves that $B_3(x) = x(x - \frac{1}{2})(x - 1)$. Thus it is positive for $0 < x < \frac{1}{2}$ and negative for $\frac{1}{2} < x < 1$. These positive and negative "bumps" are symmetrical because $B_3(1 - x) = (1 - x)(1 - x - \frac{1}{2})(1 - x - 1) = -B_3(x)$, so the graph of $\bar{B}_3(x)$

is a wave consisting of positive bumps on $(n, n+\frac{1}{2})$ and symmetrically negative bumps on $(n+\frac{1}{2}, n+1)$. Since x^{-5} decreases, this shows immediately that

$$0 \le 4\int_{10}^{100} \frac{\bar{B}_3(x)\,dx}{x^5} \le 4\int_{10}^{10.5} \frac{\bar{B}_3(x)\,dx}{x^5}.$$

Now

$$\begin{aligned} 4\int_{10}^{10.5} \frac{\bar{B}_3(x)\,dx}{x^5} &\le \frac{4}{10^5}\int_{10}^{10.5} \bar{B}_3(x)\,dx = \frac{4}{10^5}\int_0^{1/2} B_3(t)\,dt \\ &= \frac{4}{10^5}\left[\frac{1}{4}t^4 - \frac{1}{2}t^3 + \frac{1}{4}t^2\right]\Big|_0^{1/2} = \frac{4}{10^5}\left[\frac{1}{64} - \frac{1}{16} + \frac{1}{16}\right] \\ &= \frac{1}{10^5 \cdot 16} = 6.25 \times 10^{-7}. \end{aligned}$$

Thus

$$S = 0.09505 + 1.665 \times 10^{-4} - 4\int_{10}^{100} \frac{\bar{B}_3(x)\,dx}{x^5}$$

lies in the range

$$0.095215875 \le S \le 0.0952165$$

which gives S to six places.

The integration by parts can of course be carried farther to put (3) in the form

$$\begin{aligned} (8) \qquad \sum_{n=M}^{N} f(n) &= \int_M^N f(x)\,dx + \frac{1}{2}[f(M) + f(N)] + \frac{B_2(0)}{2}f'(x)\Big|_M^N \\ &\quad - \frac{B_3(0)}{2\cdot 3}f''(x)\Big|_M^N + \cdots + (-1)^k \frac{B_k(0)}{k!}f^{(k-1)}(x)\Big|_M^N \\ &\quad + (-1)^{k+1}\frac{1}{k!}\int_M^N \bar{B}_k(x) f^{(k)}(x)\,dx. \end{aligned}$$

If k is odd, then $\bar{B}_k(x)$ is an oscillating function similar to $\bar{B}_3(x)$ and the integral in this formula can be estimated, provided $f^{(k)}(x)$ is monotone, by an alternating series technique like the one used above. To prove that $\bar{B}_{2\nu+1}(x)$ has this oscillating character, note that the identities

$$\begin{aligned} \int_x^{x+1} B_n(1-t)\,dt &= -\int_{1-x}^{-x} B_n(u)\,du = \int_{-x}^{-x+1} B_n(u)\,du \\ &= (-x)^n = (-1)^n \int_x^{x+1} B_n(t)\,dt \end{aligned}$$

and

$$\begin{aligned} \int_x^{x+(1/2)} B_n(2t)\,dt &= \tfrac{1}{2}\int_{2x}^{2x+1} B_n(u)\,du = \tfrac{1}{2}(2x)^n = 2^{n-1}\int_x^{x+1} B_n(t)\,dt \\ &= 2^{n-1}\int_x^{x+(1/2)} [B_n(t) + B_n(t+\tfrac{1}{2})]\,dt \end{aligned}$$

imply

$$B_n(1-x) = (-1)^n B_n(x),$$
$$B_n(2x) = 2^{n-1}[B_n(x) + B_n(x+\tfrac{1}{2})]. \tag{9}$$

Thus $B_{2\nu+1}(1-x) = -B_{2\nu+1}(x)$, $B_{2\nu+1}(\frac{1}{2}) = 0$, $B_{2\nu+1}(0) = 2^{2\nu}[B_{2\nu+1}(0) + 0]$, $B_{2\nu+1}(0) = 0$; that is,

$$B_{2\nu+1}(0) = B_{2\nu+1}(\tfrac{1}{2}) = 0 \qquad (\nu = 1, 2, \ldots). \tag{10}$$

Now there are no zeros of $B_{2\nu+1}(x)$ between 0 and $\frac{1}{2}$ because such a zero would imply two zeros of the derivative $(2\nu+1)B_{2\nu}(x)$ between 0 and $\frac{1}{2}$, which would in turn imply a zero of $(2\nu+1)2\nu B_{2\nu-1}(x)$ and hence a zero of $B_{2\nu-1}(x)$; repeating this process one would ultimately arrive at the conclusion that $B_3(x)$ had a zero between 0 and $\frac{1}{2}$, which it does not; hence neither does $B_{2\nu+1}(x)$. Thus $B_{2\nu+1}(x)$ has one sign on $(0, \frac{1}{2})$; by $B_{2\nu+1}(1-x) = -B_{2\nu+1}(x)$ it has the opposite sign on $(\frac{1}{2}, 1)$. Therefore $\bar{B}_{2\nu+1}(x)$ oscillates as was to be shown.

This also shows, in passing, that many of the terms of (8) are zero, namely, the terms containing $B_{2\nu+1}(0)$. To apply formula (8), one must of course find the values of the constants $B_{2\nu}(0)$. This is accomplished by proving that *the constants $B_{2\nu}(0)$ coincide with the Bernoulli numbers $B_{2\nu}$* defined in Section 1.5. To see this, apply formula (8) in the case $f(x) = e^{-hx}$, $M = 0$, $N = \infty$ to find

$$1 + e^{-h} + e^{-2h} + \cdots = \int_0^\infty e^{-hx}\,dx + \frac{1}{2} - \frac{B_2(0)}{2}(-h) + \frac{B_3(0)}{3!}(-h)^2 + \cdots - (-1)^k \frac{B_k(0)}{k!}(-h)^{k-1} + (-1)^{k+1}(-h)^k \frac{1}{k!}\int_0^\infty \bar{B}_k(x)e^{-hx}\,dx,$$

$$\frac{1}{1-e^{-h}} = \frac{1}{h} + \frac{1}{2} + \frac{B_2(0)}{2}h + \frac{B_3(0)}{3!}h^2 + \cdots + \frac{B_k(0)}{k!}h^{k-1} - \frac{h^k}{k!}\int_0^\infty \bar{B}_k(x)e^{-hx}\,dx,$$

$$\sum_{n=0}^{\infty} \frac{B_n}{n!}(-h)^n = \frac{-h}{e^{-h}-1} = h \cdot \frac{1}{1-e^{-h}} = 1 + \frac{1}{2}h + \frac{B_2(0)}{2}h^2 + \frac{B_3(0)}{3!}h^3 + \cdots + \frac{B_k(0)}{k!}h^k - \frac{h^{k+1}}{k!}\int_0^\infty \bar{B}_k(x)e^{-hx}\,dx.$$

This formula is valid for small positive h, and, provided k is odd, the absolute value of the last term is at most $(h^{k+1}/k!)\left|\int_0^{1/2} B_k(t)\,dt\right| \leq \text{const}\, h^{k+1}$. Thus the polynomial

$$\sum_{n=0}^{k} [(-1)^n B_n - B_n(0)]\frac{h^n}{n!} = p(h)$$

has the property that $p(h)/h^{k+1}$ is bounded as $h \downarrow 0$, which proves $p(h) \equiv 0$, $B_n(0) \equiv (-1)^n B_n$ for $n \leq k$. Since k was arbitrary and since only even† values of n are at issue, this proves the theorem.

Putting these facts together gives the *Euler–Maclaurin summation formula*

$$\sum_{n=M}^{N} f(n) = \int_M^N f(x)\,dx + \frac{1}{2}[f(M) + f(N)] + \frac{B_2}{2} f'(x)\Big|_M^N + \frac{B_4}{4!} f'''(x)\Big|_M^N + \cdots + \frac{B_{2\nu}}{(2\nu)!} f^{(2\nu-1)}(x)\Big|_M^N + R_{2\nu},$$

where the B_n are the Bernoulli numbers, where $f(x)$ is any function which has $2\nu + 1$ continuous derivatives on $[N, M]$, and where $R_{2\nu}$ is given by either of the formulas

$$R_{2\nu} = \frac{-1}{(2\nu)!} \int_M^N \bar{B}_{2\nu}(x) f^{(2\nu)}(x)\,dx$$

or

$$R_{2\nu} = \frac{1}{(2\nu+1)!} \int_M^N \bar{B}_{2\nu+1}(x) f^{(2\nu+1)}(x)\,dx$$

in which $\bar{B}_k(x) = B_k(x - [x])$. [The two forms of $R_{2\nu}$ are obtained by setting $k = 2\nu$ and $k = 2\nu + 1$, respectively, in (8).]

Continuing with the example of the sum S in (1), the next few terms are

$$\frac{B_4}{4!} \cdot \frac{(-4)(-3)(-2)}{x^5}\Big|_{10}^{100} = -\frac{1}{30}\frac{10^5 - 1}{10^{10}} = -\frac{33333}{10^{11}} = -3.3333 \times 10^{-7}$$

$$\frac{B_6}{6!} \frac{(-6)(-5)\cdots(-2)}{x^7}\Big|_{10}^{100} = \frac{1}{42}\frac{10^7 - 1}{10^{14}} = (2.3809\cdots) \times 10^{-9}$$

$$\frac{B_8}{8!} \frac{(-1)8!}{x^9}\Big|_{10}^{100} = -\frac{1}{30}\frac{10^9 - 1}{10^{18}} = (-3.33\cdots) \times 10^{-11}$$

which gives the approximation 0.095 216 169 017 6 to the value of S. The

0.095 05	−0.000 000 333 33
0.000 166 5	−0. 33 3
0.000 000 002 380 9	−0.000 000 333 363 3
0.095 216 502 380 9	
−0.000 000 333 363 3	
$S \sim$ 0.095 216 169 017 6	

Computation of the approximation to S.

†For odd values of n this gives an alternative proof that $B_3, B_5, B_7, \ldots$ are all zero. See Section 1.6.

magnitude of the error in this approximation is

$$\left|\frac{1}{9!}\int_{10}^{100} \bar{B}_9(x)\frac{(-1)10!}{x^{11}}\,dx\right| \le \frac{1}{10^{11}}\left|\int_0^{1/2} 10B_9(t)\,dt\right|$$
$$= 10^{-11}\left|B_{10}\left(\frac{1}{2}\right) - B_{10}(0)\right|$$
$$= 10^{-11}\left|B_{10}\left(\frac{1}{2}\right) + B_{10}(0) - 2B_{10}(0)\right|$$
$$= 10^{-11}\left|2^{-9}B_{10}(0) - 2B_{10}(0)\right|$$
$$\le 2B_{10}\cdot 10^{-11}.$$

Rather than use this error estimate directly, however, it is more effective to note that it says that the magnitude of the error cannot be more than about *twice the size of the first term omitted*, and that the same estimate would apply no matter where the series was truncated. Since the terms are still getting smaller—$B_{10} = 5/66$, $B_{12} = -691/2730$ so the next two are about 8 in the thirteenth place and 3 in the fourteenth place—this implies that *the error is of the order of magnitude of the first term omitted* because when this term is included, the error is reduced to an amount much smaller than this term (namely, to an amount comparable to the next term). Thus one can be confident that the error in the above approximation to S is *less than one in the twelfth place*, that is, $S = 0.095\ 216\ 169\ 018 \pm 1$.

It is a general rule of thumb in applying the Euler–Maclaurin summation formula that *as long as the terms are decreasing rapidly in size, the bulk of the error is in the first term omitted.* In order to give a rigorous proof of this fact in specific cases it is not necessary to estimate $|R_{2\nu}|$ in any refined way but merely to give a crude estimate showing that it is of an order of magnitude comparable to the first term omitted at most. Then the error has absolute value at most

$$|R_{2\nu}| = \left|\frac{B_2}{2!}f^{(2\nu-1)}\Big|_N^M + R_{2\nu+2}\right| \le |\text{first term omitted}| + |R_{2\nu+2}|$$

and $|R_{2\nu+2}|$ is comparable to the second term omitted, hence much smaller than the first term omitted.

However, it must be observed that the terms do *not* continue to decrease indefinitely [except for very special functions f such as $f(x) = e^{-hx}$ for small h] and that in fact they ultimately grow without bound. To see this, it suffices to combine Euler's formula

$$\zeta(2\nu) = (2\pi)^{2\nu}(-1)^{\nu+1}B_{2\nu}/2\cdot(2\nu)!$$

[see (2) of Section 1.5] with the trivial observation that $\zeta(2\nu) = 1 + 2^{-2\nu} + 3^{-2\nu} + \cdots$ is only slightly larger than one for ν large; hence

$$B_{2\nu} \sim \pm 2\cdot(2\nu)!(2\pi)^{-2\nu},$$

and in the example the term containing B_{2v} is roughly $\pm 2\cdot(2v)!(2\pi)^{-2v}\cdot 10^{-2v-1}$ which decreases quite rapidly at first but which ultimately grows without bound. This shows that there is a limit to the degree of accuracy with which one can evaluate S by extending the above process, and that this limit is roughly determined by the minimum value of $\Pi(x)(20\pi)^{-x}\cdot\frac{1}{5}$ for $x \geq 0$. If for any reason greater accuracy is required, then the first few terms can be summed separately, say $10^{-2} + 11^{-2} + 12^{-2} + 13^{-2} + 14^{-2}$, and the remaining sum evaluated by Euler–Maclaurin summation, which is now much more accurate because the denominators 10^{2v+1} are replaced by 15^{2v+1}. [This observation is of crucial importance in the evaluation of $\zeta(s)$ by Euler–Maclaurin summation in Section 6.4. After all, the sum S is essentially $\zeta(2)$ except that the first nine terms and all terms past the hundredth are missing.]

In summary, the Euler–Maclaurin summation formula says that

$$\sum_{n=M}^{N} f(n) \sim \int_M^N f(x)\,dx + \frac{1}{2}[f(M) + f(N)] + \sum_{j=1}^{v} \frac{B_{2j}}{(2j)!} f^{(2j-1)}(x)\Big|_M^N.$$

In many examples the terms of the series on the right are at first rapidly decreasing in size and the bulk of the error in the approximation is accounted for by the first term omitted. In any case the error is precisely equal to

$$R_{2v} = \frac{1}{(2v+1)!}\int_M^N \bar{B}_{2v+1}(x) f^{(2v+1)}(x)\,dx,$$

and when $f^{(2v+1)}$ is monotone this leads to a simple estimate of $|R_{2v}|$ using the fact that $\bar{B}_{2v+1}(x)$ alternates in sign.

6.3 EVALUATION OF Π BY EULER–MACLAURIN SUMMATION. STIRLING'S SERIES

To evaluate $\Pi(s)$, it of course suffices to evaluate $\log \Pi(s)$. Now if s is a positive integer, say $s = N$, then $\log \Pi(N) = \sum_1^N \log n$ which, by Euler–Maclaurin summation (with $v = 0$), is

$$\int_1^N \log x\,dx + \frac{1}{2}[\log 1 + \log N] + \int_1^N \frac{\bar{B}_1(x)\,dx}{x}.$$

The first integral can be evaluated using $\int \log x\,dx = x\log x - x$. The second integral approaches a limit as $N \longrightarrow \infty$ (by the alternating series test), so it can be written in the form

$$\int_1^N \frac{\bar{B}_1(x)\,dx}{x} = \int_1^\infty \frac{\bar{B}_1(x)\,dx}{x} - \int_N^\infty \frac{\bar{B}_1(x)\,dx}{x}$$

in which the first term is constant and the second term approaches zero as $N \to \infty$. This gives

$$(1) \qquad \log \Pi(N) = \left(N + \frac{1}{2}\right) \log N - N + A - \int_N^\infty \frac{\bar{B}_1(x)\,dx}{x},$$

where A is the constant

$$A = 1 + \int_1^\infty \frac{\bar{B}_1(x)\,dx}{x}.$$

Except for the fact that the constant A must still be evaluated, this gives a simple approximate formula for $\log \Pi(N)$, and it is natural to ask whether there is a similar formula for $\log \Pi(s)$ for other values of s.

As it stands, formula (1) cannot possibly be valid for all real numbers N because the derivative of its right side with respect to N is discontinuous at all integers [because $\bar{B}_1(x)$ is discontinuous]. However, if (1) is rewritten in the form

$$(2) \qquad \log \Pi(s) = \left(s + \frac{1}{2}\right) \log s - s + A - \int_0^\infty \frac{\bar{B}_1(t)\,dt}{t + s},$$

then both sides are well-behaved functions of s for all positive real numbers and, since equality holds whenever s is an integer, it is reasonable to expect that equality will hold for all real $s > 0$. The fact that it does hold follows quite easily from the application of Euler–Maclaurin summation (with $\nu = 0$) to the definition of $\Pi(s)$. Explicitly, from the definition

$$\Pi(s) = \lim_{N\to\infty} \frac{1 \cdot 2 \cdot 3 \cdots N}{(s+1)(s+2)\cdots(s+N)} (N+1)^s$$

of $\Pi(s)$ [see (3) of Section 1.3] it follows that

$$\begin{aligned}
\log \Pi(s) &= \lim_{N\to\infty} \left\{ s \log(N+1) + \sum_{n=1}^{N} \log n - \sum_{n=1}^{N} \log(s+n) \right\} \\
&= \lim_{N\to\infty} \left\{ s \log(N+1) + \int_1^N \log x\,dx + \frac{1}{2} \log N + \int_1^N \frac{\bar{B}_1(x)\,dx}{x} \right. \\
&\quad - \int_1^N \log(s+x)\,dx - \frac{1}{2}[\log(s+1) + \log(s+N)] \\
&\quad \left. - \int_1^N \frac{\bar{B}_1(x)\,dx}{s+x} \right\} \\
&= \lim_{N\to\infty} \left\{ s \log(N+1) + N \log N - N + 1 + \frac{1}{2} \log N + \int_1^N \frac{\bar{B}_1(x)\,dx}{x} \right. \\
&\quad - (s+N) \log(s+N) + (s+N) + (s+1) \log(s+1) \\
&\quad \left. - (s+1) - \frac{1}{2} \log(s+1) - \frac{1}{2} \log(s+N) - \int_1^N \frac{\bar{B}_1(x)\,dx}{s+x} \right\}
\end{aligned}$$

$$= \left(s + \tfrac{1}{2}\right)\log(s+1) + \int_1^\infty \frac{\bar{B}_1(x)\,dx}{x} - \int_1^\infty \frac{\bar{B}_1(x)\,dx}{s+x}$$
$$+ \lim_{N\to\infty}\left\{s\log(N+1) + \left(N+\tfrac{1}{2}\right)\log N - \left(s+N+\tfrac{1}{2}\right)\log(s+N)\right\}$$
$$= \left(s + \tfrac{1}{2}\right)\log(s+1) + (A-1) - \int_1^\infty \frac{\bar{B}_1(x)\,dx}{s+x}$$
$$+ \lim_{N\to\infty}\left\{s\log\frac{N+1}{N+s} - \left(N+\tfrac{1}{2}\right)\log\left(1+\frac{s}{N}\right)\right\}$$
$$= \left(s + \tfrac{1}{2}\right)\log(s+1) + A - 1 - \int_1^\infty \frac{\bar{B}_1(x)\,dx}{s+x} - s.$$

This differs from the right side of (2) by

$$\left(s+\tfrac{1}{2}\right)\log\left(\frac{s+1}{s}\right) - 1 + \int_0^1 \frac{\bar{B}_1(t)\,dt}{s+t}$$
$$= \left(s+\tfrac{1}{2}\right)\log\left(\frac{s+1}{s}\right) - 1 + \int_0^1 \frac{t-\frac{1}{2}}{s+t}\,dt$$
$$= \left(s+\tfrac{1}{2}\right)\log\left(\frac{s+1}{s}\right) - 1 + \int_0^1 \frac{t+s-s-\frac{1}{2}}{s+t}\,dt$$
$$= \left(s+\tfrac{1}{2}\right)\log\left(\frac{s+1}{s}\right) - 1 + \int_0^1 dt - \left(s+\tfrac{1}{2}\right)\int_0^1 \frac{dt}{s+t}$$
$$= 0$$

which shows that (2) is indeed true for all positive real numbers s.

Formula (2) can be combined with the Legendre relation

$$\Pi(2s) = \frac{1}{\pi^{1/2}}2^{2s}\Pi(s)\Pi\left(s-\tfrac{1}{2}\right)$$

[see (7) of Section 1.3] to give† the value of the constant A. To this end let (2) be rewritten in the form

$$\Pi(s) = s^{s+(1/2)}e^{-s}e^{A}r(s),$$

where

$$r(s) = \exp\left[-\int_0^\infty \frac{\bar{B}_1(t)\,dt}{s+t}\right].$$

†An alternative method of proving $A = \frac{1}{2}\log 2\pi$, not relying on the Legendre relation, is given later in this section.

Then $r(s) \longrightarrow 1$ as $s \longrightarrow \infty$, and the Legendre relation says

$$(2s)^{2s+(1/2)}e^{-2s}e^{A}r(2s)$$
$$= \pi^{-1/2}2^{2s}s^{s+(1/2)}e^{-s}e^{A}r(s)\left(s-\frac{1}{2}\right)^{s}e^{-s+(1/2)}e^{A}r\left(s-\frac{1}{2}\right)$$
$$2^{1/2}\left(1-\frac{1}{2s}\right)^{-s}e^{-1/2}\pi^{1/2}\frac{r(2s)}{r(s)r(s-\frac{1}{2})} = e^{A}.$$

As $s \longrightarrow \infty$ the left side approaches $2^{1/2}e^{1/2}e^{-1/2}\pi^{1/2}\cdot 1 = (2\pi)^{1/2}$, so $A = \frac{1}{2}\log 2\pi$. Thus (2) says

$$\log \Pi(s) = \left(s+\frac{1}{2}\right)\log s - s + \frac{1}{2}\log 2\pi - \int_0^\infty \frac{\bar{B}_1(x)\,dx}{s+x}$$

or, if the last term is integrated by parts a number of times,

$$\log \Pi(s) = \left(s+\frac{1}{2}\right)\log s - s + \frac{1}{2}\log 2\pi + \frac{B_2}{2s} + \frac{B_4}{4\cdot 3\cdot s^3} + \cdots + \frac{B_{2\nu}}{2\nu(2\nu-1)s^{2\nu-1}} + R_{2\nu}, \tag{3}$$

where

$$R_{2\nu} = -\int_0^\infty \frac{\bar{B}_{2\nu}(x)\,dx}{2\nu(s+x)^{2\nu}} = -\int_0^\infty \frac{\bar{B}_{2\nu+1}(x)\,dx}{(2\nu+1)(s+x)^{2\nu+1}}.$$

This formula, which is known as *Stirling's series*†, is very effective for finding the approximate numerical value of $\log \Pi(s)$.

For example, consider the case $s = 10$. Then

$$\frac{B_2}{2s} = \frac{1}{6\cdot 2\cdot 10} = \frac{1}{120} = 8.3333\cdots \times 10^{-3},$$
$$\frac{B_4}{4\cdot 3\cdot s^3} = \frac{-1}{30\cdot 4\cdot 3\cdot 10^3} = \frac{-1}{36\cdot 10^4} = -2.7777\cdots \times 10^{-6},$$
$$\frac{B_6}{6\cdot 5\cdot s^5} = \frac{1}{42\cdot 6\cdot 5\cdot 10^5} = \frac{1}{126\cdot 10^6} = 7.936507\cdots \times 10^{-9},$$
$$\frac{B_8}{8\cdot 7\cdot s^7} = \frac{-1}{30\cdot 8\cdot 7\cdot 10^7} = \frac{-1}{168\cdot 10^8} = -5.9523\cdots \times 10^{-11},$$
$$\frac{B_{10}}{10\cdot 9\cdot s^9} = \frac{5}{66\cdot 10\cdot 9\cdot 10^9} = \frac{1}{1188\cdot 10^9} \sim 9 \times 10^{-13}.$$

The terms are still decreasing rapidly and the "rule of thumb" of the previous section would lead one to believe that if the B_{10} term is the first one

†It is named for James Stirling, who published it [S7] in 1730. The integral formula for the remainder $R_{2\nu}$ was published by Stieltjes [S5] in 1889 and does not seem to have been known, or at least used, earlier than that.

omitted, then the error will not be much larger than 9×10^{-13}, so the answer will be correct to 12 places. Now in fact

$$\begin{aligned}|R_{10}| &= \left|\int_0^\infty \frac{\bar{B}_{11}(x)\,dx}{11\cdot(10+x)^{11}}\right| \le \left|\int_0^{1/2} \frac{B_{11}(x)\,dx}{11\cdot(10+x)^{11}}\right| \\ &\le \frac{1}{12\cdot 11\cdot 10^{11}}\left|\int_0^{1/2} 12B_{11}(x)\,dx\right| \\ &= \frac{1}{12\cdot 11\cdot 10^{11}}\left|B_{12}\left(\frac{1}{2}\right) - B_{12}(0)\right| = \frac{|B_{12}|(2-2^{-11})}{12\cdot 11\cdot 10^{11}} \\ &\le \frac{2\cdot 691}{12\cdot 11\cdot 10^{11}\cdot 2730},\end{aligned}$$

so $|R_{10}|$ is very much smaller than the B_{10} term, and hence $|R_8| \sim 9 \times 10^{-13}$ as expected. Thus $\log \Pi(10) \sim 15.10441\ 25730\ 7470$ is accurate to 12 places.

(10.5) log 10 =	24.17714 34764 3748
−10 =	−10.
$+\frac{1}{2}\log \pi$ =	+0.57236 49429 2470
$+\frac{1}{2}\log 2$ =	+0.34657 35902 7997
B_2 term =	+0.00833 33333 3333
B_4 term =	−0.00000 27777 7777
B_6 term =	+0.00000 00079 3651
B_8 term =	−0.00000 00000 5952
$\log \Pi(10) \sim$	15.10441 25730 7470

Computation of the approximation to $\log \Pi(10)$.

Of course, if $s > 10$, then the terms of Stirling's series decrease even faster at first and it is even easier to compute $\log \Pi(s)$ with 10- or 12-place accuracy. In fact, Stirling's series is so effective a means of computation that one tends to forget that the series does not converge and that there is a limit to the accuracy with which $\log \Pi(s)$ can be computed using it. Nonetheless, the terms of Stirling's series do ultimately grow without bound as $\nu \longrightarrow \infty$, and one cannot expect to reduce the error in the approximation to less than the size of the smallest term of the series. If it is desired to compute $\log \Pi(s)$ with greater accuracy than is possible with Stirling's series, then the formula

$$\begin{aligned}\log \Pi(s) = \log \Pi(s+N) &- \log(s+1) \\ &- \log(s+2) - \cdots - \log(s+N)\end{aligned}$$

can be used. Given $\epsilon > 0$ there is an N such that Stirling's series can be used to compute $\log \Pi(s+N)$ with an error of less than ϵ; since the other loga-

rithms can then be computed by elementary means, this makes possible the evaluation of $\log \Pi(s)$ with any prescribed degree of accuracy. In the numerical analysis of the roots ρ it will not be necessary to use this technique, however, because Stirling's series itself gives the needed values with sufficient accuracy.

Once $\log \Pi(s)$ is found, $\Pi(s)$ can of course be found simply by exponentiating. Note that if $\log \Pi(s)$ is found to 12 decimal places, then $\Pi(s)$ is known to within a *factor* of $\exp(\pm 10^{-12}) \sim 1 \pm 10^{-12}$; so it is the *relative* error which is small, that is, the error divided by the value is less than 10^{-12}. Since $\Pi(s)$ is very large for large s, it is well to keep in mind in evaluating $\Pi(s)$ using Stirling's series that a small relative error may still mean a large absolute error. However, in the numerical analysis of the roots ρ only $\log \Pi(s)$—and in fact only $\operatorname{Im} \log \Pi(s)$—will need to be evaluated, so these considerations will not be necessary.

What *will* be necessary in the numerical analysis of the roots ρ is the use of Stirling's series for complex values of s. Let the "slit plane" be the set of all complex numbers other than the negative reals and zero. Then all terms of Stirling's series (3) are defined throughout the slit plane and are analytic functions of s. (This is obvious for all terms except the integral for R_0. However, as will be shown below, this integral too is convergent for all s in the slit plane.) Since (3) is true for positive real s, the theory of analytic continuation implies that it is true throughout the slit plane—in Riemann's terminology formula (3) for $\log \Pi(s)$ remains valid throughout the slit plane. However, the alternating series method of estimating $|R_{2\nu}|$ cannot be used when s is not real, so some other method of estimating $|R_{2\nu}|$ is needed if Stirling's series is to be used to compute $\log \Pi(s)$ for complex s. The following estimate was given by Stieltjes [S5]:

The objective is to show that the error at any stage is comparable in magnitude to the first term omitted. To this end, let the $B_{2\nu}$ term be the first term omitted and consider the resulting error

$$R_{2\nu-2} = -\int_0^\infty \frac{\bar{B}_{2\nu-1}(x)\,dx}{(2\nu-1)(s+x)^{2\nu-1}}.$$

The function $B_{2\nu} - \bar{B}_{2\nu}(x)$ is an antiderivative of $-2\nu\bar{B}_{2\nu-1}(x)$ which is zero at $x = 0$; hence by integration by parts

$$R_{2\nu-2} = \int_0^\infty \frac{[B_{2\nu} - \bar{B}_{2\nu}(x)]\,dx}{2\nu(s+x)^{2\nu}}.$$

[This integration by parts can be justified, even in the case $\nu = 1$ where $\bar{B}_{2\nu-1}(x)$ is discontinuous, by writing the integral for $R_{2\nu-2}$ as a sum over n of

Riemann integrals, as in the previous section.] Thus

$$|R_{2\nu-2}| \le \int_0^\infty \frac{|B_{2\nu} - \bar{B}_{2\nu}(x)|\,dx}{2\nu|s+x|^{2\nu}}.$$

Now $B_{2\nu} - B_{2\nu}(x)$ is zero at $x = 0$ and has only one extremum in the interval $0 < x < 1$, namely, at $x = \frac{1}{2}$ where its derivative $-2\nu B_{2\nu-1}(x)$ is zero. This implies that $B_{2\nu} - \bar{B}_{2\nu}(x)$ never changes sign. Since $\bar{B}_{2\nu}(x)$ has zeros—namely, at the extrema of $\bar{B}_{2\nu+1}$—the sign of $B_{2\nu} - \bar{B}_{2\nu}(x)$ is always the same as the sign of $B_{2\nu}$, which by Euler's formula for $\zeta(2\nu)$ [(2) of Section 1.5] is $(-1)^{\nu+1}$. Therefore the numerator of the above integral can be rewritten

$$|B_{2\nu} - \bar{B}_{2\nu}(x)| = (-1)^{\nu+1}[B_{2\nu} - \bar{B}_{2\nu}(x)].$$

On the other hand, it is a simple calculus problem to show that $|s+x|(|s|+x)^{-1}$ for $s = re^{i\theta}$ in the slit plane $\{-\pi < \theta < \pi\}$ assumes its minimum value for $x \ge 0$ at $x = |s|$ where it is $\cos(\theta/2)$. Thus

$$\frac{1}{|s+x|} = \frac{1}{|s|+x} \cdot \frac{|s|+x}{|s+x|} \le \frac{1}{(|s|+x)\cos(\theta/2)}$$

and

$$\begin{aligned}
|R_{2\nu-2}| &\le \int_0^\infty \frac{(-1)^{\nu+1}[B_{2\nu} - \bar{B}_{2\nu}(x)]\,dx}{2\nu\cos^{2\nu}(\theta/2)(|s|+x)^{2\nu}} \\
&= \int_0^\infty \frac{(-1)^{\nu+1}B_{2\nu}\,dx}{2\nu\cos^{2\nu}(\theta/2)(|s|+x)^{2\nu}} \\
&\quad - \int_0^\infty \frac{(-1)^{\nu+1}\bar{B}_{2\nu}(x)\,dx}{2\nu\cos^{2\nu}(\theta/2)(|s|+x)^{2\nu}} \\
&= \frac{1}{\cos^{2\nu}(\theta/2)} \cdot \frac{|B_{2\nu}|}{2\nu(2\nu-1)|s|^{2\nu-1}} \\
&\quad - \frac{1}{\cos^{2\nu}(\theta/2)} \int_0^\infty \frac{(-1)^{\nu+1}\bar{B}_{2\nu+1}(x)\,dx}{(2\nu+1)(|s|+x)^{2\nu+1}}.
\end{aligned}$$

By the alternating series method, the second integral is easily seen to have the same sign as $(-1)^{\nu+1}\bar{B}_{2\nu+1}(x)$ on $\{0 < x < \frac{1}{2}\}$. Since this function starts at $x = 0$ with the value zero and the derivative $(-1)^{\nu+1}(2\nu+1)B_{2\nu} > 0$, this sign is $+$ and the inequality becomes stronger if the last term is deleted

$$|R_{2\nu-2}| \le \left(\frac{1}{\cos(\theta/2)}\right)^{2\nu} \left|\frac{B_{2\nu}}{(2\nu)(2\nu-1)s^{2\nu-1}}\right|$$

which is the desired inequality. In words, *if the $B_{2\nu}$ term is the first term omitted in Stirling's series, then the magnitude of the error is at most* $\cos^{-2\nu}(\theta/2)$ *times the magnitude of the first term omitted.*

In the special case $\nu = 1$, $\operatorname{Re} s \ge 0$, $\cos(\theta/2) \ge \sqrt{2}/2$, this gives the estimate used in Section 3.4, namely, that the magnitude of the error in the approximation $\log \Pi(s/2) \sim (s+\frac{1}{2})\log s - s + \frac{1}{2}\log 2\pi$ is at most $(6|s|)^{-1}$ in the halfplane $\operatorname{Re} s \ge 0$. More generally, it shows that if s is a real number $s > 0$, the magnitude of the error is at most the magnitude of the first term

omitted (each new term overshoots the mark and the true value lies between any two consecutive partial sums of Stirling's series) and if s is any complex number in the slit plane, then the magnitude of the error is *comparable* to the magnitude of the first term omitted unless s is quite near the negative real axis. This implies that, unless s is near the negative real axis, one is reasonably safe in using the rule of thumb that when the terms are rapidly decreasing in magnitude, the first term omitted accounts for the bulk of the error (because when it is included, the error is reduced to the next lower order of magnitude).

The constant A in Stirling's formula can be evaluated quite easily by applying Stirling's formula on the imaginary axis and combining the result with the estimate of $\operatorname{Re} \log \Pi$ on the imaginary axis which follows from

$$\sin \pi s = \pi s/\Pi(s)\Pi(-s)$$

[see (6) of Section 1.3]. Since Π is real on the real axis, the reflection principle implies $\Pi(\bar{s}) = \overline{\Pi(s)}$, and with $s = it$, the above gives

$$|\Pi(it)|^2 = \frac{\pi it}{\sin \pi it} = \frac{2\pi t}{e^{\pi t} - e^{-\pi t}},$$

$$2 \log |\Pi(it)| = \log 2\pi + \log t - \log e^{\pi t} - \log(1 - e^{-2\pi t}),$$

$$\operatorname{Re} \log \Pi(it) = \frac{1}{2} \log 2\pi + \frac{1}{2} \log t - \frac{\pi t}{2} - \frac{1}{2} \log(1 - e^{-2\pi t}).$$

On the other hand by Stirling's formula (2)

$$\operatorname{Re} \log \Pi(it) = \operatorname{Re}\left\{\left(it + \frac{1}{2}\right) \log it - it + A - \int_0^\infty \frac{\bar{B}_1(u)\, du}{u + it}\right\}$$

$$= \frac{1}{2} \log t - t\frac{\pi}{2} + A - \int_0^\infty \frac{u\bar{B}_1(u)\, du}{u^2 + t^2};$$

hence

$$A = \frac{1}{2} \log 2\pi - \frac{1}{2} \log(1 - e^{-2\pi t}) + \int_0^\infty \frac{u\bar{B}_1(u)\, du}{u^2 + t^2}$$

and the desired result $A = \frac{1}{2} \log 2\pi$ follows by taking the limit as $t \longrightarrow \infty$.

The derivative of Stirling's series is the series

$$\frac{\Pi'(s)}{\Pi(s)} = \log s + \frac{1}{2s} - \frac{B_2}{2s^2} - \frac{B_4}{4s^4} - \cdots - \frac{B_{2\nu}}{2\nu s^{2\nu}} + R'_{2\nu}, \tag{4}$$

where

$$R'_{2\nu} = \int_0^\infty \frac{\bar{B}_{2\nu}(x)\, dx}{(s + x)^{2\nu+1}} = \int_0^\infty \frac{\bar{B}_{2\nu+1}(x)\, dx}{(s + x)^{2\nu+2}}.$$

This gives a precise form of the estimate $\Pi'(s)/\Pi(s) \sim \log s$ which was used in de la Vallée Poussin's proof in Section 5.2. Specifically, to prove that there is a $K > 0$ such that

$$\operatorname{Re} \frac{\Pi'[(\sigma + it)/2]}{\Pi[(\sigma + it)/2]} \leq 2 \log t$$

in the region $1 \leq \sigma \leq 2$, $t \geq K$ as was claimed in Section 5.2, it suffices to estimate R_0' to find

$$|R_0'| \leq \frac{1}{\cos^3(\theta/2)} \cdot \left|\frac{B_2}{2s^2}\right|,$$

$$\left|\frac{\Pi'(s)}{\Pi(s)} - \log s\right| \leq \frac{1}{2|s|} + \frac{B_2}{\cos^3(\theta/2)2|s|^2}.$$

For s in the halfplane $\operatorname{Re} s \geq 0$ and for $|s|$ sufficiently large, this gives

$$\left|\operatorname{Re}\frac{\Pi'(s)}{\Pi(s)} - \operatorname{Re}\log s\right| \leq \frac{1}{|s|},$$

$$\operatorname{Re}\frac{\Pi'[(\sigma+it)/2]}{\Pi[(\sigma+it)/2]} \leq \log\left|\frac{\sigma+it}{2}\right| + \frac{1}{|s|}$$

from which the desired inequality follows.

6.4 EVALUATION OF ζ BY EULER–MACLAURIN SUMMATION

Euler–Maclaurin summation applied directly to the series $\zeta(s) = \sum_1^\infty n^{-s}$ ($\operatorname{Re} s \geq 1$) does not give a workable method of evaluating $\zeta(s)$ because the remainders are not at all small. [This is the case $N = 1$ of formula (1) below.] However, if Euler–Maclaurin is applied instead to the series $\sum_{n=N}^\infty n^{-s}$, it gives a quite workable means of approximating numerically the sum of this series and hence, since the terms $\sum_{n=1}^{N-1} n^{-s}$ can be summed directly, a workable means of evaluating $\zeta(s)$. Explicitly, if $\operatorname{Re} s > 1$, then

$$\sum_{n=1}^{\infty} n^{-s} - \sum_{n=1}^{N-1} n^{-s} = \sum_{n=N}^{\infty} n^{-s},$$

$$\zeta(s) - \sum_{n=1}^{N-1} n^{-s} = \int_N^\infty x^{-s}\,dx + \tfrac{1}{2}N^{-s} + \int_N^\infty \bar{B}_1(x)(-s)x^{-s-1}\,dx,$$

$$\zeta(s) = \sum_{n=1}^{N-1} n^{-s} + \frac{N^{1-s}}{s-1} + \frac{1}{2}N^{-s} + \frac{B_2}{2}sN^{-s-1} + \cdots + \frac{B_{2\nu}}{(2\nu)!}s(s+1)\cdots(s+2\nu-2)N^{-s-2\nu+1} + R_{2\nu}, \tag{1}$$

where

$$R_{2\nu} = -\frac{s(s+1)\cdots(s+2\nu-1)}{(2\nu)!}\int_N^\infty \bar{B}_{2\nu}(x)x^{-s-2\nu}\,dx$$

$$= -\frac{s(s+1)\cdots(s+2\nu)}{(2\nu+1)!}\int_N^\infty \bar{B}_{2\nu+1}(x)x^{-s-2\nu-1}\,dx.$$

If N is at all large, say N is about the same size as $|s|$, then the terms of the series (1) decrease quite rapidly at first and it is natural to expect that the re-

mainder $R_{2\nu}$ will be quite small. The same method by which Stieltjes estimated the remainder in Stirling's series can be applied to estimate the remainder $R_{2\nu}$ above. It gives

$$\begin{aligned}
|R_{2\nu-2}| &= \left|\frac{s(s+1)\cdots(s+2\nu-2)}{(2\nu-1)!}\int_N^\infty \bar{B}_{2\nu-1}(x)x^{-s-2\nu+1}\,dx\right| \\
&= \left|\frac{s(s+1)\cdots(s+2\nu-1)}{(2\nu)!}\int_N^\infty [B_{2\nu}-\bar{B}_{2\nu}(x)]x^{-s-2\nu}\,dx\right| \\
&\le \left|\frac{s(s+1)\cdots(s+2\nu-1)}{(2\nu)!}\right|\int_N^\infty (-1)^{\nu+1}[B_{2\nu}-\bar{B}_{2\nu}(x)]x^{-\sigma-2\nu}\,dx \\
&= \left|\frac{s(s+1)\cdots(s+2\nu-1)}{(2\nu)!}\right||B_{2\nu}|\int_N^\infty x^{-\sigma-2\nu}\,dx \\
&\quad -\left|\frac{s(s+1)\cdots(s+2\nu-1)}{(2\nu)!}\right|\int_N^\infty \frac{(-1)^{\nu+1}\bar{B}_{2\nu+1}(x)}{(2\nu+1)} \\
&\quad \times(\sigma+2\nu)x^{-\sigma-2\nu-1}\,dx \\
&\le \left|\frac{s(s+1)\cdots(s+2\nu-1)B_{2\nu}N^{-\sigma-2\nu+1}}{(2\nu)!(\sigma+2\nu-1)}\right| \\
&= \left|\frac{s+2\nu-1}{\sigma+2\nu-1}\right||B_{2\nu}\text{ term of (1)}|,
\end{aligned}$$

where $\sigma = \operatorname{Re} s$. In words, *if the $B_{2\nu}$ term is the first term omitted, then the magnitude of the remainder in the series* (1) *is at most* $|s+2\nu-1|(\sigma+2\nu-1)^{-1}$ *times the magnitude of the first term omitted.* This estimate is due to Backlund [B2]. In particular, if s is real, the remainder is less than the first term omitted, every term overshoots, and the actual value always lies between two consecutive partial sums of the series (1).

Although formula (1) is derived from the formula $\zeta(s) = \Sigma n^{-s}$ which is valid only for $\operatorname{Re} s > 1$, it obviously "remains valid," in Riemann's terminology, as long as the integral for $R_{2\nu}$ converges, which is true throughout the halfplane $\operatorname{Re}(s+2\nu+1) > 1$. Since ν is arbitrary, this gives an alternate proof of the fact that $\zeta(s)$ can be continued analytically throughout the s-plane with just one simple pole at $s = 1$ and no other singularities.

For some idea of how formula (1) works in actual practice, consider the case $s = 2$. A good way to proceed is to compute the first several numbers in the sequence $s(s+1)\cdots(s+2\nu-2)B_{2\nu}/(2\nu)!$ and to see how large N must be in order to make the terms of the series (1) decrease rapidly in size. Now

$$\frac{B_2}{2}\cdot 2 = \frac{1}{6} = 0.16666\ldots,$$

$$\frac{B_4}{4!}\cdot 2\cdot 3\cdot 4 = -\frac{1}{30} = -0.03333\ldots,$$

$$\frac{B_6}{6!}\cdot 2\cdot 3\cdot 4\cdot 5\cdot 6 = \frac{1}{42} = 0.0238095\ldots.$$

$1^{-2} =$	1.000 000 00
$2^{-2} =$	0.250 000 00
$3^{-2} =$	0.111 111 11
$4^{-2} =$	0.062 500 00
$5^{-1}(2-1)^{-1} =$	0.200 000 00
$\frac{1}{2}5^{-2} =$	0.020 000 00
B_2 term $=$	0.001 333 33
B_4 term $=$	−0.000 010 66
$\zeta(2) \sim$	1.644 933 78

Computation of approximation to $\zeta(2)$.

With $N = 5$ the last term is divided by $5^7 > 70{,}000$ so it is less than about 3 in the seventh place. Thus the approximation $\zeta(2) \sim 1.644\,933\,78$ is correct to six decimal places. For $s = \frac{1}{2}$ the terms decrease even more rapidly and with N only equal to 4 the B_6 term is just

$$\frac{B_6}{6!} \cdot \left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\left(\frac{7}{2}\right)\left(\frac{9}{2}\right)4^{-11/2} = \frac{1}{2^{21}} = \frac{1}{2}(2^{-10})^2 < \frac{1}{2}(10^{-3})^2,$$

so it does not affect the sixth decimal place and the approximation† $\zeta(\frac{1}{2}) \sim 1.460\,354\,96$ is correct to six decimal places.

$1^{-1/2} =$	1.000 000 00
$2^{-1/2} =$	0.707 106 78
$3^{-1/2} =$	0.577 350 27
$4^{1/2}(\frac{1}{2}-1)^{-1} =$	−4.000 000 00
$\frac{1}{2}4^{-1/2} =$	0.250 000 00
B_2 term $=$	0.005 208 33
B_4 term $=$	−0.000 020 34
$\zeta(\frac{1}{2}) \sim$	−1.460 354 96

Computation of approximation to $\zeta(\frac{1}{2})$.

Consider finally a case in which s is not real, the case $s = \frac{1}{2} + 18i$. Since $|s|$ is considerably larger in this case than in the two previous ones, it is clear that it will be necessary to use a much larger value of N in order to achieve comparable accuracy. On the other hand, the numbers $2^{-s}, 3^{-s}, \ldots$ are quite a bit more difficult to compute when s is complex (since $n^{-s} = n^{-1/2}e^{-i18\log n} = n^{-1/2}[\cos(18\log n) - i\sin(18\log n)]$ each one involves computing a square root, a logarithm, and two trigonometric functions), so this involves a great deal of computation. Rather than carry through all this computation, it will

†The square roots $(0.5)^{1/2}$, $(0.333\ldots)^{1/2}$ can be computed very easily using the usual iteration $x_{n+1} = \frac{1}{2}(x_n + Ax_n^{-1})$ for $A^{1/2}$.

be more illustrative of the technique to settle for less accuracy and to use a smaller value of N. Since $4^{-s} = (2^{-s})^2$ and $6^{-s} = 2^{-s}\cdot 3^{-s}$, the three values 2^{-s}, 3^{-s}, and 5^{-s} are the only values of n^{-s} which need to be computed in order to use (1) with $N = 6$. For this reason, the value $N = 6$ will be used in the computation below.

The first step is to estimate the size of the $B_{2\nu}$ terms in order to determine the degree of accuracy with which the calculations should be carried out. The B_2 term has modulus about $\frac{1}{2}|B_2|\cdot 18\cdot 6^{-3/2} = (2\cdot 2\cdot\sqrt{6})^{-1} \sim 0.1$, the B_4 term modulus about $18\cdot 18\cdot 18/24\cdot 30\cdot 6^{7/2} = 3/8\cdot 10\sqrt{6} \sim 3/196 \sim 0.015$, and the B_6 term modulus about $18\cdot 18\cdot 18\cdot 18\cdot 20/6!\cdot 42\cdot 6^{11/2} = (2^3\cdot 14\cdot\sqrt{6})^{-1} \sim 1/270 \sim 0.0037$. The modulus of the B_8 term is about 0.0037 times

$$\left|\frac{B_8}{B_6\cdot 8\cdot 7}\cdot\left(\frac{11}{2}+18i\right)\left(\frac{13}{2}+18i\right)6^{-2}\right| \sim \frac{42}{30\cdot 8\cdot 7\cdot 6\cdot 6}|6+18i|^2 = \frac{1}{4},$$

so it is about 9 in the fourth place. Then the modulus of the B_{10} term is about 0.0009 times

$$\left|\frac{B_{10}}{B_8\cdot 10\cdot 9}\cdot\left(\frac{15}{2}+18i\right)\left(\frac{17}{2}+18i\right)6^{-2}\right| \sim \frac{5\cdot 30(8^2+18^2)}{66\cdot 10\cdot 9\cdot 6^2}$$
$$\sim \frac{5^2\cdot 2^2(4^2+9^2)}{11\cdot 10\cdot 9\cdot 6^2} \sim \frac{100\cdot 97}{1{,}000\cdot 36}$$
$$\sim 0.27,$$

so it is less than 3 in the fourth place. Thus, by Backlund's estimate of the remainder, if the B_{10} term is the first one omitted, the error is less than $|s+9|(\frac{1}{2}+9)^{-1}\cdot 3\cdot 10^{-4} \sim |1+2i|\cdot 3\cdot 10^{-4} < 7\cdot 10^{-4}$ so the answer is correct to three decimal places at least. The terms are now decreasing fairly slowly; two more terms would be required to obtain one more place, so it is reasonable to quit with the B_8 term. [If more than three-place accuracy were required, the easiest way to achieve it would probably be to compute 7^{-s}, after which $8^{-s} = 2^{-s}\cdot 4^{-s}$, $9^{-s} = (3^{-s})^2$, $10^{-s} = 2^{-s}\cdot 5^{-s}$ are easy and one can use $N = 10$ in (1).] Therefore the calculations will be carried out to five places with the intention of retaining three places in the final answer.

To compute 2^{-s} $(s = \frac{1}{2} + 18i)$ with five-place accuracy, one must have accurate values of $\log 2$ and π, say $\log 2 = 0.693\ 147\ 18$ and $\pi = 3.141\ 592\ 65$. Then $18\log 2 = 4\pi - \theta$, where θ is 0.08972 to five places. The power series for $\cos\theta$ and $\sin\theta$ then give $\cos\theta = 0.99598$, $\sin\theta = 0.08960$ which, combined with the value $2^{-1/2} = 0.707\ 11$ found above, gives $2^{-s} = (0.704\ 27) + (0.063\ 36)i$. Similarly, using accurate values of $\log 3$ and π, one can find $18\log 3 = 6\pi + (\pi/3) - \theta$ where $\theta = 0.12174$ to five places, from which

$$3^{-s} = 3^{-1/2}e^{-i\pi/3}e^{i\theta} = \frac{1}{\sqrt{3}}\left(\frac{1}{2}-\frac{\sqrt{3}}{2}i\right)(\cos\theta + i\sin\theta)$$
$$= 0.34726 - 0.46124i$$

can be computed. Finally, $18 \log 5 = 9\pi + (\pi/4) - \theta$, where $\theta = 0.08985$ so that

$$5^{-s} = -\frac{1}{\sqrt{5}}\left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)(\cos\theta + i\sin\theta) = -0.34333 + 0.28657i.$$

Using these values one can then compute $4^{-s} = 0.49199 + 0.08924i$ and $6^{-s} = 0.27379 - 0.30284i$ with (almost) five-place accuracy. Now

$$\frac{B_2}{2}s\cdot 6^{-s-1} = \left(\frac{1}{144} + \frac{1}{4}i\right)6^{-s} = 0.07761 + 0.06634i,$$

$$\begin{aligned}\frac{B_4}{4!}s(s+1)(s+2)6^{-s-3} &= -\frac{(\frac{1}{2}+18i)(\frac{3}{2}+18i)(\frac{5}{2}+18i)}{30\cdot 24\cdot 6^3}6^{-s}\\ &= \frac{1456\frac{1}{8} + 5728\frac{1}{2}i}{720\cdot 6^3}\cdot 6^{-s}\\ &= 0.01372 + 0.00725i.\end{aligned}$$

The B_6 term is the B_4 term times

$$\frac{B_6(\frac{7}{2}+18i)(\frac{9}{2}+18i)}{B_4\cdot 6\cdot 5\cdot 6^2} = \frac{308\frac{1}{4} - 144i}{1512} = 0.20387 - 0.09524i$$

so it is $0.00349 + 0.00017i$. Similarly the B_8 term is the B_6 term times

$$\frac{B_8(\frac{11}{2}+18i)(\frac{13}{2}+18i)}{B_6\cdot 8\cdot 7\cdot 6^2} = \frac{288\frac{1}{4} - 216i}{1440}$$

which gives $0.00072 - 0.00053i$ as the value of the B_8 term. Adding these values up then gives $\zeta(\frac{1}{2} + 18i) \sim 2.329\,22 - 0.188\,65i$ as the value of $\zeta(\frac{1}{2} + 18i)$ to three decimal places.

$1^{-s} =$	1.000 00
$2^{-s} =$	+0.704 27 + 0.063 36i
$3^{-s} =$	+0.347 26 − 0.461 24i
$4^{-s} =$	+0.491 99 + 0.089 24i
$5^{-s} =$	−0.343 33 + 0.286 57i
$6^{1-s}(s-1)^{-1} =$	−0.103 40 − 0.088 39i
$\frac{1}{2}6^{-s} =$	+0.136 89 − 0.151 42i
B_2 term =	+0.077 61 + 0.066 34i
B_4 term =	+0.013 72 + 0.007 25i
B_6 term =	+0.003 49 + 0.000 17i
B_8 term =	+0.000 72 − 0.000 53i
$\zeta(\frac{1}{2} + 18i) \sim$	2.329 22 − 0.188 65i

Computation of approximation to $\zeta(\frac{1}{2} + 18i)$.

6.5 TECHNIQUES FOR LOCATING ROOTS ON THE LINE

The roots ρ are by definition the zeros of the function $\xi(s) = \Pi(s/2)\pi^{-s/2}(s-1)\zeta(s)$. For any given s, the value of $\xi(s)$ can be computed to any prescribed degree of accuracy by combining the techniques of the preceding two sections. Since $\xi(s)$ is *real valued* on the line $\operatorname{Re} s = \frac{1}{2}$, it can be shown to have zeros on the line by showing that it *changes sign*. This, in a nutshell, is the method by which roots ρ on the line $\operatorname{Re} s = \frac{1}{2}$ will be located in this section.

Consider, then, the problem of determining the sign of $\xi(\frac{1}{2} + it)$. If this function of t is rewritten in the form

$$\begin{aligned}\xi\left(\frac{1}{2}+it\right) &= \frac{s}{2}\Pi\left(\frac{s}{2}-1\right)\pi^{-s/2}(s-1)\zeta(s)\\ &= e^{\log \Pi[(s/2)-1]}\pi^{-s/2}\cdot\frac{s(s-1)}{2}\cdot\zeta(s)\\ &= \left[e^{\operatorname{Re}\log \Pi[(s/2)-1]}\pi^{-1/4}\cdot\frac{-t^2-\frac{1}{4}}{2}\right]\\ &\quad\times\left[e^{i\operatorname{Im}\log \Pi[(s/2)-1]}\pi^{-it/2}\zeta\left(\frac{1}{2}+it\right)\right]\end{aligned}$$

(where $s = \frac{1}{2} + it$), then the determination of its sign can be simplified by the observation that the factor in the first set of brackets is a negative real number; hence that the sign of $\xi(\frac{1}{2} + it)$ is opposite to the sign of the factor in the second set of brackets. The standard notation for this second factor is $Z(t)$, that is,

$$Z(t) = e^{i\vartheta(t)}\zeta(\tfrac{1}{2} + it)$$

where $\vartheta(t)$ is defined by

$$\vartheta(t) = \operatorname{Im}\log\Pi\left(\frac{it}{2}-\frac{3}{4}\right) - \frac{t}{2}\log\pi.$$

If $\vartheta(t)$ and $Z(t)$ are so defined, then *the sign of* $\xi(\frac{1}{2} + it)$ *is opposite to the sign*† *of* $Z(t)$. Thus, to determine the sign of $\xi(\frac{1}{2} + it)$ it suffices to compute $\vartheta(t)$, $\zeta(\frac{1}{2} + it)$ by the methods of the preceding two sections and to combine them to find $Z(t)$.

The computation of $\vartheta(t)$ can be simplified as follows:

$$\begin{aligned}\vartheta(t) &= \operatorname{Im}\left[\log\Pi\left(\frac{it}{2}+\frac{1}{4}\right) - \log\left(\frac{it}{2}+\frac{1}{4}\right)\right] - \frac{t}{2}\log\pi\\ &= \operatorname{Im}\left[\left(\frac{it}{2}-\frac{1}{4}\right)\log\left(\frac{it}{2}+\frac{1}{4}\right) - \left(\frac{it}{2}+\frac{1}{4}\right) + \frac{1}{2}\log 2\pi\right.\\ &\quad\left. + \frac{1}{12\left(\frac{it}{2}+\frac{1}{4}\right)} - \frac{1}{360\left(\frac{it}{2}+\frac{1}{4}\right)^3} + \cdots\right] - \frac{t}{2}\log\pi\end{aligned}$$

†In particular, $Z(t)$ is real when, of course, t is real.

$$= \frac{t}{2}\operatorname{Re}\log\left(\frac{it}{2}+\frac{1}{4}\right) - \frac{1}{4}\operatorname{Im}\log\left(\frac{it}{2}+\frac{1}{4}\right) - \frac{t}{2}$$
$$+ \frac{-\frac{t}{2}}{12\left(\frac{t^2}{4}+\frac{1}{16}\right)} - \frac{\operatorname{Im}\left(-\frac{it}{2}+\frac{1}{4}\right)^3}{360\left(\frac{t^2}{4}+\frac{1}{16}\right)^3} + \cdots - \frac{t}{2}\log\pi$$
$$= \frac{t}{2}\log\left[\left(\frac{t}{2}\right)^2\left(1+\frac{1}{4t^2}\right)\right]^{1/2} - \frac{1}{4}\left[\frac{\pi}{2} - \operatorname{Arctan}\left(\frac{1}{4}\Big/\frac{t}{2}\right)\right] - \frac{t}{2}$$
$$- \frac{1}{6t\left(1+\frac{1}{4t^2}\right)} - \frac{\frac{t^3}{8}+3\left(-\frac{t}{2}\right)\left(\frac{1}{4}\right)^2}{360\left(\frac{t^2}{4}\right)^3\left(1+\frac{1}{4t^2}\right)^3} + \cdots - \frac{t}{2}\log\pi$$
$$= \frac{t}{2}\log\frac{t}{2} + \frac{t}{4}\log\left(1+\frac{1}{4t^2}\right) - \frac{\pi}{8} + \frac{1}{4}\operatorname{Arctan}\left(\frac{1}{2t}\right)$$
$$- \frac{t}{2} - \frac{1}{6t}\left(1+\frac{1}{4t^2}\right)^{-1} - \frac{1}{45t^3}\left(1+\frac{1}{4t^2}\right)^{-3}$$
$$+ \frac{1}{60t^5}\left(1+\frac{1}{4t^2}\right)^{-3} + \cdots - \frac{t}{2}\log\pi$$
$$= \frac{t}{2}\log\frac{t}{2\pi} + \frac{t}{4}\left[\frac{1}{4t^2} - \frac{1}{2}\left(\frac{1}{4t^2}\right)^2 + \cdots\right]$$
$$- \frac{\pi}{8} + \frac{1}{4}\left[\left(\frac{1}{2t}\right) - \frac{1}{3}\left(\frac{1}{2t}\right)^3 + \cdots\right]$$
$$- \frac{t}{2} - \frac{1}{6t}\left[1 - \frac{1}{4t^2} + \cdots\right] - \frac{1}{45t^3}\left[1 - \frac{3}{4t^2} + \cdots\right]$$
$$+ \frac{1}{60t^5}\left(1+\frac{1}{4t^2}\right)^{-3} + \cdots.$$

This gives finally

$$\vartheta(t) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \frac{7}{5760t^3} + \cdots. \tag{1}$$

Since the terms decrease very rapidly for t at all large and since the error is comparable to the first term omitted, it is clear that the error in the approximation

$$\vartheta(t) \sim \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} \tag{2}$$

is very slight. Specifically, the error involved in the above use of Stirling's series is less than

$$\frac{1}{\cos^6\frac{\theta}{2}} \cdot \left|\frac{1}{42\cdot 6\cdot 5\left(\frac{it}{2}+\frac{1}{4}\right)^5}\right| \leq \frac{1}{t^5}\frac{2^3}{42\cdot 6\cdot\left(\frac{1}{2}\right)^5\left(1+\frac{1}{4t^2}\right)^{5/2}}$$
$$\leq \frac{1}{t^5}\cdot\frac{64}{63},$$

and the errors resulting from truncating the alternating series are less than the first term omitted, which gives a total error of less than

$$\frac{t}{4}\cdot\frac{1}{3}\left(\frac{1}{4t^2}\right)^3+\frac{1}{4}\cdot\frac{1}{5}\left(\frac{1}{2t}\right)^5+\frac{1}{6t}\left(\frac{1}{4t^2}\right)^2+\frac{1}{45t^3}\cdot\frac{1}{60t^5}<\frac{1}{t^5},$$

so the total error in the approximation (2) is comfortably less than

$$\frac{7}{5760t^3}+\frac{2}{t^5}.$$

The second term in this estimate is very crude and could be much improved by using more terms of Stirling's series, finding further terms of the series (1), and estimating the error in terms of t^{-7} or t^{-9} to show that the first term $7/5760t^3$ contains the bulk of the error when t is at all large. However, this form of the estimate is quite adequate for the numerical analysis of the roots ρ.

Thus to find the sign of $\xi(\frac{1}{2}+18i)$ one can simply compute

$$\begin{aligned}\vartheta(18) &\sim 9\log\frac{9}{\pi}-9-\frac{\pi}{8}+\frac{1}{48\cdot 18}\\ &= 9.472452-9-0.392699+0.001158\\ &= 0.080911\end{aligned}$$

and combine it with the value of $\zeta(\frac{1}{2}+18i)$ computed in the previous section to find that

$$Z(18)=e^{0.0809i}(2.329-0.189i)=2.337+0.000i$$

is positive and that $\xi(\frac{1}{2}+18i)$ is therefore *negative*. Since $\xi(\frac{1}{2})$ is positive (ξ is positive on the entire real axis), it follows that *there is at least one root ρ on the line segment from $\frac{1}{2}$ to $\frac{1}{2}+18i$*. By computing further values of Z one could obtain more detailed information on the sign of $\xi(\frac{1}{2}+it)$ and therefore more precise information on the location of roots ρ on $\operatorname{Re} s=\frac{1}{2}$. However, the evaluation of Z requires *both* an evaluation of ζ *and* an evaluation of ϑ, and in order to locate roots ρ on $\operatorname{Re} s=\frac{1}{2}$, it suffices to evaluate just ζ—and in fact just $\operatorname{Im}\zeta$—provided one analyzes the result with a certain amount of ingenuity. Consider, for example, in Table IV the values of $\zeta(\frac{1}{2}+it)$ computed with two-place accuracy at intervals of 0.2 from $t=0$ to $t=50$.

Although these values were taken from Haselgrove's tables [H8], it would not be too lengthy to compute them from scratch using Euler–Maclaurin, particularly in view of the fact that there are economies of scale in computing many values of n^{it} ($n=2,3,4,\ldots$) which is the major operation in the Euler–Maclaurin evaluation of ζ. Now examination of this table leads to a few very elementary but very useful observations.

In the first place, the real part of ζ has a strong tendency to be positive. There are brief intervals, 11 in all, where $\operatorname{Re}\zeta$ is negative, but apart from the first one, which is clearly atypical, the longest of them is from 47.2 to 48.0

TABLE IV[a]

t	$\zeta(\frac{1}{2}+it)$	t	$\zeta(\frac{1}{2}+it)$	t	$\zeta(\frac{1}{2}+it)$
0.0	-1.46	9.0	$+1.45+0.19i$	18.0	$+2.33-0.19i$
0.2	$-1.18-0.67i$	9.2	$+1.48+0.14i$	18.2	$+2.27-0.43i$
0.4	$-0.68-0.94i$	9.4	$+1.51+0.08i$	18.4	$+2.17-0.66i$
0.6	$-0.28-0.94i$	9.6	$+1.53+0.02i$	18.6	$+2.02-0.86i$
0.8	$-0.02-0.84i$	9.8	$+1.54-0.04i$	18.8	$+1.84-1.03i$
1.0	$+0.14-0.72i$	10.0	$+1.54-0.12i$	19.0	$+1.62-1.16i$
1.2	$+0.25-0.62i$	10.2	$+1.54-0.19i$	19.2	$+1.38-1.24i$
1.4	$+0.32-0.52i$	10.4	$+1.53-0.26i$	19.4	$+1.13-1.28i$
1.6	$+0.37-0.44i$	10.6	$+1.50-0.34i$	19.6	$+0.88-1.26i$
1.8	$+0.41-0.37i$	10.8	$+1.47-0.42i$	19.8	$+0.65-1.18i$
2.0	$+0.44-0.31i$	11.0	$+1.42-0.49i$	20.0	$+0.43-1.06i$
2.2	$+0.46-0.26i$	11.2	$+1.36-0.56i$	20.2	$+0.25-0.90i$
2.4	$+0.48-0.21i$	11.4	$+1.29-0.62i$	20.4	$+0.11-0.70i$
2.6	$+0.50-0.16i$	11.6	$+1.21-0.67i$	20.6	$+0.02-0.48i$
2.8	$+0.52-0.12i$	11.8	$+1.12-0.71i$	20.8	$-0.02-0.25i$
3.0	$+0.53-0.08i$	12.0	$+1.02-0.75i$	21.0	$-0.01-0.02i$
3.2	$+0.55-0.04i$	12.2	$+0.91-0.76i$	21.2	$+0.06+0.19i$
3.4	$+0.56-0.01i$	12.4	$+0.79-0.76i$	21.4	$+0.18+0.38i$
3.6	$+0.58+0.03i$	12.6	$+0.68-0.75i$	21.6	$+0.34+0.52i$
3.8	$+0.59+0.06i$	12.8	$+0.56-0.71i$	21.8	$+0.52+0.62i$
4.0	$+0.61+0.09i$	13.0	$+0.44-0.66i$	22.0	$+0.72+0.67i$
4.2	$+0.62+0.12i$	13.2	$+0.33-0.58i$	22.2	$+0.92+0.66i$
4.4	$+0.64+0.15i$	13.4	$+0.23-0.49i$	22.4	$+1.11+0.60i$
4.6	$+0.66+0.18i$	13.6	$+0.15-0.38i$	22.6	$+1.26+0.49i$
4.8	$+0.68+0.21i$	13.8	$+0.07-0.25i$	22.8	$+1.38+0.34i$
5.0	$+0.70+0.23i$	14.0	$+0.02-0.10i$	23.0	$+1.45+0.16i$
5.2	$+0.73+0.26i$	14.2	$-0.01+0.05i$	23.2	$+1.46-0.03i$
5.4	$+0.75+0.28i$	14.4	$-0.01+0.21i$	23.4	$+1.41-0.21i$
5.6	$+0.78+0.30i$	14.6	$+0.01+0.38i$	23.6	$+1.30-0.38i$
5.8	$+0.81+0.32i$	14.8	$+0.07+0.55i$	23.8	$+1.14-0.50i$
6.0	$+0.84+0.34i$	15.0	$+0.15+0.70i$	24.0	$+0.95-0.58i$
6.2	$+0.87+0.36i$	15.2	$+0.26+0.85i$	24.2	$+0.73-0.60i$
6.4	$+0.91+0.37i$	15.4	$+0.39+0.98i$	24.4	$+0.51-0.55i$
6.6	$+0.94+0.38i$	15.6	$+0.56+1.09i$	24.6	$+0.30-0.43i$
6.8	$+0.98+0.39i$	15.8	$+0.74+1.17i$	24.8	$+0.13-0.25i$
7.0	$+1.02+0.40i$	16.0	$+0.94+1.22i$	25.0	$\pm 0.00-0.01i$
7.2	$+1.06+0.40i$	16.2	$+1.15+1.23i$	25.2	$-0.05+0.26i$
7.4	$+1.11+0.40i$	16.4	$+1.36+1.20i$	25.4	$-0.04+0.55i$
7.6	$+1.15+0.39i$	16.6	$+1.57+1.14i$	25.6	$+0.06+0.85i$
7.8	$+1.20+0.38i$	16.8	$+1.77+1.04i$	25.8	$+0.25+1.11i$
8.0	$+1.24+0.36i$	17.0	$+1.95+0.90i$	26.0	$+0.50+1.34i$
8.2	$+1.29+0.34i$	17.2	$+2.10+0.72i$	26.2	$+0.82+1.49i$
8.4	$+1.33+0.31i$	17.4	$+2.22+0.52i$	26.4	$+1.17+1.56i$
8.6	$+1.37+0.28i$	17.6	$+2.30+0.29i$	26.6	$+1.55+1.54i$
8.8	$+1.41+0.24i$	17.8	$+2.34+0.06i$	26.8	$+1.92+1.42i$

[a]Values from Haselgrove [H8].